

8. Command

8.1. Command List

Regular Command Set													
Command Function	D/CX	RDX	WRX	D[15:8]	D7	D6	D5	D4	D3	D2	D1	D0	Hex
NOP	0	1	↑	XXXXXXXX	0	0	0	0	0	0	0	0	00h
Soft Reset	0	1	↑	XXXXXXXX	0	0	0	0	0	0	0	1	01h
Read display identification information	0	1	↑	XXXXXXXX	0	0	0	0	0	1	0	0	04h
	1	↑	1	XXXXXXXX	X	X	X	X	X	X	X	X	XX
	1	↑	1	XXXXXXXX	ID1 [7:0]							XX	
	1	↑	1	XXXXXXXX	ID2 [7:0]							XX	
	1	↑	1	XXXXXXXX	ID3 [7:0]							XX	
Read Number of the Errors on DSI	0	↑	1	XXXXXXXX	0	0	0	0	0	1	0	1	05h
	1	↑	1	XXXXXXXX	X	X	X	X	X	X	X	X	XX
	1	↑	1	XXXXXXXX	P[7:0]							XX	
Read Display Status	0	↑	1	XXXXXXXX	0	0	0	0	1	0	0	1	09h
	1	↑	1	XXXXXXXX	X	X	X	X	X	X	X	X	XX
	1	↑	1	XXXXXXXX	D[31:24]							XX	
	1	↑	1	XXXXXXXX	D[23:16]							XX	
	1	↑	1	XXXXXXXX	D[15:8]							XX	
	1	↑	1	XXXXXXXX	D[7:0]							XX	
Read Display Power Mode	0	↑	1	XXXXXXXX	0	0	0	0	1	0	1	0	0Ah
	1	↑	1	XXXXXXXX	X	X	X	X	X	X	X	X	XX
	1	↑	1	XXXXXXXX	D[7:2]							0	0
Read Display MADCTL	0	↑	1	XXXXXXXX	0	0	0	0	1	0	1	1	0Bh
	1	↑	1	XXXXXXXX	X	X	X	X	X	X	X	X	XX
	1	↑	1	XXXXXXXX	D[7:2]							0	0
Read Pixel Format	0	↑	1	XXXXXXXX	0	0	0	0	1	1	0	0	0Ch
	1	↑	1	XXXXXXXX	X	X	X	X	X	X	X	X	XX
	1	↑	1	XXXXXXXX	DPI[3:0]				0	DBI[2:0]			XX
Read Display Image Mode	0	↑	1	XXXXXXXX	0	0	0	0	1	1	0	1	0Dh
	1	↑	1	XXXXXXXX	X	X	X	X	X	X	X	X	XX
	1	↑	1	XXXXXXXX	D[7:0]							XX	
Read Display signal Mode	0	↑	1	XXXXXXXX	0	0	0	0	1	1	1	0	0Eh
	1	↑	1	XXXXXXXX	X	X	X	X	X	X	X	X	XX
	1	↑	1	XXXXXXXX	D7	D6	D5	D4	D3	D2	D1	D0	XX
Read Display Self-Diagnostic Result	0	↑	1	XXXXXXXX	0	0	0	0	1	1	1	1	0Fh
	1	↑	1	XXXXXXXX	X	X	X	X	X	X	X	X	XX
	1	↑	1	XXXXXXXX	D7	D6	0	0	0	0	0	D0	XX
Sleep IN	0	1	↑	XXXXXXXX	0	0	0	1	0	0	0	0	10h
Sleep OUT	0	1	↑	XXXXXXXX	0	0	0	1	0	0	0	1	11h
Partial Mode ON	0	1	↑	XXXXXXXX	0	0	0	1	0	0	1	0	12h
Normal Display Mode ON	0	1	↑	XXXXXXXX	0	0	0	1	0	0	1	1	13h
Display Inversion OFF	0	1	↑	XXXXXXXX	0	0	1	0	0	0	0	0	20h
Display Inversion ON	0	1	↑	XXXXXXXX	0	0	1	0	0	0	0	1	21h
Display OFF	0	1	↑	XXXXXXXX	0	0	1	0	1	0	0	0	28h
Display ON	0	1	↑	XXXXXXXX	0	0	1	0	1	0	0	1	29h
Column Address Set	0	1	↑	XXXXXXXX	0	0	1	0	1	0	1	0	2Ah
	1	1	↑	XXXXXXXX	SC[15:8]							XX	
	1	1	↑	XXXXXXXX	SC[7:0]							XX	
	1	1	↑	XXXXXXXX	EC[15:8]							XX	
	1	1	↑	XXXXXXXX	EC[7:0]							XX	
Page Address Set	0	1	↑	XXXXXXXX	0	0	1	0	1	0	1	1	2Bh
	1	1	↑	XXXXXXXX	SP[15:8]							XX	
	1	1	↑	XXXXXXXX	SP[7:0]							XX	
	1	1	↑	XXXXXXXX	EP[15:8]							XX	
	1	1	↑	XXXXXXXX	EP[7:0]							XX	
Memory Write	0	1	↑	XXXXXXXX	0	0	1	0	1	1	0	0	2Ch
	1	1	↑	D1[15:0]							XX		
	1	1	↑	Dx[15:0]							XX		
	1	1	↑	Dn[15:0]							XX		
Memory Read	0	↑	1	XXXXXXXX	0	0	1	0	1	1	1	0	2Eh
	1	↑	1	XXXXXXXX	X	X	X	X	X	X	X	X	XX

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	1	↑	1	D1[15:0]									XX
	1	↑	1	Dx[15:0]									XX
	1	↑	1	Dn[15:0]									XX
Partial Area	1	↑	1	XXXXXXXX	Pn[7:0]								XX
	0	1	↑	XXXXXXXX	0	0	1	1	0	0	0	0	30h
	1	1	↑	XXXXXXXX	SR[15:8]								XX
	1	1	↑	XXXXXXXX	SR[7:0]								XX
	1	1	↑	XXXXXXXX	ER[15:8]								XX
	1	1	↑	XXXXXXXX	ER[7:0]								XX
Vertical Scrolling Definition	0	1	↑	XXXXXXXX	0	0	1	1	0	0	1	1	33h
	1	1	↑	XXXXXXXX	TFA[15:8]								XX
	1	1	↑	XXXXXXXX	TFA[7:0]								XX
	1	1	↑	XXXXXXXX	VSA[15:8]								XX
	1	1	↑	XXXXXXXX	VSA[7:0]								XX
	1	1	↑	XXXXXXXX	BFA[15:8]								XX
	1	1	↑	XXXXXXXX	BFA[7:0]								XX
Tearing Effect Line OFF	0	1	↑	XXXXXXXX	0	0	1	1	0	1	0	0	34h
Tearing Effect Line ON	0	1	↑	XXXXXXXX	0	0	1	1	0	1	0	1	35h
Memory Access Control	0	1	↑	XXXXXXXX	0	0	1	1	0	1	1	0	36h
	1	1	↑	XXXXXXXX	MY	MX	MV	ML	BGR	MH	X	X	XX
Vertical Scrolling Start Address	0	1	↑	XXXXXXXX	0	0	1	1	0	1	1	1	37h
	1	1	↑	XXXXXXXX	VSP[15:8]								XX
	1	1	↑	XXXXXXXX	VSP[7:0]								XX
Idle Mode OFF	0	1	↑	XXXXXXXX	0	0	1	1	1	0	0	0	38h
Idle Mode ON	0	1	↑	XXXXXXXX	0	0	1	1	1	0	0	1	39h
Interface Pixel Format	0	1	↑	XXXXXXXX	0	0	1	1	1	0	1	0	3Ah
	1	1	↑	XXXXXXXX	0	DPI[6:4]			0	DBI[2:0]			XX
Memory Write Continue	0	1	↑	XXXXXXXX	0	0	1	1	1	1	0	0	3Ch
	1	1	↑	D1[15:0]									XX
	1	1	↑	Dx[15:0]									XX
	1	1	↑	Dn[15:0]									XX
Memory Read Continue	0	↑	1	XXXXXXXX	0	0	1	1	1	1	1	0	3Eh
	1	↑	1	XXXXXXXX	X	X	X	X	X	X	X	X	XX
	1	↑	1	D1[15:0]									XX
	1	↑	1	Dx[15:0]									XX
	1	↑	1	Dn[15:0]									XX
Write Tear Scan line	0	1	↑	XXXXXXXX	0	1	0	0	0	1	0	0	44h
	1	1	↑	XXXXXXXX	N[15:8]								XX
	1	1	↑	XXXXXXXX	N[7:0]								XX
Read Tear Scan Line	0	↑	1	XXXXXXXX	0	1	0	0	0	1	0	1	45h
	1	↑	1	XXXXXXXX	X	X	X	X	X	X	X	X	XX
	1	↑	1	XXXXXXXX	N[15:8]								XX
	1	↑	1	XXXXXXXX	N[7:0]								XX
Write Display Brightness value	0	1	↑	XXXXXXXX	0	1	0	1	0	0	0	1	51h
	1	1	↑	XXXXXXXX	DBV[7:0]								XX
Read Display Brightness Value	0	1	↑	XXXXXXXX	0	1	0	1	1	0	1	0	52h
	1	↑	1	XXXXXXXX	X	X	X	X	X	X	X	X	XX
	1	↑	1	XXXXXXXX	DBV[7:0]								XX
Write CTRL Display value	0	1	↑	XXXXXXXX	0	1	0	1	0	0	1	1	53h
	1	1	↑	XXXXXXXX	0	0	BCTRL	0	DD	BL	0	0	XX
Read CTRL Display value	0	1	↑	XXXXXXXX	0	1	0	1	0	1	0	0	54h
	1	↑	1	XXXXXXXX	X	X	X	X	X	X	X	X	XX
	1	↑	1	XXXXXXXX	0	0	BCTRL	0	DD	BL	0	0	XX
Write Content Adaptive Brightness Control value	0	1	↑	XXXXXXXX	0	1	0	1	0	1	0	1	55h
	1	1	↑	XXXXXXXX	0	0	0	0	0	0	C[1:0]		XX
Read Content Adaptive Brightness Control value	0	1	↑	XXXXXXXX	0	1	0	1	0	1	1	0	56h
	1	↑	1	XXXXXXXX	X	X	X	X	X	X	X	X	XX
	1	↑	1	XXXXXXXX	0	0	0	0	0	0	C[1:0]		XX
Write CABC Minimum Brightness	0	1	↑	XXXXXXXX	0	1	0	1	1	1	1	0	5Eh
	1	1	↑	XXXXXXXX	CMB[7:0]								XX
Read CABC Minimum Brightness	0	1	↑	XXXXXXXX	0	1	0	1	1	1	1	1	5Fh
	1	↑	1	XXXXXXXX	X	X	X	X	X	X	X	X	XX
	1	↑	1	XXXXXXXX	CMB[7:0]								XX
Read First Checksum	0	1	↑	XXXXXXXX	1	0	1	0	1	0	1	0	AAh
	1	↑	1	XXXXXXXX	X	X	X	X	X	X	X	X	XX

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	1	↑	1	XXXXXXXX	FCS[7:0]								XX
Read Continue Checksum	0	1	↑	XXXXXXXX	1	0	1	0	1	1	1	1	AFh
	1	↑	1	XXXXXXXX	X	X	X	X	X	X	X	X	XX
	1	↑	1	XXXXXXXX	CCS[7:0]								XX
Read ID1	0	1	↑	XXXXXXXX	1	1	0	1	1	0	1	0	DAh
	1	↑	1	XXXXXXXX	X	X	X	X	X	X	X	X	XX
	1	↑	1	XXXXXXXX	ID1[7:0]								XX
Read ID2	0	1	↑	XXXXXXXX	1	0	1	0	1	0	1	1	DBh
	1	↑	1	XXXXXXXX	X	X	X	X	X	X	X	X	XX
	1	↑	1	XXXXXXXX	ID2[7:0]								XX
Read ID3	0	1	↑	XXXXXXXX	1	0	1	0	1	1	0	0	DCh
	1	↑	1	XXXXXXXX	X	X	X	X	X	X	X	X	XX
	1	↑	1	XXXXXXXX	ID3[7:0]								XX

Extended Command Set													
Command Function	D/CX	RDX	WRX	D[15:8]	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Interface Mode Control	0	1	↑	XXXXXXXX	1	0	1	1	0	0	0	0	B0h
	1	1	↑	XXXXXXXX	SDA_EN	0	0	0	VSPL	HSPL	DPL	EPL	XX
Frame Rate Control (In Normal Mode/Full Colors)	0	1	↑	XXXXXXXX	1	0	1	1	0	0	0	1	B1h
	1	1	↑	XXXXXXXX	FRS[3:0]				0	0	DIVA[1:0]		XX
	1	1	↑	XXXXXXXX	0	0	0	RTNA[4:0]					XX
Frame Rate Control (In Idle Mode/8 colors)	0	1	↑	XXXXXXXX	1	0	1	1	0	0	1	0	B2h
	1	1	↑	XXXXXXXX	0	0	0	0	0	0	DIVB[1:0]		XX
	1	1	↑	XXXXXXXX	0	0	0	RTNB[4:0]					XX
Frame Rate Control (In Partial Mode/Full colors)	0	1	↑	XXXXXXXX	1	0	1	1	0	0	1	1	B3h
	1	1	↑	XXXXXXXX	0	0	0	0	0	0	DIVC[1:0]		XX
	1	1	↑	XXXXXXXX	0	0	0	RTN[4:0]					XX
Display Inversion Control	0	1	↑	XXXXXXXX	1	0	1	1	0	1	0	0	B4h
	1	1	↑	XXXXXXXX	0	0	0	ZINV	0	0	DINV[1:0]		XX
Blanking Porch Control	0	1	↑	XXXXXXXX	1	0	1	1	0	1	0	1	B5h
	1	1	↑	XXXXXXXX	VFP[7:0]								XX
	1	1	↑	XXXXXXXX	VBP[7:0]								XX
	1	1	↑	XXXXXXXX	0	0	0	HFP[4:0]				XX	
	1	1	↑	XXXXXXXX	HBP[7:0]								XX
Display Function Control	0	1	↑	XXXXXXXX	1	0	1	1	0	1	1	0	B6h
	1	1	↑	XXXXXXXX	BYPASS	0	RM	DM	PTG[1:0]		PT[1:0]		XX
	1	1	↑	XXXXXXXX	0	GS	SS	SM	ISC[3:0]				XX
	1	1	↑	XXXXXXXX	0	0	NL[5:0]						XX
Entry Mode Set	0	1	↑	XXXXXXXX	1	0	1	1	0	1	1	1	B7h
	1	1	↑	XXXXXXXX	EPF[1:0]		0	0	DSTB	GON	DTE	GAS	XX
Power Control 1	0	1	↑	XXXXXXXX	1	1	0	0	0	0	0	0	C0h
	1	1	↑	XXXXXXXX	0	0	0	VRH1[4:0]				XX	
	1	1	↑	XXXXXXXX	0	0	0	VRH2[4:0]				XX	
Power Control 2	0	1	↑	XXXXXXXX	1	1	0	0	0	0	0	1	C1h
	1	1	↑	XXXXXXXX	0	SAP[2:0]			BT[2:0]				XX
	1	1	↑	XXXXXXXX	0	0	0	0	0	VC[2:0]			XX
Power Control 3	0	1	↑	XXXXXXXX	1	1	0	0	0	0	1	0	C2h
	1	1	↑	XXXXXXXX	0	DCA1[2:0]			0	DCA0[2:0]			XX
Power Control 4	0	1	↑	XXXXXXXX	1	1	0	0	0	0	1	1	C3h
	1	1	↑	XXXXXXXX	0	DCB1[2:0]			0	DCB0[2:0]			XX
Power Control 5	0	1	↑	XXXXXXXX	1	1	0	0	0	1	0	0	C4h
	1	1	↑	XXXXXXXX	0	DCC2[2:0]			0	DCC0[2:0]			XX
VCOM Control 1	0	↑	1	XXXXXXXX	1	1	0	0	0	1	0	1	C5h
	1	1	↑	XXXXXXXX	0	0	0	0	0	0	0	nVM	XX
	1	1	↑	XXXXXXXX	VCM_REG[7:0]								XX
	1	1	↑	XXXXXXXX	VCM_REG_EN	0	0	0	0	0	0	0	XX
	1	↑	1	XXXXXXXX	VCM_OUT[7:0]								XX
CABC Control 1	0	1	↑	XXXXXXXX	1	1	0	0	0	1	1	0	C6h

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	1	1	↑	XXXXXXXX	SCD_VLINE[7:0]								XX	
	1	1	↑	XXXXXXXX	0	0	0	0	0	SCD_VLINE[10:8]			XX	
	0	1	↑	XXXXXXXX	1	1	0	0	1	0	0		C8h	
	1	1	↑	XXXXXXXX	0	0	0	0	0	LEDONR	LEDONPOL	PWMPOL	XX	
	1	1	↑	XXXXXXXX	PWM_DIV[7:0]								XX	
	0	1	↑	XXXXXXXX	1	1	0	0	1	0	0	1	C9h	
	1	1	↑	XXXXXXXX	THRES_MOV[3:0]				THRES_STILL[3:0]				XX	
	0	1	↑	XXXXXXXX	1	1	0	0	1	0	1	0	CAh	
	1	1	↑	XXXXXXXX	0	0	0	0	THRES_UI[3:0]				XX	
	0	1	↑	XXXXXXXX	1	1	0	0	1	0	1	1	CBh	
	1	1	↑	XXXXXXXX	DTH_MOV[3:0]				DTH_STILL[3:0]				XX	
	0	1	↑	XXXXXXXX	1	1	0	0	1	1	0	0	CCh	
	1	1	↑	XXXXXXXX	0	0	0	0	DTH_UI[3:0]				XX	
	0	1	↑	XXXXXXXX	1	1	0	0	1	1	0	1	CDh	
	1	1	↑	XXXXXXXX	0	DIM_MOV[2:0]			0	DIM_STILL[2:0]			XX	
	0	1	↑	XXXXXXXX	1	1	0	0	1	1	1	0	CEh	
	1	1	↑	XXXXXXXX	DIM_MIN[3:0]				0	DIM_UI[2:0]			XX	
	0	1	↑	XXXXXXXX									CFh	
	1	1	↑	XXXXXXXX	PWM_DIV[7:0]								XX	
		0	1	↑	XXXXXXXX	1	1	0	1	0	0	0	0	D0h
1		1	↑	XXXXXXXX	0	0	0	PGM_ADR[4:0]					XX	
1		1	↑	XXXXXXXX	PGM_DATA[7:0]								XX	
		0	1	↑	XXXXXXXX	1	1	0	1	0	0	0	1	D1h
	1	1	↑	XXXXXXXX	KEY[23:16]								XX	
	0	↑	1	XXXXXXXX	KEY[15:8]								XX	
	1	↑	1	XXXXXXXX	KEY[7:0]								XX	
	1	↑	1	XXXXXXXX	1	1	0	1	0	0	1	0	D2h	
	1	↑	1	XXXXXXXX	X	X	X	X	X	X	X	X	XX	
	0	1	↑	XXXXXXXX	ID2_CNT[3:0]				ID1_CNT[3:0]				XX	
	1	1	↑	XXXXXXXX	VMF_CNT[3:0]				ID3_CNT[3:0]				XX	
	1	1	↑	XXXXXXXX	BUSY	0	0	0	0	0	0	MDDI_V12	XX	
	1	1	↑	XXXXXXXX	OTP_DATA[7:0]								XX	
	0	↑	1	XXXXXXXX	1	1	0	1	0	0	1	1	D3h	
	1	↑	1	XXXXXXXX	X	X	X	X	X	X	X	X	XX	
	1	↑	1	XXXXXXXX	ID41[7:0]								XX	
	1	↑	1	XXXXXXXX	ID42[7:0]								XX	
	1	↑	1	XXXXXXXX	ID43[7:0]								XX	
PGAMCTRL (Positive Gamma Control)	0	1	↑	XXXXXXXX	1	1	1	0	0	0	0	0	E0h	
	1	1	↑	XXXXXXXX	0	0	0	0	VP0[3:0]					XX
	1	1	↑	XXXXXXXX	0	0	VP1[5:0]						XX	
	1	1	↑	XXXXXXXX	0	0	VP2[5:0]						XX	
	1	1	↑	XXXXXXXX	0	0	0	0	VP4[3:0]				XX	
	1	1	↑	XXXXXXXX	0	0	0	VP6[4:0]					XX	
	1	1	↑	XXXXXXXX	0	0	0	0	VP13[3:0]				XX	
	1	1	↑	XXXXXXXX	0	VP20[6:0]							XX	
	1	1	↑	XXXXXXXX	VP36[3:0]				VP27[3:0]				XX	
	1	1	↑	XXXXXXXX	0	VP43[6:0]							XX	
	1	1	↑	XXXXXXXX	0	0	0	0	VP50[3:0]				XX	
	1	1	↑	XXXXXXXX	0	0	0	VP57[4:0]					XX	
	1	1	↑	XXXXXXXX	0	0	0	0	VP59[3:0]				XX	
	1	1	↑	XXXXXXXX	0	0	VP61[5:0]						XX	
	1	1	↑	XXXXXXXX	0	0	VP62[5:0]						XX	
	1	1	↑	XXXXXXXX	0	0	0	0	VP63[3:0]				XX	
NGAMCTRL (Negative Gamma Control)	0	1	↑	XXXXXXXX	1	1	1	0	0	0	0	1	E1h	
	1	1	↑	XXXXXXXX	0	0	0	0	VN0[3:0]				XX	
	1	1	↑	XXXXXXXX	0	0	VN1[5:0]						XX	
	1	1	↑	XXXXXXXX	0	0	VN2[5:0]						XX	

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	1	1	↑	XXXXXXXX	0	0	0	0	VN4[3:0]				XX
	1	1	↑	XXXXXXXX	0	0	0		VN6[4:0]				XX
	1	1	↑	XXXXXXXX	0	0	0		VN13[3:0]				XX
	1	1	↑	XXXXXXXX	0	VN20[6:0]							XX
	1	1	↑	XXXXXXXX	VN36[3:0]				VN27[3:0]				XX
	1	1	↑	XXXXXXXX	0	VN43[6:0]							XX
	1	1	↑	XXXXXXXX	0	0	0	0	VN50[3:0]				XX
	1	1	↑	XXXXXXXX	0	0	0		VN57[4:0]				XX
	1	1	↑	XXXXXXXX	0	0	0	0	VN59[3:0]				XX
	1	1	↑	XXXXXXXX	0	0	VN61[5:0]						XX
	1	1	↑	XXXXXXXX	0	0	VN62[5:0]						XX
	1	1	↑	XXXXXXXX	0	0	0	0	VN63[3:0]				XX
Digital Gamma Control 1	0	1	↑	XXXXXXXX	1	1	1	0	0	0	0	1	E2h
	1	1	↑	XXXXXXXX	RCA0[3:0]				BCA0[3:0]				XX
	1	1	↑	XXXXXXXX	RCAx[3:0]				BCAx[3:0]				XX
	1	1	↑	XXXXXXXX	RCA63[3:0]				BCA63[3:0]				XX
Digital Gamma Control 2	0	1	↑	XXXXXXXX	1	1	1	0	0	0	0	1	E3h
	1	1	↑	XXXXXXXX	RFA0[3:0]				BFA0[3:0]				XX
	1	1	↑	XXXXXXXX	RFAx[3:0]				BFAx[3:0]				XX
	1	1	↑	XXXXXXXX	RFA255[3:0]				BFA255[3:0]				XX
SPI Read Command Setting	0	1	↑	XXXXXXXX	1	1	1	1	1	0	1	1	FBh
	1	1	↑	XXXXXXXX	0	0	0	SPI_READ_EN		SPI_CNT[3:0]			

8.2. Command Description

8.2.1. NOP (00h)

00h	NOP (No Operation)																								
	D/CX	RDX	WRX	D[15:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX												
Command	0	1	↑	XXXXXXXX	0	0	0	0	0	0	0	0	00h												
Parameter	No parameter																								
Description	This command is an empty command; it does not have any effect on ILI9486. However it can be used to terminate Frame Memory Write or Read as described in RAMWR (Memory Write) and RAMRD (Memory Read) Commands. X = Don't care.																								
Restriction	None																								
Register Availability	<table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table>													Status	Availability	Normal Mode On, Idle Mode Off, Sleep Out	Yes	Normal Mode On, Idle Mode On, Sleep Out	Yes	Partial Mode On, Idle Mode Off, Sleep Out	Yes	Partial Mode On, Idle Mode On, Sleep Out	Yes	Sleep In	Yes
Status	Availability																								
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Normal Mode On, Idle Mode On, Sleep Out	Yes																								
Partial Mode On, Idle Mode Off, Sleep Out	Yes																								
Partial Mode On, Idle Mode On, Sleep Out	Yes																								
Sleep In	Yes																								
Default	<table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>N/A</td></tr><tr><td>SW Reset</td><td>N/A</td></tr><tr><td>HW Reset</td><td>N/A</td></tr></table>													Status	Default Value	Power On Sequence	N/A	SW Reset	N/A	HW Reset	N/A				
Status	Default Value																								
Power On Sequence	N/A																								
SW Reset	N/A																								
HW Reset	N/A																								
Flow Chart	None																								

8.2.2. Soft Reset (01h)

01h	SWRESET (Soft Reset)																								
	D/CX	RDX	WRX	D[15:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX												
Command	0	1	↑	XXXXXXXX	0	0	0	0	0	0	0	1	01h												
Parameter	No parameter																								
Description	When the Software Reset command is written, it causes software reset. It resets the commands and parameters to their S/W Reset default values. (See default tables in each command description.) The display is blank immediately Note: The Frame Memory contents is kept or not by this command. X = Don't care																								
Restriction	It will be necessary to wait 5msec before sending new command following software reset. The display module loads all display supplier factory default values to the registers during this 5msec. If Software Reset is applied during Sleep Out mode, it will be necessary to wait 120msec before sending Sleep out command. Software Reset Command cannot be sent during Sleep Out sequence.																								
Register Availability	<table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table>													Status	Availability	Normal Mode On, Idle Mode Off, Sleep Out	Yes	Normal Mode On, Idle Mode On, Sleep Out	Yes	Partial Mode On, Idle Mode Off, Sleep Out	Yes	Partial Mode On, Idle Mode On, Sleep Out	Yes	Sleep In	Yes
Status	Availability																								
Normal Mode On, Idle Mode Off, Sleep Out	Yes																								
Normal Mode On, Idle Mode On, Sleep Out	Yes																								
Partial Mode On, Idle Mode Off, Sleep Out	Yes																								
Partial Mode On, Idle Mode On, Sleep Out	Yes																								
Sleep In	Yes																								
Default	<table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>N/A</td></tr><tr><td>SW Reset</td><td>N/A</td></tr><tr><td>HW Reset</td><td>N/A</td></tr></table>													Status	Default Value	Power On Sequence	N/A	SW Reset	N/A	HW Reset	N/A				
Status	Default Value																								
Power On Sequence	N/A																								
SW Reset	N/A																								
HW Reset	N/A																								
Flow Chart	SWRESET(01h) Display whole blank screen Set Commands to S/W Default Values Sleep In Mode Legend Command Parameter Display Action Mode Sequential transfer																								

8.2.3. Read display identification information (04h)

04h	RDDIDIF (Read Display Identification Information)																								
	D/CX	RDX	WRX	D[17:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX												
Command	0	1	↑	XX	0	0	0	0	0	1	0	0	04h												
1 st Parameter	1	↑	1	XX	X	X	X	X	X	X	X	X	X												
2 nd Parameter	1	↑	1	XX	ID1 [7:0]								XX												
3 rd Parameter	1	↑	1	XX	ID2 [7:0]								XX												
4 th Parameter	1	↑	1	XX	ID3 [7:0]								XX												
Description	This read byte returns 24 bits display identification information.																								
	The 1 st parameter is dummy data.																								
	The 2 nd parameter (ID1 [7:0]): LCD module's manufacturer ID.																								
	The 3 rd parameter (ID2 [7:0]): LCD module/driver version ID.																								
	The 4 th parameter (ID3 [7:0]): LCD module/driver ID.																								
Restriction																									
Register Availability	<table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table>													Status	Availability	Normal Mode On, Idle Mode Off, Sleep Out	Yes	Normal Mode On, Idle Mode On, Sleep Out	Yes	Partial Mode On, Idle Mode Off, Sleep Out	Yes	Partial Mode On, Idle Mode On, Sleep Out	Yes	Sleep In	Yes
	Status	Availability																							
	Normal Mode On, Idle Mode Off, Sleep Out	Yes																							
	Normal Mode On, Idle Mode On, Sleep Out	Yes																							
	Partial Mode On, Idle Mode Off, Sleep Out	Yes																							
	Partial Mode On, Idle Mode On, Sleep Out	Yes																							
Sleep In	Yes																								
Default	<table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>See description</td></tr><tr><td>SW Reset</td><td>See description</td></tr><tr><td>HW Reset</td><td>See description</td></tr></table>													Status	Default Value	Power On Sequence	See description	SW Reset	See description	HW Reset	See description				
	Status	Default Value																							
	Power On Sequence	See description																							
	SW Reset	See description																							
HW Reset	See description																								
Flow Chart	RDDIDIF(04h) ↓ Host ----- Driver 1st Parameter: Dummy Read 2nd Parameter: Send LCD module's manufacturer information 3rd Parameter: Send panel type and LCM/driver version information 4th Parameter: Send module/driver information Legend Command Parameter Display Action Mode Sequential transfer																								

8.2.4. Read Number of the Errors on DSI (05h)

05h	RDNUMED (Read Number of the Errors on DSI)																								
	D/CX	RDX	WRX	D[15:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX												
Command	0	1	↑	XXXXXXXX	0	0	0	0	0	1	0	1	05h												
1 st Parameter	1	↑	1	XXXXXXXX	X	X	X	X	X	X	X	X	XX												
2 nd Parameter	1	↑	1	XXXXXXXX	P[7:0]								XX												
Description	The second parameter is telling a number of the errors on DSI. The more detailed description of the bits is below. P [6..0] bits are telling a number of the errors. P [7] is set to '1' if there is overflow with P[6..0] bits. P [7...0] bits are set to '0's (as well as RDDSM(0Eh)'s D0 is set '0' at the same time) after there is sent the second parameter information (= The read function is completed). This function is always returning P [7...0] = 00h if the parallel MCU interface is selected. X = can be '0' or '1'																								
Restriction	ILI9486 is sending 2nd parameter value on the data lines if the MCU wants to read more than one parameter (= more than 2 RDX cycle) on parallel MCU interface. Only 2nd parameter is sent on DSI (The 1st parameter is not sent).																								
Register Availability	<table><thead><tr><th>Status</th><th>Availability</th></tr></thead><tbody><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></tbody></table>													Status	Availability	Normal Mode On, Idle Mode Off, Sleep Out	Yes	Normal Mode On, Idle Mode On, Sleep Out	Yes	Partial Mode On, Idle Mode Off, Sleep Out	Yes	Partial Mode On, Idle Mode On, Sleep Out	Yes	Sleep In	Yes
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Partial Mode On, Idle Mode Off, Sleep Out	Yes																								
Partial Mode On, Idle Mode On, Sleep Out	Yes																								
Sleep In	Yes																								
Default	<table><thead><tr><th>Status</th><th>Default Value</th></tr></thead><tbody><tr><td>Power On Sequence</td><td>08_{HEX}</td></tr><tr><td>SW Reset</td><td>08_{HEX}</td></tr><tr><td>HW Reset</td><td>08_{HEX}</td></tr></tbody></table>													Status	Default Value	Power On Sequence	08 _{HEX}	SW Reset	08 _{HEX}	HW Reset	08 _{HEX}				
Status	Default Value																								
Power On Sequence	08 _{HEX}																								
SW Reset	08 _{HEX}																								
HW Reset	08 _{HEX}																								
Flow Chart	Read number of the Errors on DSI ↓ 1st Parameter: Dummy Read 2nd Parameter: Read ↓ P[7:0] = 00h RDDSM(0Eh)' s D0 = '0' Host ILI9486 Legend Command Parameter Display Action Mode Sequential transfer																								

8.2.5. Read Display Status (09h)

09h	RDDST (Read Display Status)												
	D/CX	RDX	WRX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	0	0	0	0	1	0	0	1	09h
1 st Parameter	1	↑	1	XX	X	X	X	X	X	X	X	X	XX
2 nd Parameter	1	↑	1	XX	D [31:25]							0	XX
3 rd Parameter	1	↑	1	XX	0	D [22:20]			D [19:16]				XX
4 th Parameter	1	↑	1	XX	D15	0	D13	0	0	D [10:8]			XX
5 th Parameter	1	↑	1	XX	D [7:5]			0	0	0	0	0	XX
Description	This command indicates the current status of the display as described in the table below:												
	Bit	Description			Value	Status							
	D31	Booster voltage status		0	Booster OFF								
				1	Booster ON								
	D30	Row address order		0	Top to Bottom (When MADCTL B7='0')								
				1	Bottom to Top (When MADCTL B7='1')								
	D29	Column address order		0	Left to Right (When MADCTL B6='0').								
				1	Right to Left (When MADCTL B6='1').								
	D28	Row/column exchange		0	Normal Mode (When MADCTL B5='0').								
				1	Reverse Mode (When MADCTL B5='1').								
	D27	Vertical refresh		0	LCD Refresh Top to Bottom (When MADCTL B4='0')								
				1	LCD Refresh Bottom to Top (When MADCTL B4='1').								
	D26	RGB/BGR order		0	RGB (When MADCTL B3='0')								
				1	BGR (When MADCTL B3='1')								
	D25	Horizontal refresh order		0	LCD Refresh Left to Right (When MADCTL B2='0')								
				1	LCD Refresh Right to Left (When MADCTL B2='1')								
	D24	Not used		0	---								
	D23	Not used		0	---								
	D22	Interface color pixel format definition		011	12-bit/pixel								
	D21			101	16-bit/pixel								
	D20			110	18-bit/pixel								
	D19	Idle mode ON/OFF		0	Idle Mode OFF								
				1	Idle Mode ON								
	D18	Partial mode ON/OFF		0	Partial Mode OFF								
				1	Partial Mode ON.								
	D17	Sleep IN/OUT		0	Sleep IN Mode								
				1	Sleep OUT Mode.								
	D16	Display normal mode ON/OFF		0	Display Normal Mode OFF.								
				1	Display Normal Mode ON.								
	D15	Vertical scrolling status		0	Vertical Scroll OFF								
				1	Vertical Scroll ON								
	D14	Not used		0	---								
	D13	Inversion status		0	Inversion OFF								
				1	Inversion ON								
	D12	All pixel ON		0	Not defined								
	D11	All pixel OFF		0	Not defined								
	D10	Display ON/OFF		0	Display is OFF								
				1	Display is ON								
	D9	Tearing effect line ON/OFF		0	Tearing Effect Line OFF								
				1	Tearing Effect ON								
	D[8:6]	Gamma curve selection		000	GC0								
				001	GC1								
				010	GC2								
				011	GC3								
				other	Not defined								

		D5	Tearing effect line mode	0	Mode 1, V-Blanking only												
				1	Mode 2, both H-Blanking and V-Blanking.												
		D4	Not used	0	---												
		D3	Not used	0	---												
		D2	Not used	0	---												
		D1	Not used	0	---												
		D0	Not used	0	---												
X = Don't care																	
Restriction																	
Register Availability	<table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table>					Status	Availability	Normal Mode On, Idle Mode Off, Sleep Out	Yes	Normal Mode On, Idle Mode On, Sleep Out	Yes	Partial Mode On, Idle Mode Off, Sleep Out	Yes	Partial Mode On, Idle Mode On, Sleep Out	Yes	Sleep In	Yes
Status	Availability																
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Normal Mode On, Idle Mode On, Sleep Out	Yes																
Partial Mode On, Idle Mode Off, Sleep Out	Yes																
Partial Mode On, Idle Mode On, Sleep Out	Yes																
Sleep In	Yes																
Default	<table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>32'h00610000h</td></tr><tr><td>SW Reset</td><td>32'h00610000h</td></tr><tr><td>HW Reset</td><td>32'h00610000h</td></tr></table>					Status	Default Value	Power On Sequence	32'h00610000h	SW Reset	32'h00610000h	HW Reset	32'h00610000h				
Status	Default Value																
Power On Sequence	32'h00610000h																
SW Reset	32'h00610000h																
HW Reset	32'h00610000h																
Flow Chart	RDDST(09h) Host Driver 1st Parameter: Dummy Read 2nd Parameter: Send D[31:25] display status 3rd Parameter: Send D[19:16] display status 4th Parameter: Send D[10:8] display status 5th Parameter: Send D[7:5] display status Legend Command Parameter Display Action Mode Sequential transfer																

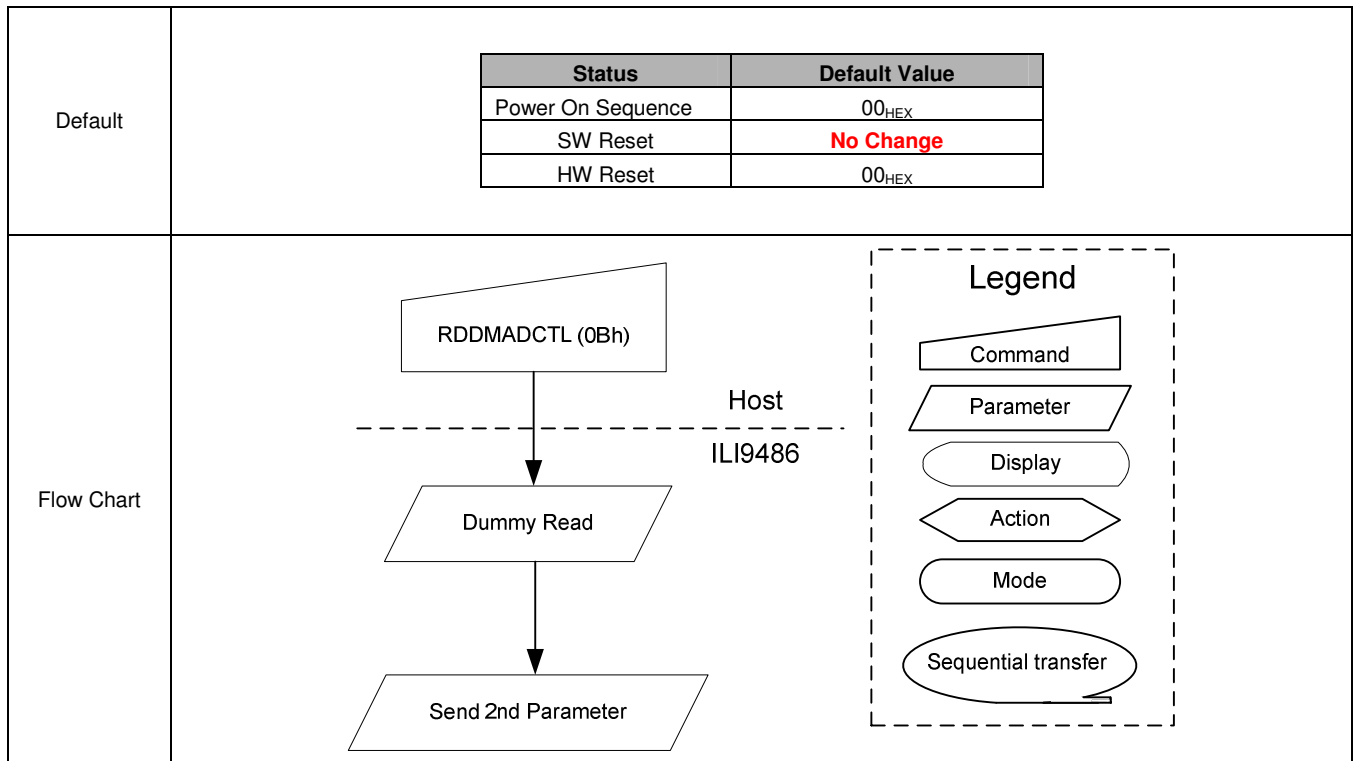
8.2.6. Read Display Power Mode (0Ah)

0Ah	RDDPM (Read Display Power Mode)																																							
	D/CX	RDX	WRX	D[15:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX																											
Command	0	1	↑	XXXXXXXX	0	0	0	0	1	0	1	0	0Ah																											
1 st Parameter	1	↑	1	XXXXXXXX	X	X	X	X	X	X	X	X	XX																											
2 nd Parameter	1	↑	1	XXXXXXXX	D[7:2]						0	0	XX																											
Description	This command indicates the current status of the display as described in the table below:																																							
	<table><tr><th>Bit</th><th>Description</th><th>Comment</th></tr><tr><td>D7</td><td>Booster Voltage Status</td><td></td></tr><tr><td>D6</td><td>Idle Mode On/Off</td><td></td></tr><tr><td>D5</td><td>Partial Mode On/Off</td><td></td></tr><tr><td>D4</td><td>Sleep In/Out</td><td></td></tr><tr><td>D3</td><td>Display Normal Mode On/Off</td><td></td></tr><tr><td>D2</td><td>Display On/Off</td><td></td></tr><tr><td>D1</td><td>Not Defined</td><td>Set to '0'</td></tr><tr><td>D0</td><td>Not Defined</td><td>Set to '0'</td></tr></table>													Bit	Description	Comment	D7	Booster Voltage Status		D6	Idle Mode On/Off		D5	Partial Mode On/Off		D4	Sleep In/Out		D3	Display Normal Mode On/Off		D2	Display On/Off		D1	Not Defined	Set to '0'	D0	Not Defined	Set to '0'
	Bit	Description	Comment																																					
	D7	Booster Voltage Status																																						
	D6	Idle Mode On/Off																																						
	D5	Partial Mode On/Off																																						
	D4	Sleep In/Out																																						
	D3	Display Normal Mode On/Off																																						
	D2	Display On/Off																																						
	D1	Not Defined	Set to '0'																																					
	D0	Not Defined	Set to '0'																																					
	Bit D7 – Booster Voltage Status																																							
	'0' = Booster Off or has a fault.																																							
	'1' = Booster On and working OK.																																							
	Bit D6 - Idle Mode On/Off																																							
	'0' = Idle Mode Off.																																							
	'1' = Idle Mode On.																																							
	Bit D5 – Partial Mode On/Off																																							
	'0' = Partial Mode Off.																																							
	'1' = Partial Mode On.																																							
Bit D4 – Sleep In/Out																																								
'0' = Sleep In Mode.																																								
'1' = Sleep Out Mode.																																								
Bit D3 – Display Normal Mode On/Off																																								
'0' = Display Normal Mode Off.																																								
'1' = Display Normal Mode On.																																								
Bit D2 – Display On/Off																																								
'0' = Display is Off.																																								
'1' = Display is On.																																								
Bit D1 – Not Defined																																								
'This bit is not applicable for this project, so it is set to '0'																																								
Bit D0 – Not Defined																																								
'This bit is not applicable for this project, so it is set to '0'																																								
X = Don't care																																								

Restriction	ILI9486 is sending 2nd parameter value on the data lines if the MCU wants to read more than one parameter (= more than 2 RDX cycle) on parallel MCU interface. Only 2nd parameter is sent on DSI (The 1st parameter is not sent).												
Register Availability	<table border="1"> <thead> <tr> <th>Status</th><th>Availability</th></tr> </thead> <tbody> <tr> <td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr> <tr> <td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr> <tr> <td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr> <tr> <td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr> <tr> <td>Sleep In</td><td>Yes</td></tr> </tbody> </table>	Status	Availability	Normal Mode On, Idle Mode Off, Sleep Out	Yes	Normal Mode On, Idle Mode On, Sleep Out	Yes	Partial Mode On, Idle Mode Off, Sleep Out	Yes	Partial Mode On, Idle Mode On, Sleep Out	Yes	Sleep In	Yes
Status	Availability												
Normal Mode On, Idle Mode Off, Sleep Out	Yes												
Normal Mode On, Idle Mode On, Sleep Out	Yes												
Partial Mode On, Idle Mode Off, Sleep Out	Yes												
Partial Mode On, Idle Mode On, Sleep Out	Yes												
Sleep In	Yes												
Default	<table border="1"> <thead> <tr> <th>Status</th><th>Default Value</th></tr> </thead> <tbody> <tr> <td>Power On Sequence</td><td>08_{HEX}</td></tr> <tr> <td>SW Reset</td><td>08_{HEX}</td></tr> <tr> <td>HW Reset</td><td>08_{HEX}</td></tr> </tbody> </table>	Status	Default Value	Power On Sequence	08 _{HEX}	SW Reset	08 _{HEX}	HW Reset	08 _{HEX}				
Status	Default Value												
Power On Sequence	08 _{HEX}												
SW Reset	08 _{HEX}												
HW Reset	08 _{HEX}												
Flow Chart	<pre> graph TD subgraph Host RDDPM[RDDPM (0Ah)] end subgraph ILI9486 DummyRead[/Dummy Read/] SendParam[/Send 2nd Parameter/] end RDDPM --> DummyRead DummyRead --> SendParam </pre> Legend - Command - Parameter - Display - Action - Mode - Sequential transfer												

8.2.7. Read Display MADCTL (0Bh)

0Bh	RDDMADCTL (Read Display MADCTL)																																							
	D/CX	RDX	WRX	D[15:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX																											
Command	0	1	↑	XXXXXXXX	0	0	0	0	1	0	1	1	0Bh																											
1 st Parameter	1	↑	1	XXXXXXXX	X	X	X	X	X	X	X	X	XX																											
2 nd Parameter	1	↑	1	XXXXXXXX	D[7:2]						0	0	XX																											
Description	This command indicates the current status of the display as described in the table below:																																							
	<table><tr><th>Bit</th><th>Description</th><th>Comment</th></tr><tr><td>D7</td><td>Page Address Order</td><td></td></tr><tr><td>D6</td><td>Column Address Order</td><td></td></tr><tr><td>D5</td><td>Page/Column Order</td><td></td></tr><tr><td>D4</td><td>Line Address Order</td><td></td></tr><tr><td>D3</td><td>RGB/BGR Order</td><td></td></tr><tr><td>D2</td><td>Display Data Latch Data Order</td><td></td></tr><tr><td>D1</td><td>Reserved</td><td>Set to '0'</td></tr><tr><td>D0</td><td>Reserved</td><td>Set to '0'</td></tr></table>													Bit	Description	Comment	D7	Page Address Order		D6	Column Address Order		D5	Page/Column Order		D4	Line Address Order		D3	RGB/BGR Order		D2	Display Data Latch Data Order		D1	Reserved	Set to '0'	D0	Reserved	Set to '0'
	Bit	Description	Comment																																					
	D7	Page Address Order																																						
	D6	Column Address Order																																						
	D5	Page/Column Order																																						
	D4	Line Address Order																																						
	D3	RGB/BGR Order																																						
	D2	Display Data Latch Data Order																																						
	D1	Reserved	Set to '0'																																					
D0	Reserved	Set to '0'																																						
◆ Bit D7 – Page Address Order ‘0’ = Top to Bottom ‘1’ = Bottom to Top																																								
◆ Bit D6 – Column Address Order ‘0’ = Left to Right ‘1’ = Right to Left																																								
◆ Bit D5 - Page/Column Order ‘0’ = Normal Mode ‘1’ = Reverse Mode Note: For Bits D7 to D5, also refer to Section 9.3 MCU to memory write/read direction.																																								
◆ Bit D4 – Line Address Order ‘0’ = LCD Refresh Top to Bottom ‘1’ = LCD Refresh Bottom to Top																																								
◆ Bit D3 – RGB/BGR Order ‘0’ = RGB ‘1’ = BGR																																								
◆ Bit D2 – Display Data Latch Data Order ‘0’ = LCD Refresh Left to Right ‘1’ = LCD Refresh Right to Left																																								
Restriction	ILI9486 is sending 2nd parameter value on the data lines if the MCU wants to read more than one parameter (= more than 2 RDX cycle) on parallel MCU interface. Only 2nd parameter is sent on DSI (The 1st parameter is not sent).																																							
Register Availability	<table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table>													Status	Availability	Normal Mode On, Idle Mode Off, Sleep Out	Yes	Normal Mode On, Idle Mode On, Sleep Out	Yes	Partial Mode On, Idle Mode Off, Sleep Out	Yes	Partial Mode On, Idle Mode On, Sleep Out	Yes	Sleep In	Yes															
Status	Availability																																							
Normal Mode On, Idle Mode Off, Sleep Out	Yes																																							
Normal Mode On, Idle Mode On, Sleep Out	Yes																																							
Partial Mode On, Idle Mode Off, Sleep Out	Yes																																							
Partial Mode On, Idle Mode On, Sleep Out	Yes																																							
Sleep In	Yes																																							

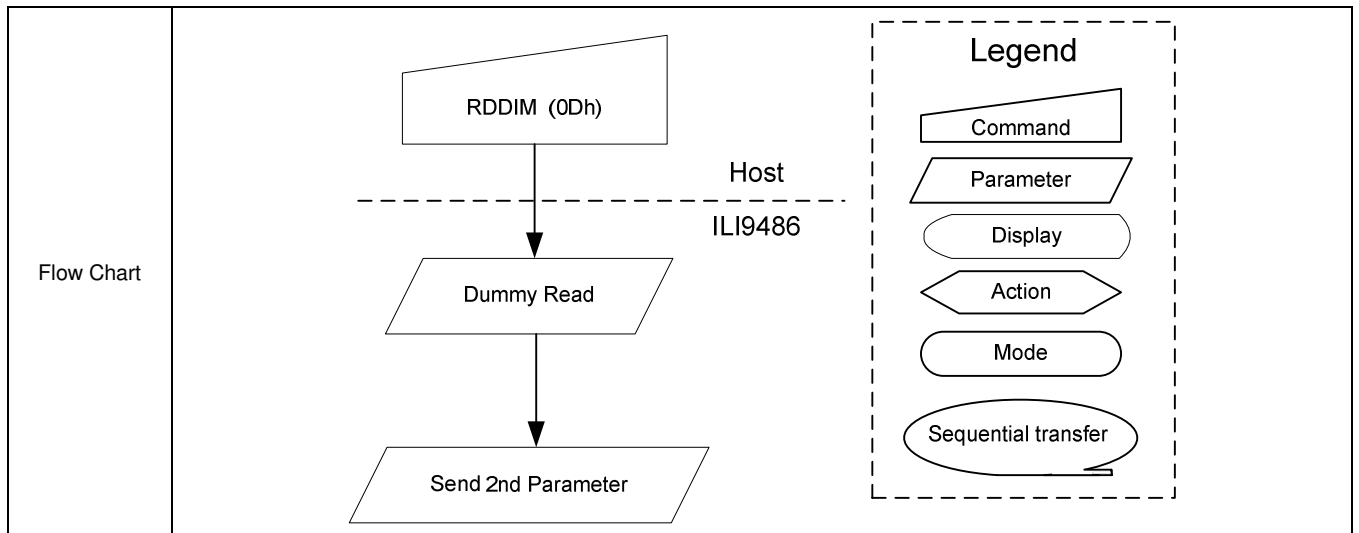


8.2.8. Read Display Pixel Format (0Ch)

0Ch	RDDCOLMOD (Read Display COLMOD)																																																																																														
	D/CX	RDX	WRX	D[15:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX																																																																																		
Command	0	1	↑	XXXXXXXX	0	0	0	0	1	1	0	0	0Ch																																																																																		
1 st Parameter	1	↑	1	XXXXXXXX	X	X	X	X	X	X	X	X	XX																																																																																		
2 nd Parameter	1	↑	1	XXXXXXXX	DPI[3:0]				0	DBI[2:0]			XX																																																																																		
Description	This command indicates the current status of the display as described in the table below: <table><tr><th colspan="4">DPI[3:0]</th><th>RGB Interface Format</th><th colspan="4">DBI[2:0]</th><th>CPU Interface Format</th></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td><td>Reserved</td><td>0</td><td>0</td><td>0</td><td>Reserved</td></tr><tr><td>0</td><td>0</td><td>0</td><td>1</td><td>Reserved</td><td>0</td><td>0</td><td>1</td><td>Reserved</td></tr><tr><td>0</td><td>0</td><td>1</td><td>0</td><td>Reserved</td><td>0</td><td>1</td><td>0</td><td>Reserved</td></tr><tr><td>0</td><td>0</td><td>1</td><td>1</td><td>Reserved</td><td>0</td><td>1</td><td>1</td><td>Reserved</td></tr><tr><td>0</td><td>1</td><td>0</td><td>0</td><td>Reserved</td><td>1</td><td>0</td><td>0</td><td>Reserved</td></tr><tr><td>0</td><td>1</td><td>0</td><td>1</td><td>16 bits / pixel</td><td>1</td><td>0</td><td>1</td><td>16 bits / pixel</td></tr><tr><td>0</td><td>1</td><td>1</td><td>0</td><td>18 bits / pixel</td><td>1</td><td>1</td><td>0</td><td>18 bits / pixel</td></tr><tr><td>0</td><td>1</td><td>1</td><td>1</td><td>Reserved</td><td>1</td><td>1</td><td>1</td><td>Reserved</td></tr></table>													DPI[3:0]				RGB Interface Format	DBI[2:0]				CPU Interface Format	0	0	0	0	Reserved	0	0	0	Reserved	0	0	0	1	Reserved	0	0	1	Reserved	0	0	1	0	Reserved	0	1	0	Reserved	0	0	1	1	Reserved	0	1	1	Reserved	0	1	0	0	Reserved	1	0	0	Reserved	0	1	0	1	16 bits / pixel	1	0	1	16 bits / pixel	0	1	1	0	18 bits / pixel	1	1	0	18 bits / pixel	0	1	1	1	Reserved	1	1	1	Reserved
DPI[3:0]				RGB Interface Format	DBI[2:0]				CPU Interface Format																																																																																						
0	0	0	0	Reserved	0	0	0	Reserved																																																																																							
0	0	0	1	Reserved	0	0	1	Reserved																																																																																							
0	0	1	0	Reserved	0	1	0	Reserved																																																																																							
0	0	1	1	Reserved	0	1	1	Reserved																																																																																							
0	1	0	0	Reserved	1	0	0	Reserved																																																																																							
0	1	0	1	16 bits / pixel	1	0	1	16 bits / pixel																																																																																							
0	1	1	0	18 bits / pixel	1	1	0	18 bits / pixel																																																																																							
0	1	1	1	Reserved	1	1	1	Reserved																																																																																							
Restriction	ILI9486 is sending 2nd parameter value on the data lines if the MCU wants to read more than one parameter (= more than 2 RDX cycle) on parallel MCU interface. Only 2nd parameter is sent on DSI (The 1st parameter is not sent).																																																																																														
Register Availability	<table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table>													Status	Availability	Normal Mode On, Idle Mode Off, Sleep Out	Yes	Normal Mode On, Idle Mode On, Sleep Out	Yes	Partial Mode On, Idle Mode Off, Sleep Out	Yes	Partial Mode On, Idle Mode On, Sleep Out	Yes	Sleep In	Yes																																																																						
Status	Availability																																																																																														
Normal Mode On, Idle Mode Off, Sleep Out	Yes																																																																																														
Normal Mode On, Idle Mode On, Sleep Out	Yes																																																																																														
Partial Mode On, Idle Mode Off, Sleep Out	Yes																																																																																														
Partial Mode On, Idle Mode On, Sleep Out	Yes																																																																																														
Sleep In	Yes																																																																																														
Default	<table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>06_{HEX}</td></tr><tr><td>SW Reset</td><td>No Change</td></tr><tr><td>HW Reset</td><td>06_{HEX}</td></tr></table>													Status	Default Value	Power On Sequence	06 _{HEX}	SW Reset	No Change	HW Reset	06 _{HEX}																																																																										
Status	Default Value																																																																																														
Power On Sequence	06 _{HEX}																																																																																														
SW Reset	No Change																																																																																														
HW Reset	06 _{HEX}																																																																																														
Flow Chart	RDDCOLMOD (0Ch) ↓ Dummy Read ↓ Send 2nd Parameter Host ILI9486 Legend Command Parameter Display Action Mode Sequential transfer																																																																																														

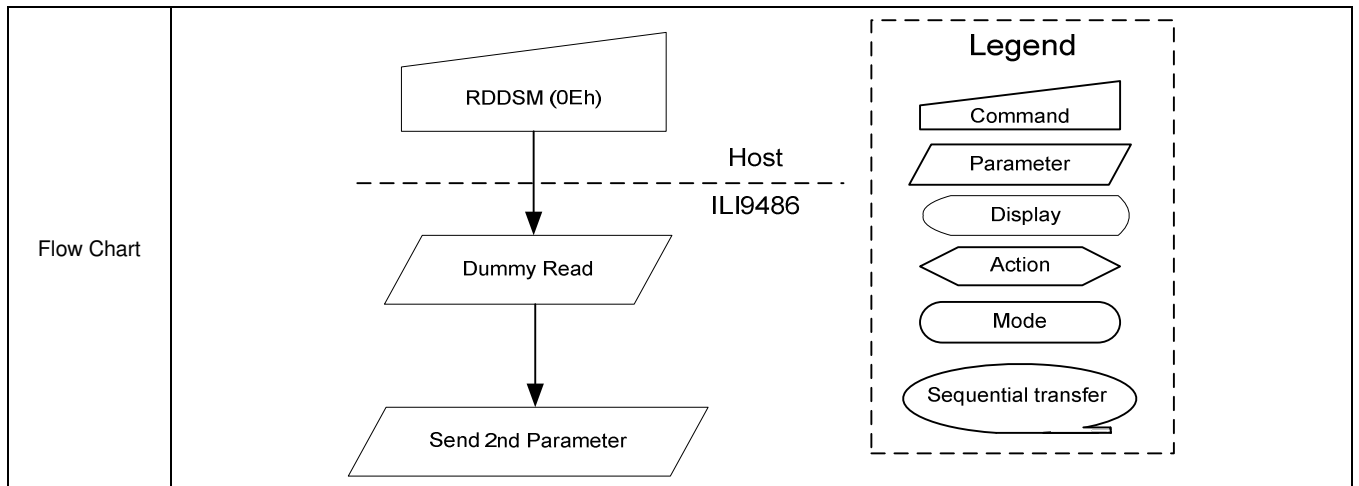
8.2.9. Read Display Image Mode (0Dh)

0Dh	RDDIM (Read Display Image Mode)																														
	D/CX	RDX	WRX	D[15:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX																		
Command	0	1	↑	XXXXXXXX	0	0	0	0	1	1	0	1	0Dh																		
1 st Parameter	1	↑	1	XXXXXXXX	X	X	X	X	X	X	X	X	XX																		
2 nd Parameter	1	↑	1	XXXXXXXX	D[7:0]								XX																		
Description	ILI9486 returns the Display Image Mode status.																														
	<table><tr><th>Bit</th><th>Description</th></tr><tr><td>D7</td><td>Vertical Scrolling Status</td></tr><tr><td>D6</td><td>Reserved</td></tr><tr><td>D5</td><td>Inversion On/Off</td></tr><tr><td>D4</td><td>Reserved</td></tr><tr><td>D3</td><td>Reserved</td></tr><tr><td>D2</td><td>Gamma Curve Selection</td></tr><tr><td>D1</td><td>Gamma Curve Selection</td></tr><tr><td>D0</td><td>Gamma Curve Selection</td></tr></table>													Bit	Description	D7	Vertical Scrolling Status	D6	Reserved	D5	Inversion On/Off	D4	Reserved	D3	Reserved	D2	Gamma Curve Selection	D1	Gamma Curve Selection	D0	Gamma Curve Selection
	Bit	Description																													
	D7	Vertical Scrolling Status																													
	D6	Reserved																													
	D5	Inversion On/Off																													
	D4	Reserved																													
	D3	Reserved																													
	D2	Gamma Curve Selection																													
	D1	Gamma Curve Selection																													
D0	Gamma Curve Selection																														
This command indicates the current status of the display as described in the table below:																															
◆ Bit D7 – Vertical Scrolling On/Off																															
'0' = Vertical Scrolling is Off.																															
'1' = Vertical Scrolling is On.																															
◆ Bit D6 – Reserved																															
◆ Bit D5 – Inversion On/Off																															
'0' = Inversion is Off.																															
'1' = Inversion is On.																															
◆ Bit D4 – Reserved																															
◆ Bit D3 – Reserved																															
◆ Bits D2, D1, D0 – Gamma Curve Selection																															
These bits are not applicable for this project, so they are set to '000', only support Gamma 2.2.																															
Restriction	ILI9486 is sending 2nd parameter value on the data lines if the MCU wants to read more than one parameter (= more than 2 RDX cycle) on parallel MCU interface. Only 2nd parameter is sent on DSI (The 1st parameter is not sent).																														
Register Availability	<table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table>													Status	Availability	Normal Mode On, Idle Mode Off, Sleep Out	Yes	Normal Mode On, Idle Mode On, Sleep Out	Yes	Partial Mode On, Idle Mode Off, Sleep Out	Yes	Partial Mode On, Idle Mode On, Sleep Out	Yes	Sleep In	Yes						
Status	Availability																														
Normal Mode On, Idle Mode Off, Sleep Out	Yes																														
Normal Mode On, Idle Mode On, Sleep Out	Yes																														
Partial Mode On, Idle Mode Off, Sleep Out	Yes																														
Partial Mode On, Idle Mode On, Sleep Out	Yes																														
Sleep In	Yes																														
Default	<table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>00_{HEX}</td></tr><tr><td>SW Reset</td><td>00_{HEX}</td></tr><tr><td>HW Reset</td><td>00_{HEX}</td></tr></table>													Status	Default Value	Power On Sequence	00 _{HEX}	SW Reset	00 _{HEX}	HW Reset	00 _{HEX}										
Status	Default Value																														
Power On Sequence	00 _{HEX}																														
SW Reset	00 _{HEX}																														
HW Reset	00 _{HEX}																														



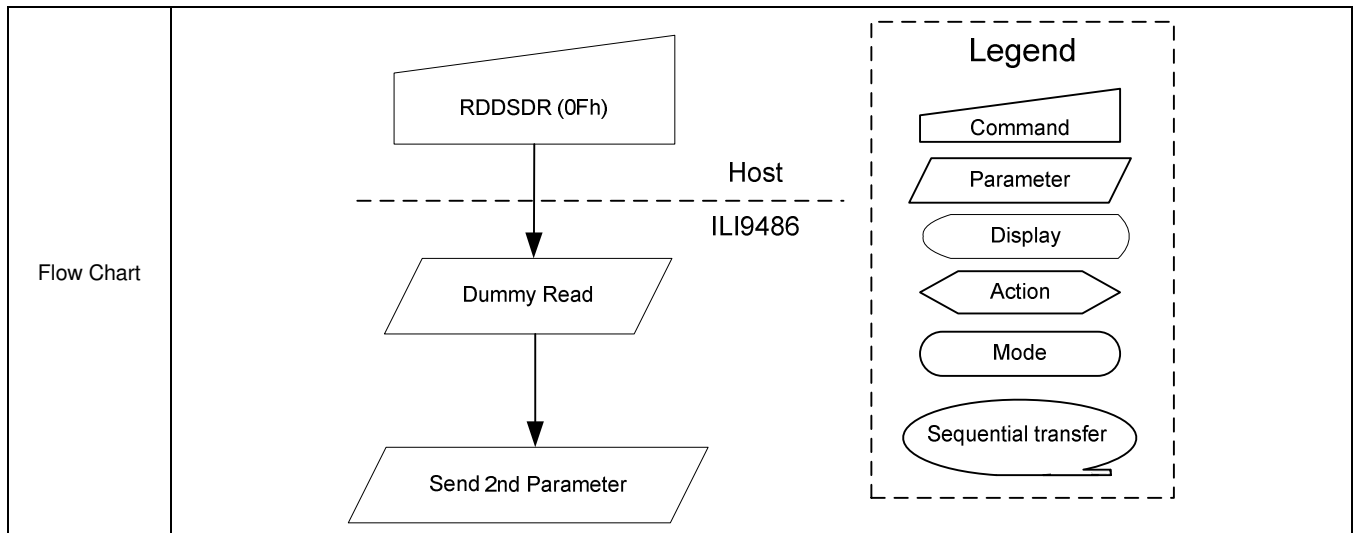
8.2.10. Read Display Signal Mode (0Eh)

0Eh	RDDSM (Read Display Signal Mode)																								
	D/CX	RDX	WRX	D[15:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX												
Command	0	1	↑	XXXXXXXX	0	0	0	0	1	1	1	0	0Eh												
1 st Parameter	1	↑	1	XXXXXXXX	X	X	X	X	X	X	X	X	XX												
2 nd Parameter	1	↑	1	XXXXXXXX	D7	D6	D5	D4	D3	D2	D1	D0	XX												
Description	This command indicates the current status of the display as described in the table below:																								
	Bit		Value		Function																				
	D7		0		Tearing Effect Line OFF																				
			1		Tearing Effect Line ON																				
	D6		0		Tearing Effect Line Mode 1																				
			1		Tearing Effect Line Mode 2																				
	D5		0		Horizontal Sync (RGB interface) OFF																				
			1		Horizontal Sync (RGB interface) ON																				
	D4		0		Vertical Sync (RGB interface) OFF																				
			1		Vertical Sync (RGB interface) ON																				
	D3		0		Pixel Clock (DOTCLK, RGB interface) OFF																				
			1		Pixel Clock (DOTCLK, RGB interface) ON																				
	D2		0		Data Enable (DE, RGB interface) OFF																				
			1		Data Enable (DE, RGB interface) ON																				
	D1		0		Reserved																				
			1		Reserved																				
	D0		0		No Error on DSI																				
			1		Error on DSI																				
Restriction	ILI9486 is sending 2nd parameter value on the data lines if the MCU wants to read more than one parameter (= more than 2 RDX cycle) on parallel MCU interface. Only 2nd parameter is sent on DSI (The 1st parameter is not sent).																								
Register Availability	<table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table>													Status	Availability	Normal Mode On, Idle Mode Off, Sleep Out	Yes	Normal Mode On, Idle Mode On, Sleep Out	Yes	Partial Mode On, Idle Mode Off, Sleep Out	Yes	Partial Mode On, Idle Mode On, Sleep Out	Yes	Sleep In	Yes
Status	Availability																								
Normal Mode On, Idle Mode Off, Sleep Out	Yes																								
Normal Mode On, Idle Mode On, Sleep Out	Yes																								
Partial Mode On, Idle Mode Off, Sleep Out	Yes																								
Partial Mode On, Idle Mode On, Sleep Out	Yes																								
Sleep In	Yes																								
Default	<table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>00_{HEX}</td></tr><tr><td>SW Reset</td><td>00_{HEX}</td></tr><tr><td>HW Reset</td><td>00_{HEX}</td></tr></table>													Status	Default Value	Power On Sequence	00 _{HEX}	SW Reset	00 _{HEX}	HW Reset	00 _{HEX}				
Status	Default Value																								
Power On Sequence	00 _{HEX}																								
SW Reset	00 _{HEX}																								
HW Reset	00 _{HEX}																								



8.2.11. Read Display Self-Diagnostic Result (0Fh)

0Fh	RDDSDR (Read Display Self-Diagnostic Result)																								
	D/CX	RDX	WRX	D[15:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX												
Command	0	1	↑	XXXXXXXX	0	0	0	0	1	1	1	1	0Fh												
1 st Parameter	1	↑	1	XXXXXXXX	X	X	X	X	X	X	X	X	XX												
2 nd Parameter	1	↑	1	XXXXXXXX	D7	D6	0	0	0	0	0	D0	XX												
Description	This command indicates the status of the display self-diagnostic results after Sleep Out -command as described in the table below:																								
	Bit	Description				Action																			
	D7	Register Loading Detection				Invert the D7 bit if register values loading work properly.																			
	D6	Functionality Detection				Invert the D6 bit if the display is functionality																			
	D5	Not Used				'0'																			
	D4	Not Used				'0'																			
	D3	Not Used				'0'																			
	D2	Not Used				'0'																			
	D1	Not Used				'0'																			
	D0	Checksums Comparison				'0' = Checksums are same '1' = Checksums are not same																			
Restriction	It will be necessary to wait 300ms after there is the last write access on User area registers before there can read Bit D0 value.																								
	ILI9486 is sending 2nd parameter value on the data lines if the MCU wants to read more than one parameter (= more than 2 RDX cycle) on parallel MCU interface.																								
	Only 2nd parameter is sent on DSI (The 1st parameter is not sent).																								
Register Availability	<table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table>													Status	Availability	Normal Mode On, Idle Mode Off, Sleep Out	Yes	Normal Mode On, Idle Mode On, Sleep Out	Yes	Partial Mode On, Idle Mode Off, Sleep Out	Yes	Partial Mode On, Idle Mode On, Sleep Out	Yes	Sleep In	Yes
Status	Availability																								
Normal Mode On, Idle Mode Off, Sleep Out	Yes																								
Normal Mode On, Idle Mode On, Sleep Out	Yes																								
Partial Mode On, Idle Mode Off, Sleep Out	Yes																								
Partial Mode On, Idle Mode On, Sleep Out	Yes																								
Sleep In	Yes																								
Default	<table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>00_{HEX}</td></tr><tr><td>SW Reset</td><td>00_{HEX}</td></tr><tr><td>HW Reset</td><td>00_{HEX}</td></tr></table>													Status	Default Value	Power On Sequence	00 _{HEX}	SW Reset	00 _{HEX}	HW Reset	00 _{HEX}				
Status	Default Value																								
Power On Sequence	00 _{HEX}																								
SW Reset	00 _{HEX}																								
HW Reset	00 _{HEX}																								



8.2.12. Sleep IN (10h)

10h	SLPIN (Sleep IN)																								
	D/CX	RDX	WRX	D[15:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX												
Command	0	1	↑	XXXXXXXX	0	0	0	1	0	0	0	0	10h												
Parameter	No parameter																								
Description	This command causes ILI9486 to enter the minimum power consumption mode. In this mode e.g. the DC/DC converter is stopped, Internal oscillator is stopped, and panel scanning is stopped. Out Blank STOP MCU interface and memory are still working and the memory keeps its contents. Dimming function does not work when there is changing mode from Sleep OUT to Sleep IN. X = Don't care																								
Restriction	This command has no effect when module is already in sleep in mode. Sleep In Mode can only be left by the Sleep Out Command (11h). It will be necessary to wait 5msec before sending next command; this is to allow time for the supply voltages and clock circuits to stabilize. It will be necessary to wait 120msec after sending Sleep Out command (when in Sleep In Mode) before Sleep In command can be sent.																								
Register Availability	<table><thead><tr><th>Status</th><th>Availability</th></tr></thead><tbody><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></tbody></table>													Status	Availability	Normal Mode On, Idle Mode Off, Sleep Out	Yes	Normal Mode On, Idle Mode On, Sleep Out	Yes	Partial Mode On, Idle Mode Off, Sleep Out	Yes	Partial Mode On, Idle Mode On, Sleep Out	Yes	Sleep In	Yes
Status	Availability																								
Normal Mode On, Idle Mode Off, Sleep Out	Yes																								
Normal Mode On, Idle Mode On, Sleep Out	Yes																								
Partial Mode On, Idle Mode Off, Sleep Out	Yes																								
Partial Mode On, Idle Mode On, Sleep Out	Yes																								
Sleep In	Yes																								
Default	<table><thead><tr><th>Status</th><th>Default Value</th></tr></thead><tbody><tr><td>Power On Sequence</td><td>Sleep IN Mode</td></tr><tr><td>SW Reset</td><td>Sleep IN Mode</td></tr><tr><td>HW Reset</td><td>Sleep IN Mode</td></tr></tbody></table>													Status	Default Value	Power On Sequence	Sleep IN Mode	SW Reset	Sleep IN Mode	HW Reset	Sleep IN Mode				
Status	Default Value																								
Power On Sequence	Sleep IN Mode																								
SW Reset	Sleep IN Mode																								
HW Reset	Sleep IN Mode																								
Flow Chart	Any Mode SLPIN (10h) Blank Display Device Power OFF Device Stop Power Supply Stop Internal Oscillator Sleep Mode ON Legend Command Parameter Display Action Mode Sequential transfer																								

8.2.13. Sleep OUT (11h)

11h	SLPOUT (Sleep OUT)																								
	D/CX	RDX	WRX	D[15:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX												
Command	0	1	↑	XXXXXXXX	0	0	0	1	0	0	0	1	11h												
Parameter	No parameter																								
Description	This command turns off sleep mode. In this mode e.g. the DC/DC converter is enabled, Internal oscillator is started, and panel scanning is started. OUT STOP Blank Memory Contents X = Don't care																								
Restriction	Sleep Out Mode can only be left by the Sleep In Command (10h). It will be necessary to wait 5msec before sending next command; this is to allow time for the supply voltages and clock circuits to stabilize. ILI9486 loads all display supplier's factory default values to the registers during this 5msec and there cannot be any abnormal visual effect on the display image if factory default and register values are same when this load is done and when ILI9486 is already Sleep Out –mode. ILI9486 is doing self-diagnostic functions during this 5msec. It will be necessary to wait 120msec after sending Sleep In command (when in Sleep Out mode) before Sleep Out command can be sent.																								
Register Availability	<table><thead><tr><th>Status</th><th>Availability</th></tr></thead><tbody><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></tbody></table>													Status	Availability	Normal Mode On, Idle Mode Off, Sleep Out	Yes	Normal Mode On, Idle Mode On, Sleep Out	Yes	Partial Mode On, Idle Mode Off, Sleep Out	Yes	Partial Mode On, Idle Mode On, Sleep Out	Yes	Sleep In	Yes
Status	Availability																								
Normal Mode On, Idle Mode Off, Sleep Out	Yes																								
Normal Mode On, Idle Mode On, Sleep Out	Yes																								
Partial Mode On, Idle Mode Off, Sleep Out	Yes																								
Partial Mode On, Idle Mode On, Sleep Out	Yes																								
Sleep In	Yes																								
Default	<table><thead><tr><th>Status</th><th>Default Value</th></tr></thead><tbody><tr><td>Power On Sequence</td><td>Sleep IN Mode</td></tr><tr><td>SW Reset</td><td>Sleep IN Mode</td></tr><tr><td>HW Reset</td><td>Sleep IN Mode</td></tr></tbody></table>													Status	Default Value	Power On Sequence	Sleep IN Mode	SW Reset	Sleep IN Mode	HW Reset	Sleep IN Mode				
Status	Default Value																								
Power On Sequence	Sleep IN Mode																								
SW Reset	Sleep IN Mode																								
HW Reset	Sleep IN Mode																								
Flow Chart	Sleep In Mode SLPOUT (11h) Start Internal Oscillator Start Power Supply Power ON Display Device Blank Display Device Sleep Mode OFF Legend Command Parameter Display Action Mode Sequential transfer																								

8.2.14. Partial Mode ON (12h)

12h	PTLON (Partial Mode ON)																								
	D/CX	RDX	WRX	D[15:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX												
Command	0	1	↑	XXXXXXXX	0	0	0	1	0	0	1	0	12h												
Parameter	No parameter																								
Description	This command turns on partial mode The partial mode window is described by the Partial Area command (30H). To leave Partial mode, the Normal Display Mode On command (13H) should be written.																								
Restriction	This command has no effect when Partial Display Mode is already active.																								
Register Availability	<table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table>													Status	Availability	Normal Mode On, Idle Mode Off, Sleep Out	Yes	Normal Mode On, Idle Mode On, Sleep Out	Yes	Partial Mode On, Idle Mode Off, Sleep Out	Yes	Partial Mode On, Idle Mode On, Sleep Out	Yes	Sleep In	Yes
Status	Availability																								
Normal Mode On, Idle Mode Off, Sleep Out	Yes																								
Normal Mode On, Idle Mode On, Sleep Out	Yes																								
Partial Mode On, Idle Mode Off, Sleep Out	Yes																								
Partial Mode On, Idle Mode On, Sleep Out	Yes																								
Sleep In	Yes																								
Default	<table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>Sleep IN Mode</td></tr><tr><td>SW Reset</td><td>Sleep IN Mode</td></tr><tr><td>HW Reset</td><td>Sleep IN Mode</td></tr></table>													Status	Default Value	Power On Sequence	Sleep IN Mode	SW Reset	Sleep IN Mode	HW Reset	Sleep IN Mode				
Status	Default Value																								
Power On Sequence	Sleep IN Mode																								
SW Reset	Sleep IN Mode																								
HW Reset	Sleep IN Mode																								
Flow Chart	See Partial Area (30h)																								

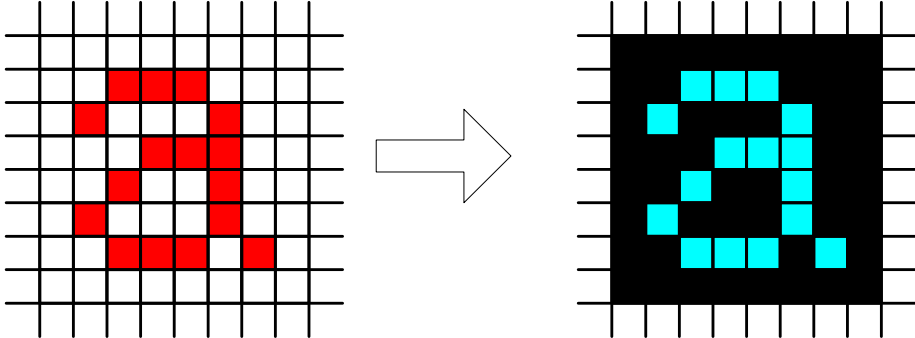
8.2.15. Normal Display Mode ON (13h)

13h	NORON (Normal Display Mode ON)																								
	D/CX	RDX	WRX	D[15:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX												
Command	0	1	↑	XXXXXXXX	0	0	0	1	0	0	1	1	13h												
Parameter	No parameter																								
Description	This command returns the display to normal mode. Normal display mode on means Partial mode off and Scroll mode off. X = Don't care																								
Restriction	This command has no effect when Normal Display mode is active.																								
Register Availability	<table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table>													Status	Availability	Normal Mode On, Idle Mode Off, Sleep Out	Yes	Normal Mode On, Idle Mode On, Sleep Out	Yes	Partial Mode On, Idle Mode Off, Sleep Out	Yes	Partial Mode On, Idle Mode On, Sleep Out	Yes	Sleep In	Yes
Status	Availability																								
Normal Mode On, Idle Mode Off, Sleep Out	Yes																								
Normal Mode On, Idle Mode On, Sleep Out	Yes																								
Partial Mode On, Idle Mode Off, Sleep Out	Yes																								
Partial Mode On, Idle Mode On, Sleep Out	Yes																								
Sleep In	Yes																								
Default	<table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>Normal Display Mode On</td></tr><tr><td>SW Reset</td><td>Normal Display Mode On</td></tr><tr><td>HW Reset</td><td>Normal Display Mode On</td></tr></table>													Status	Default Value	Power On Sequence	Normal Display Mode On	SW Reset	Normal Display Mode On	HW Reset	Normal Display Mode On				
Status	Default Value																								
Power On Sequence	Normal Display Mode On																								
SW Reset	Normal Display Mode On																								
HW Reset	Normal Display Mode On																								
Flow Chart	See Partial Area Descriptions for details of when to use this command.																								

8.2.16. Display Inversion OFF (20h)

20h	INVOFF (Display Inversion OFF)																								
	D/CX	RDX	WRX	D[15:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX												
Command	0	1	↑	XXXXXXXX	0	0	1	0	0	0	0	0	20h												
Parameter	No parameter																								
Description	This command is used to recover from display inversion mode. Output from the Frame Memory is enabled. This command makes no change of the content of frame memory. This command doesn't change any other status. Memory → Display Panel X = Don't care																								
Restriction	This command has no effect when ILI9486 is already in Inversion off mode.																								
Register Availability	<table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table>													Status	Availability	Normal Mode On, Idle Mode Off, Sleep Out	Yes	Normal Mode On, Idle Mode On, Sleep Out	Yes	Partial Mode On, Idle Mode Off, Sleep Out	Yes	Partial Mode On, Idle Mode On, Sleep Out	Yes	Sleep In	Yes
Status	Availability																								
Normal Mode On, Idle Mode Off, Sleep Out	Yes																								
Normal Mode On, Idle Mode On, Sleep Out	Yes																								
Partial Mode On, Idle Mode Off, Sleep Out	Yes																								
Partial Mode On, Idle Mode On, Sleep Out	Yes																								
Sleep In	Yes																								
Default	<table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>Display Inversion OFF</td></tr><tr><td>SW Reset</td><td>Display Inversion OFF</td></tr><tr><td>HW Reset</td><td>Display Inversion OFF</td></tr></table>													Status	Default Value	Power On Sequence	Display Inversion OFF	SW Reset	Display Inversion OFF	HW Reset	Display Inversion OFF				
Status	Default Value																								
Power On Sequence	Display Inversion OFF																								
SW Reset	Display Inversion OFF																								
HW Reset	Display Inversion OFF																								
Flow Chart	Invert Mode ON ↓ INVOFF (20h) ↓ Invert Mode OFF Legend Command Parameter Display Action Mode Sequential transfer																								

8.2.17. Display Inversion ON (21h)

21h	INVON (Display Inversion ON)																								
	D/CX	RDX	WRX	D[15:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX												
Command	0	1	↑	XXXXXXXX	0	0	1	0	0	0	0	1	21h												
Parameter	No parameter																								
Description	This command is used to enter into display inversion mode. This command makes no change of the content of frame memory. Every bit is inverted from the frame memory to the display. This command doesn't change any other status. To exit Display inversion mode, the Display inversion OFF command (20h) should be written.  X = Don't care																								
Restriction	This command has no effect when ILI9486 is already in Inversion on mode.																								
Register Availability	<table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table>													Status	Availability	Normal Mode On, Idle Mode Off, Sleep Out	Yes	Normal Mode On, Idle Mode On, Sleep Out	Yes	Partial Mode On, Idle Mode Off, Sleep Out	Yes	Partial Mode On, Idle Mode On, Sleep Out	Yes	Sleep In	Yes
Status	Availability																								
Normal Mode On, Idle Mode Off, Sleep Out	Yes																								
Normal Mode On, Idle Mode On, Sleep Out	Yes																								
Partial Mode On, Idle Mode Off, Sleep Out	Yes																								
Partial Mode On, Idle Mode On, Sleep Out	Yes																								
Sleep In	Yes																								
Default	<table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>Display Inversion OFF</td></tr><tr><td>SW Reset</td><td>Display Inversion OFF</td></tr><tr><td>HW Reset</td><td>Display Inversion OFF</td></tr></table>													Status	Default Value	Power On Sequence	Display Inversion OFF	SW Reset	Display Inversion OFF	HW Reset	Display Inversion OFF				
Status	Default Value																								
Power On Sequence	Display Inversion OFF																								
SW Reset	Display Inversion OFF																								
HW Reset	Display Inversion OFF																								
Flow Chart	Invert Mode OFF ↓ INVON (21h) ↓ Invert Mode ON Legend Command Parameter Display Action Mode Sequential transfer																								

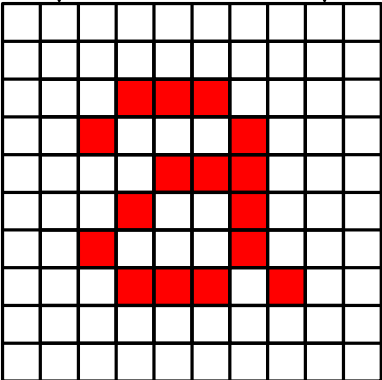
8.2.18. Display OFF (28h)

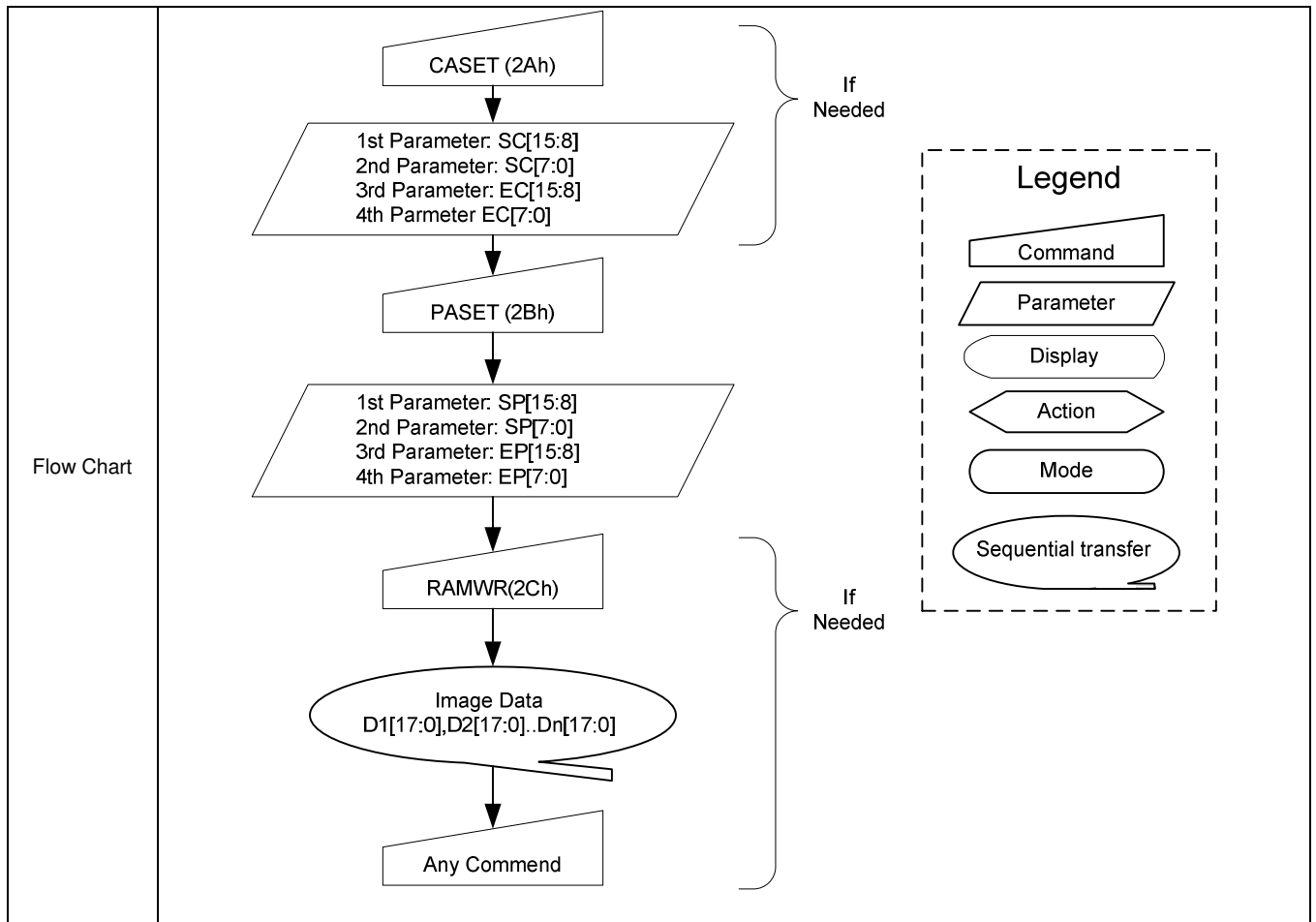
28h	DISOFF (Display OFF)																								
	D/CX	RDX	WRX	D[15:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX												
Command	0	1	↑	XXXXXXXX	0	0	1	0	1	0	0	0	28h												
Parameter	No parameter																								
Description	This command causes ILI9486 to stop displaying the image data on the display device. The frame memory contents remain unchanged. No status bits are changed. Memory X = Don't care Display Panel X = Don't care																								
Restriction	This command has no effect when ILI9486 is already in Display off mode.																								
Register Availability	<table><thead><tr><th>Status</th><th>Availability</th></tr></thead><tbody><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></tbody></table>													Status	Availability	Normal Mode On, Idle Mode Off, Sleep Out	Yes	Normal Mode On, Idle Mode On, Sleep Out	Yes	Partial Mode On, Idle Mode Off, Sleep Out	Yes	Partial Mode On, Idle Mode On, Sleep Out	Yes	Sleep In	Yes
Status	Availability																								
Normal Mode On, Idle Mode Off, Sleep Out	Yes																								
Normal Mode On, Idle Mode On, Sleep Out	Yes																								
Partial Mode On, Idle Mode Off, Sleep Out	Yes																								
Partial Mode On, Idle Mode On, Sleep Out	Yes																								
Sleep In	Yes																								
Default	<table><thead><tr><th>Status</th><th>Default Value</th></tr></thead><tbody><tr><td>Power On Sequence</td><td>Display OFF</td></tr><tr><td>SW Reset</td><td>Display OFF</td></tr><tr><td>HW Reset</td><td>Display OFF</td></tr></tbody></table>													Status	Default Value	Power On Sequence	Display OFF	SW Reset	Display OFF	HW Reset	Display OFF				
Status	Default Value																								
Power On Sequence	Display OFF																								
SW Reset	Display OFF																								
HW Reset	Display OFF																								
Flow Chart	<pre>graph TD; A([Display Panel ON]) --> B[/DISOFF (28h)/]; B --> C([Display Panel OFF]);</pre> Legend - Command - Parameter - Display - Action - Mode - Sequential transfer																								

8.2.19. Display ON (29h)

29h	DISON (Display ON)																								
	D/CX	RDX	WRX	D[15:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX												
Command	0	1	↑	XXXXXXXX	0	0	1	0	1	0	0	1	29h												
Parameter	No parameter																								
Description	This command causes ILI9486 to start displaying the image data on the display device. The frame memory contents remain unchanged. No status bits are changed. Memory X = Don't care Display Panel																								
Restriction	This command has no effect when ILI9486 is already in Display on mode.																								
Register Availability	<table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table>													Status	Availability	Normal Mode On, Idle Mode Off, Sleep Out	Yes	Normal Mode On, Idle Mode On, Sleep Out	Yes	Partial Mode On, Idle Mode Off, Sleep Out	Yes	Partial Mode On, Idle Mode On, Sleep Out	Yes	Sleep In	Yes
Status	Availability																								
Normal Mode On, Idle Mode Off, Sleep Out	Yes																								
Normal Mode On, Idle Mode On, Sleep Out	Yes																								
Partial Mode On, Idle Mode Off, Sleep Out	Yes																								
Partial Mode On, Idle Mode On, Sleep Out	Yes																								
Sleep In	Yes																								
Default	<table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>Display OFF</td></tr><tr><td>SW Reset</td><td>Display OFF</td></tr><tr><td>HW Reset</td><td>Display OFF</td></tr></table>													Status	Default Value	Power On Sequence	Display OFF	SW Reset	Display OFF	HW Reset	Display OFF				
Status	Default Value																								
Power On Sequence	Display OFF																								
SW Reset	Display OFF																								
HW Reset	Display OFF																								
Flow Chart	Display OFF Mode ↓ DISON (29h) ↓ Display ON Mode Legend - Command - Parameter - Display - Action - Mode - Sequential transfer																								

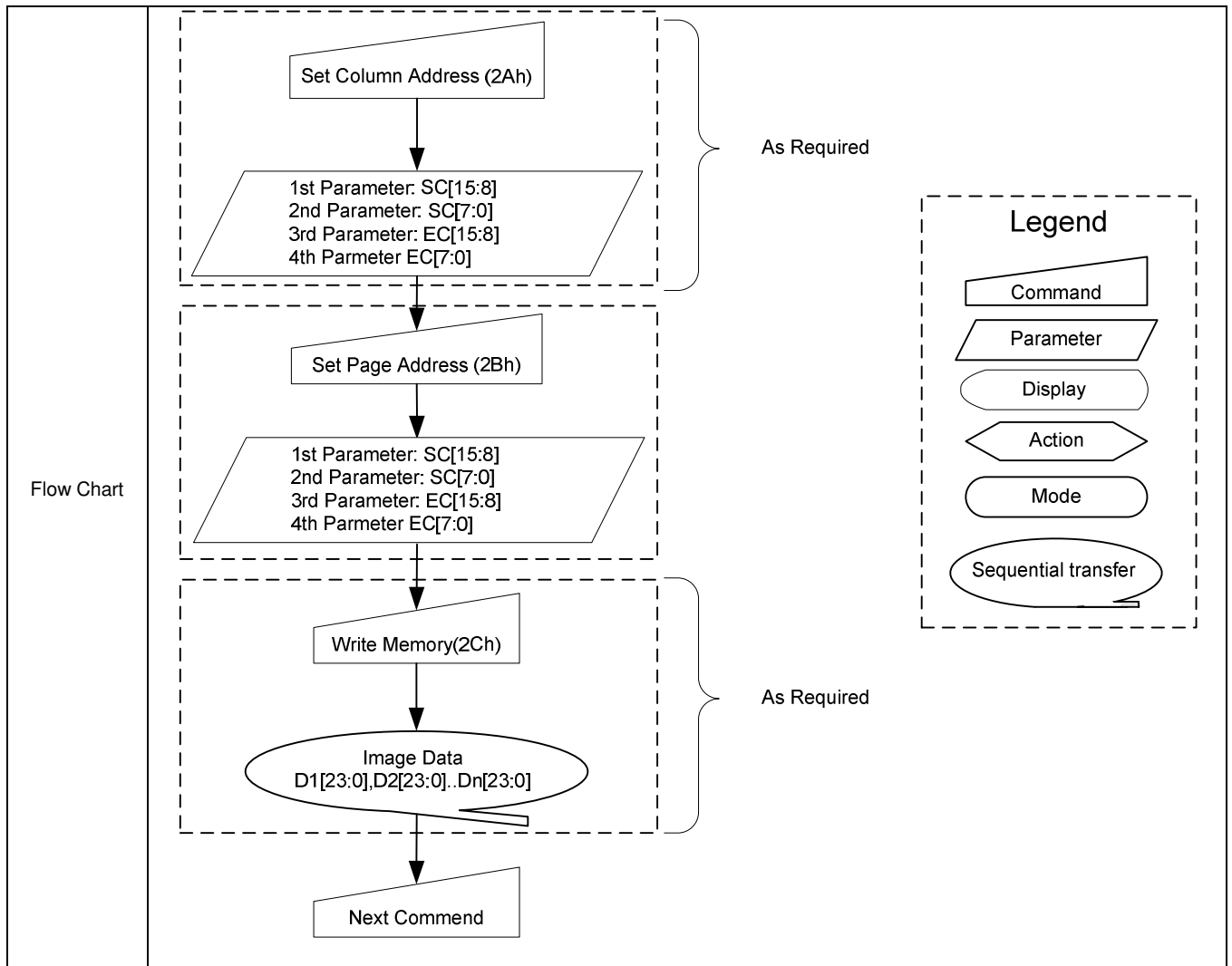
8.2.20. Column Address Set (2Ah)

2Ah	CASET (Column Address Set)																								
	D/CX	RDX	WRX	D[15:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX												
Command	0	1	↑	XXXXXXXX	0	0	1	0	1	0	1	0	2Ah												
1 st Parameter	1	1	↑	XXXXXXXX	SC[15:8]								XX												
2 nd Parameter	1	1	↑	XXXXXXXX	SC[7:0]								XX												
3 rd Parameter	1	1	↑	XXXXXXXX	EC[15:8]								XX												
4 th Parameter	1	1	↑	XXXXXXXX	EC[7:0]								XX												
Description	This command is used to define area of frame memory where MCU can access. This command makes no change on the other driver status. The values of SC[15:0] and EC[15:0] are referred when RAMWR command comes. Each value represents one column line in the Frame Memory.																								
	SC[15:0] EC[15:0] 																								
Restriction	SC[15:0] always must be equal to or less than EC[15:0]. Note 1: When SC[15:0] or EC[15:0] is greater than 013Fh (When MADCTL's B5 = 0) or 01DFh (When MADCTL's B5 = 1), data of out of range will be ignored																								
Register Availability	<table><thead><tr><th>Status</th><th>Availability</th></tr></thead><tbody><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></tbody></table>													Status	Availability	Normal Mode On, Idle Mode Off, Sleep Out	Yes	Normal Mode On, Idle Mode On, Sleep Out	Yes	Partial Mode On, Idle Mode Off, Sleep Out	Yes	Partial Mode On, Idle Mode On, Sleep Out	Yes	Sleep In	Yes
Status	Availability																								
Normal Mode On, Idle Mode Off, Sleep Out	Yes																								
Normal Mode On, Idle Mode On, Sleep Out	Yes																								
Partial Mode On, Idle Mode Off, Sleep Out	Yes																								
Partial Mode On, Idle Mode On, Sleep Out	Yes																								
Sleep In	Yes																								
Default	<table><thead><tr><th>Status</th><th colspan="2">Default Value</th></tr></thead><tbody><tr><td>Power On Sequence</td><td>SC[15:0]=0000h</td><td>EC[15:0]=00EFh</td></tr><tr><td>SW Reset</td><td>SC[15:0]=0000h</td><td>If MADCTL's B5 = 0: EC[15:0]=013Fh If MADCTL's B5 = 1: EC[15:0]=01DFh</td></tr><tr><td>HW Reset</td><td>SC[15:0]=0000h</td><td>EC[15:0]=013Fh</td></tr></tbody></table>													Status	Default Value		Power On Sequence	SC[15:0]=0000h	EC[15:0]=00EFh	SW Reset	SC[15:0]=0000h	If MADCTL's B5 = 0: EC[15:0]=013Fh If MADCTL's B5 = 1: EC[15:0]=01DFh	HW Reset	SC[15:0]=0000h	EC[15:0]=013Fh
Status	Default Value																								
Power On Sequence	SC[15:0]=0000h	EC[15:0]=00EFh																							
SW Reset	SC[15:0]=0000h	If MADCTL's B5 = 0: EC[15:0]=013Fh If MADCTL's B5 = 1: EC[15:0]=01DFh																							
HW Reset	SC[15:0]=0000h	EC[15:0]=013Fh																							



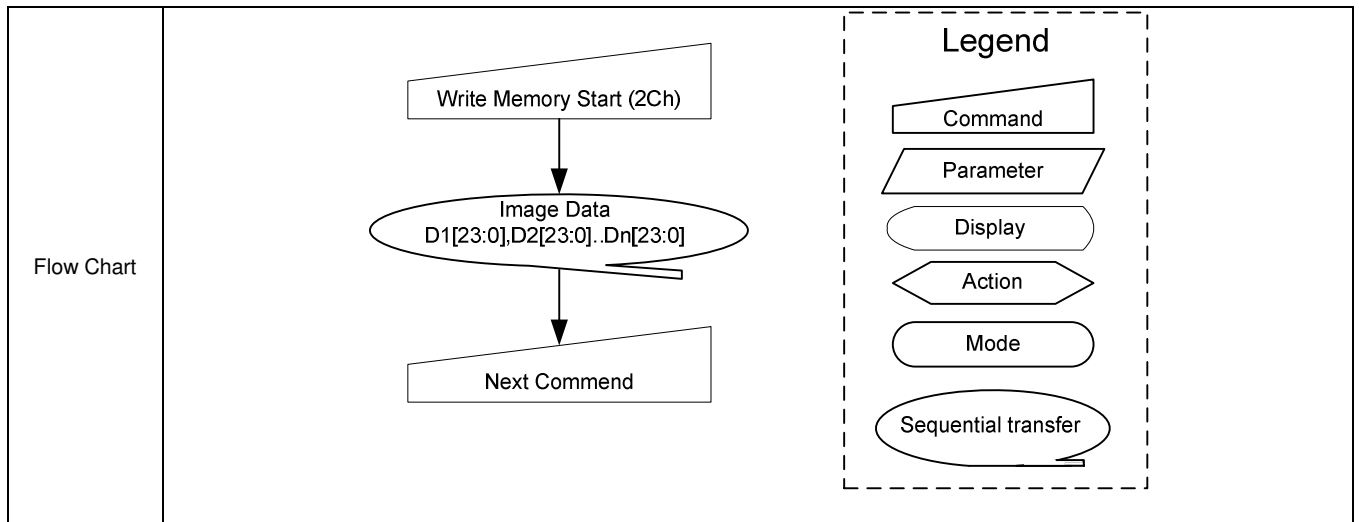
8.2.21. Page Address Set (2Bh)

2Bh	PASET (Page Address Set)																								
	D/CX	RDX	WRX	D[15:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX												
Command	0	1	↑	XXXXXXXX	0	0	1	0	1	0	1	1	2Bh												
1 st Parameter	1	1	↑	XXXXXXXX	SP[15:8]								XX												
2 nd Parameter	1	1	↑	XXXXXXXX	SP[7:0]								XX												
3 rd Parameter	1	1	↑	XXXXXXXX	EP[15:8]								XX												
4 th Parameter	1	1	↑	XXXXXXXX	EP[7:0]								XX												
Description	This command is used to define area of frame memory where MCU can access. This command makes no change on the other driver status. The values of SP[15:0] and EP[15:0] are referred when RAMWR command comes. Each value represents one Page line in the Frame Memory.																								
	SP[15:0] → EP[15:0] →																								
Restriction	SP[15:0] always must be equal to or less than EP[15:0] When SP[15:0] or EP[15:0] is greater than 01DFh (When MADCTL's B5 = 0) or 013Fh (When MADCTL's B5 = 1), data of out of range will be ignored.																								
Register Availability	<table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table>													Status	Availability	Normal Mode On, Idle Mode Off, Sleep Out	Yes	Normal Mode On, Idle Mode On, Sleep Out	Yes	Partial Mode On, Idle Mode Off, Sleep Out	Yes	Partial Mode On, Idle Mode On, Sleep Out	Yes	Sleep In	Yes
Status	Availability																								
Normal Mode On, Idle Mode Off, Sleep Out	Yes																								
Normal Mode On, Idle Mode On, Sleep Out	Yes																								
Partial Mode On, Idle Mode Off, Sleep Out	Yes																								
Partial Mode On, Idle Mode On, Sleep Out	Yes																								
Sleep In	Yes																								
Default	<table><tr><th>Status</th><th colspan="2">Default Value</th></tr><tr><td>Power On Sequence</td><td>SP[15:0]=0000h</td><td>EP[15:0]=013Fh</td></tr><tr><td>SW Reset</td><td>SP[15:0]=0000h</td><td>If MADCTL's B5 = 0: EP[15:0]=01DFh If MADCTL's B5 = 1: EP[15:0]=013Fh</td></tr><tr><td>HW Reset</td><td>SP[15:0]=0000h</td><td>EP[15:0]=01EFh</td></tr></table>													Status	Default Value		Power On Sequence	SP[15:0]=0000h	EP[15:0]=013Fh	SW Reset	SP[15:0]=0000h	If MADCTL's B5 = 0: EP[15:0]=01DFh If MADCTL's B5 = 1: EP[15:0]=013Fh	HW Reset	SP[15:0]=0000h	EP[15:0]=01EFh
Status	Default Value																								
Power On Sequence	SP[15:0]=0000h	EP[15:0]=013Fh																							
SW Reset	SP[15:0]=0000h	If MADCTL's B5 = 0: EP[15:0]=01DFh If MADCTL's B5 = 1: EP[15:0]=013Fh																							
HW Reset	SP[15:0]=0000h	EP[15:0]=01EFh																							



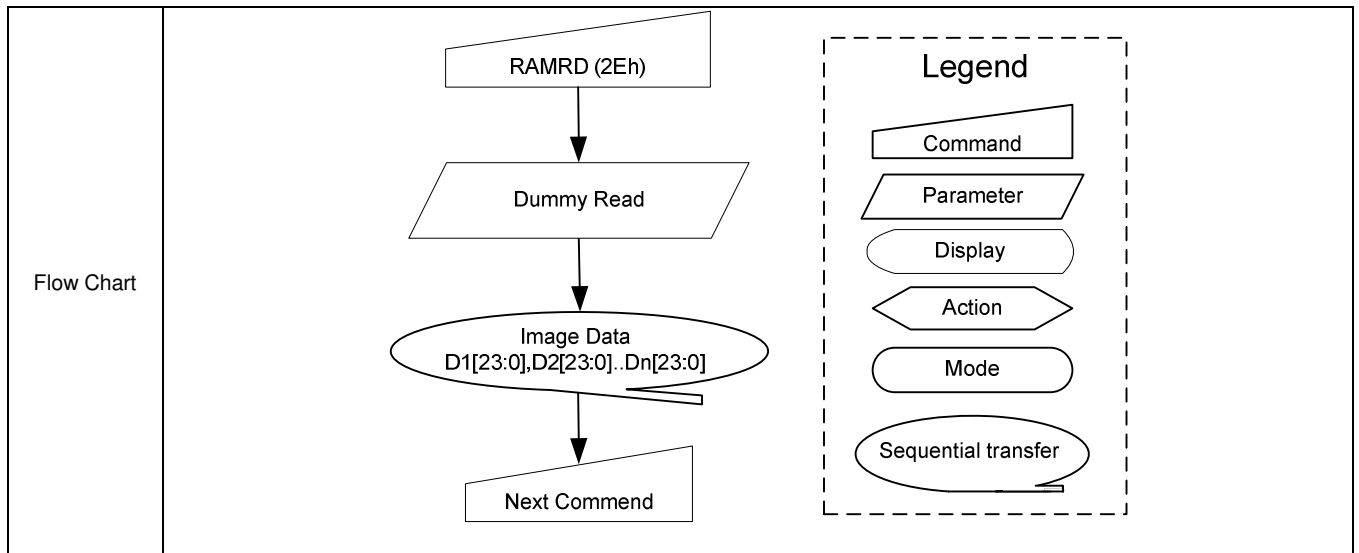
8.2.22. Memory Write (2Ch)

2Ch	RAMWR (Memory Write)																								
	D/CX	RDX	WRX	D[15:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX												
Command	0	1	↑	XXXXXXXX	0	0	1	0	1	1	0	0	2Ch												
1 st Parameter	1	1	↑	D1[15:0]									XX												
:	1	1	↑	Dx[15:0]									XX												
N th Parameter	1	1	↑	Dn[15:0]									XX												
Description	This command transfers image data from the host processor to ILI9486's frame memory starting at the pixel location specified by preceding Column Address Set (2Ah) and Page Address Set (2Bh) commands.																								
	If Memory Access Control (36h) B5 = 0:																								
	The column and page registers are reset to the Start Column (SC) and Start Page (SP), respectively. Pixel Data 1 is stored in frame memory at (SC, SP). The column register is then incremented and pixels are written to the frame memory until the column register equals the End Column (EC) value. The column register is then reset to SC and the page register is incremented. Pixels are written to the frame memory until the page register equals the End Page (EP) value or the host processor sends another command. If the number of pixels exceeds (EC – SC + 1) * (EP – SP + 1) the extra pixels are ignored.																								
	If Memory Access control (36h) B5 = 1 :																								
	The column and page registers are reset to the Start Column (SC) and Start Page (SP), respectively. Pixel Data 1 is stored in frame memory at (SC, SP). The page register is then incremented and pixels are written to the frame memory until the page register equals the End Page (EP) value. The page register is then reset to SP and the column register is incremented. Pixels are written to the frame memory until the column register equals the End column (EC) value or the host processor sends another command. If the number of pixels exceeds (EC – SC + 1) * (EP – SP + 1) the extra pixels are ignored.																								
Restriction	There is no restriction on length of parameters.																								
Register Availability	<table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table>													Status	Availability	Normal Mode On, Idle Mode Off, Sleep Out	Yes	Normal Mode On, Idle Mode On, Sleep Out	Yes	Partial Mode On, Idle Mode Off, Sleep Out	Yes	Partial Mode On, Idle Mode On, Sleep Out	Yes	Sleep In	Yes
Status	Availability																								
Normal Mode On, Idle Mode Off, Sleep Out	Yes																								
Normal Mode On, Idle Mode On, Sleep Out	Yes																								
Partial Mode On, Idle Mode Off, Sleep Out	Yes																								
Partial Mode On, Idle Mode On, Sleep Out	Yes																								
Sleep In	Yes																								
Default	<table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>Contents of memory is set randomly</td></tr><tr><td>SW Reset</td><td>Contents of memory is set randomly</td></tr><tr><td>HW Reset</td><td>Contents of memory is set randomly</td></tr></table>													Status	Default Value	Power On Sequence	Contents of memory is set randomly	SW Reset	Contents of memory is set randomly	HW Reset	Contents of memory is set randomly				
Status	Default Value																								
Power On Sequence	Contents of memory is set randomly																								
SW Reset	Contents of memory is set randomly																								
HW Reset	Contents of memory is set randomly																								

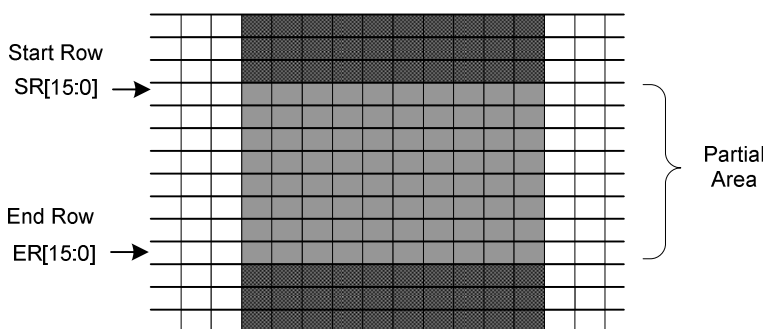
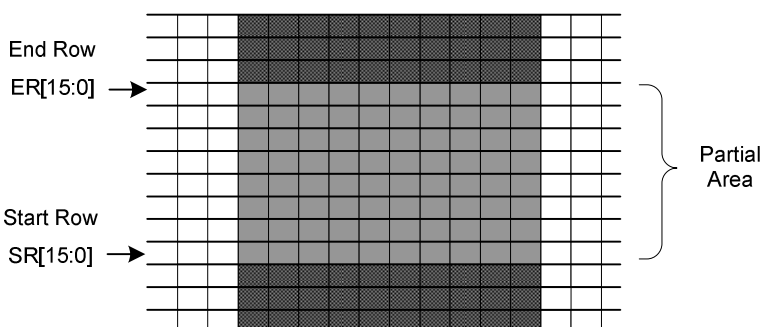
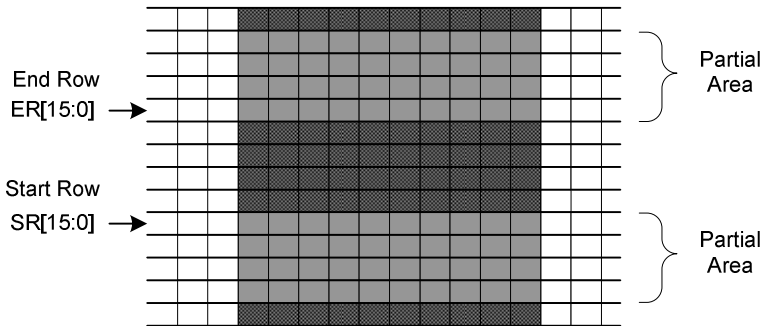


8.2.23. Memory Read (2Eh)

2Eh	RAMRD (Memory Read)																								
	D/CX	RDX	WRX	D[15:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX												
Command	0	1	↑	XXXXXXXX	0	0	1	0	1	1	1	0	2Eh												
1 st Parameter	1	1	↑	XXXXXXXX	X	X	X	X	X	X	X	X	XX												
2 nd Parameter	1	1	↑	D1[15:0]									XX												
:	1	1	↑	Dx[15:0]									XX												
(N+1) th Parameter	1	1	↑	Dn[15:0]									XX												
Description	This command transfers image data from ILI9486's frame memory to the host processor starting at the pixel location specified by preceding set_column_address and set_page_address commands.																								
	If Memory Access control B5 = 0:																								
	The column and page registers are reset to the Start Column (SC) and Start Page (SP), respectively. Pixels are read from frame memory at (SC, SP). The column register is then incremented and pixels read from the frame memory until the column register equals the End Column (EC) value. The column register is then reset to SC and the page register is incremented. Pixels are read from the frame memory until the page register equals the End Page (EP) value or the host processor sends another command.																								
	If Memory Access Control B5 = 1:																								
	The column and page registers are reset to the Start Column (SC) and Start Page (SP), respectively. Pixels are read from frame memory at (SC, SP). The page register is then incremented and pixels read from the frame memory until the page register equals the End Page (EP) value. The page register is then reset to SP and the column register is incremented. Pixels are read from the frame memory until the column register equals the End Column (EC) value or the host processor sends another command.																								
Restriction	There is no restriction on length of parameters.																								
Register Availability	<table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table>													Status	Availability	Normal Mode On, Idle Mode Off, Sleep Out	Yes	Normal Mode On, Idle Mode On, Sleep Out	Yes	Partial Mode On, Idle Mode Off, Sleep Out	Yes	Partial Mode On, Idle Mode On, Sleep Out	Yes	Sleep In	Yes
Status	Availability																								
Normal Mode On, Idle Mode Off, Sleep Out	Yes																								
Normal Mode On, Idle Mode On, Sleep Out	Yes																								
Partial Mode On, Idle Mode Off, Sleep Out	Yes																								
Partial Mode On, Idle Mode On, Sleep Out	Yes																								
Sleep In	Yes																								
Default	<table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>Contents of memory is set randomly</td></tr><tr><td>SW Reset</td><td>Contents of memory is set randomly</td></tr><tr><td>HW Reset</td><td>Contents of memory is set randomly</td></tr></table>													Status	Default Value	Power On Sequence	Contents of memory is set randomly	SW Reset	Contents of memory is set randomly	HW Reset	Contents of memory is set randomly				
Status	Default Value																								
Power On Sequence	Contents of memory is set randomly																								
SW Reset	Contents of memory is set randomly																								
HW Reset	Contents of memory is set randomly																								



8.2.24. Partial Area (30h)

30h	PLTAR (Partial Area)												
	D/CX	RDX	WRX	D[15:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XXXXXXXX	0	0	1	1	0	0	0	0	30h
1 st Parameter	1	1	↑	XXXXXXXX	SR[15:8]								XX
2 nd Parameter	1	1	↑	XXXXXXXX	SR[7:0]								XX
3 rd Parameter	1	1	↑	XXXXXXXX	ER[15:8]								XX
4 th Parameter	1	1	↑	XXXXXXXX	ER[7:0]								XX
Description	This command defines the Partial Display mode's display area. There are two parameters associated with this command, the first defines the Start Row (SR) and the second the End Row (ER), as illustrated in the following figure. SR and ER refer to the Frame Memory												
	If End Row>Start Row when MADCTL B4=0:-												
	Start Row SR[15:0] → End Row ER[15:0] → 												
	If End Row>Start Row when MADCTL B4=1:-												
	End Row ER[15:0] → Start Row SR[15:0] → 												
Description	If End Row<Start Row when MADCTL B4=0:-												
	End Row ER[15:0] → Start Row SR[15:0] → 												
	If End Row = Start Row then the Partial Area will be one row deep. X = don't care.												
Restriction	SR[15:0] and ER[15:0] cannot be 0000h nor exceed the last vertical line number (01EFh).												

Register Availability	<table><tr><th>Status</th><th colspan="2">Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td colspan="2">Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td colspan="2">Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td colspan="2">Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td colspan="2">Yes</td></tr><tr><td>Sleep In</td><td colspan="2">Yes</td></tr></table>			Status	Availability		Normal Mode On, Idle Mode Off, Sleep Out	Yes		Normal Mode On, Idle Mode On, Sleep Out	Yes		Partial Mode On, Idle Mode Off, Sleep Out	Yes		Partial Mode On, Idle Mode On, Sleep Out	Yes		Sleep In	Yes	
	Status	Availability																			
	Normal Mode On, Idle Mode Off, Sleep Out	Yes																			
	Normal Mode On, Idle Mode On, Sleep Out	Yes																			
	Partial Mode On, Idle Mode Off, Sleep Out	Yes																			
	Partial Mode On, Idle Mode On, Sleep Out	Yes																			
Sleep In	Yes																				
Default	<table><tr><th>Status</th><th colspan="2">Default Value</th></tr><tr><td>Power On Sequence</td><td>SR[15:0]=0000_{HEX}</td><td>ER[15:0]=01DF_{HEX}</td></tr><tr><td>SW Reset</td><td>SR[15:0]=0000_{HEX}</td><td>ER[15:0]=01DF_{HEX}</td></tr><tr><td>HW Reset</td><td>SR[15:0]=0000_{HEX}</td><td>ER[15:0]=01DF_{HEX}</td></tr></table>			Status	Default Value		Power On Sequence	SR[15:0]=0000 _{HEX}	ER[15:0]=01DF _{HEX}	SW Reset	SR[15:0]=0000 _{HEX}	ER[15:0]=01DF _{HEX}	HW Reset	SR[15:0]=0000 _{HEX}	ER[15:0]=01DF _{HEX}						
Status	Default Value																				
Power On Sequence	SR[15:0]=0000 _{HEX}	ER[15:0]=01DF _{HEX}																			
SW Reset	SR[15:0]=0000 _{HEX}	ER[15:0]=01DF _{HEX}																			
HW Reset	SR[15:0]=0000 _{HEX}	ER[15:0]=01DF _{HEX}																			
Flow Chart	1. To Enter Partial Mode Any Mode PTLAR (30h) 1st Parameter: SR[15:8] 2nd Parameter: SR[7:0] 3rd Parameter: ER[15:8] 4th Parameter: ER[7:0] PTLON (12h) Partial Mode ON 2. To Exit Partial Mode Partial Mode ON DISPOFF (28h) NORON (13h) Normal Mode ON Normal Mode turns Partial Mode OFF RAMWR (2Ch) Image Data D1[23:0],D2[23:0]..Dn[23:0] DSIPON (29h) Optional (To avoid Tearing Effect) Legend Command Parameter Display Action Mode Sequential transfer																				

8.2.25. Vertical Scrolling Definition (33h)

33h	VSCRDEF (Vertical Scrolling Definition)												
	D/CX	RDX	WRX	D[15:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XXXXXXXX	0	0	1	1	0	0	1	1	33h
1 st Parameter	1	1	↑	XXXXXXXX	TFA[15:8]								XX
2 nd Parameter	1	1	↑	XXXXXXXX	TFA[7:0]								XX
3 rd Parameter	1	1	↑	XXXXXXXX	VSA[15:8]								XX
4 th Parameter	1	1	↑	XXXXXXXX	VSA[7:0]								XX
5 th Parameter	1	1	↑	XXXXXXXX	BFA[15:8]								XX
6 th Parameter	1	1	↑	XXXXXXXX	BFA[7:0]								XX

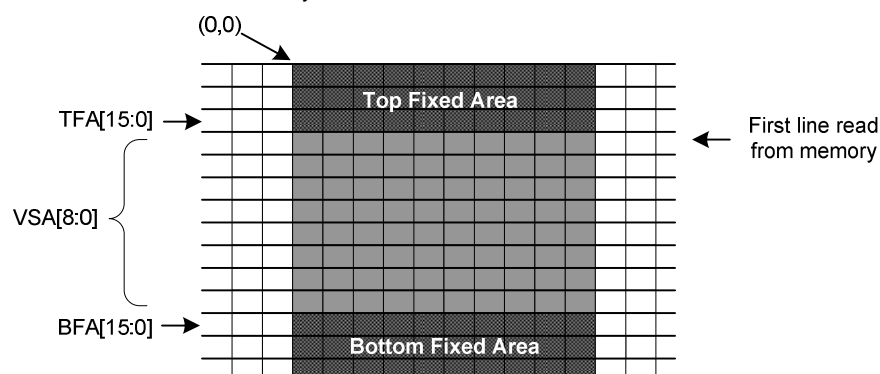
This command defines the display vertical scrolling area.

Memory Access Control (36h) B4 = 0:

The 1st & 2nd parameter, TFA[8:0], describes the Top Fixed Area in number of lines from the top of the frame memory. The top of the frame memory and top of the display device are aligned. The 3rd & 4th parameter, VSA[8:0], describes the height of the Vertical Scrolling Area in number of lines of frame memory from the Vertical Scrolling Start Address. The first line of the Vertical Scrolling Area starts immediately after the bottom most line of the Top Fixed Area. The last line of the Vertical Scrolling Area ends immediately before the top most line of the Bottom Fixed Area.

The 5th & 6th parameter, BFA[8:0], describes the Bottom Fixed Area in number of lines from the bottom of the frame memory. The bottom of the frame memory and bottom of the display device are aligned.

TFA, VSA and BFA refer to the Frame Memory Line Pointer.



Memory Access Control (36h) B4 = 1:

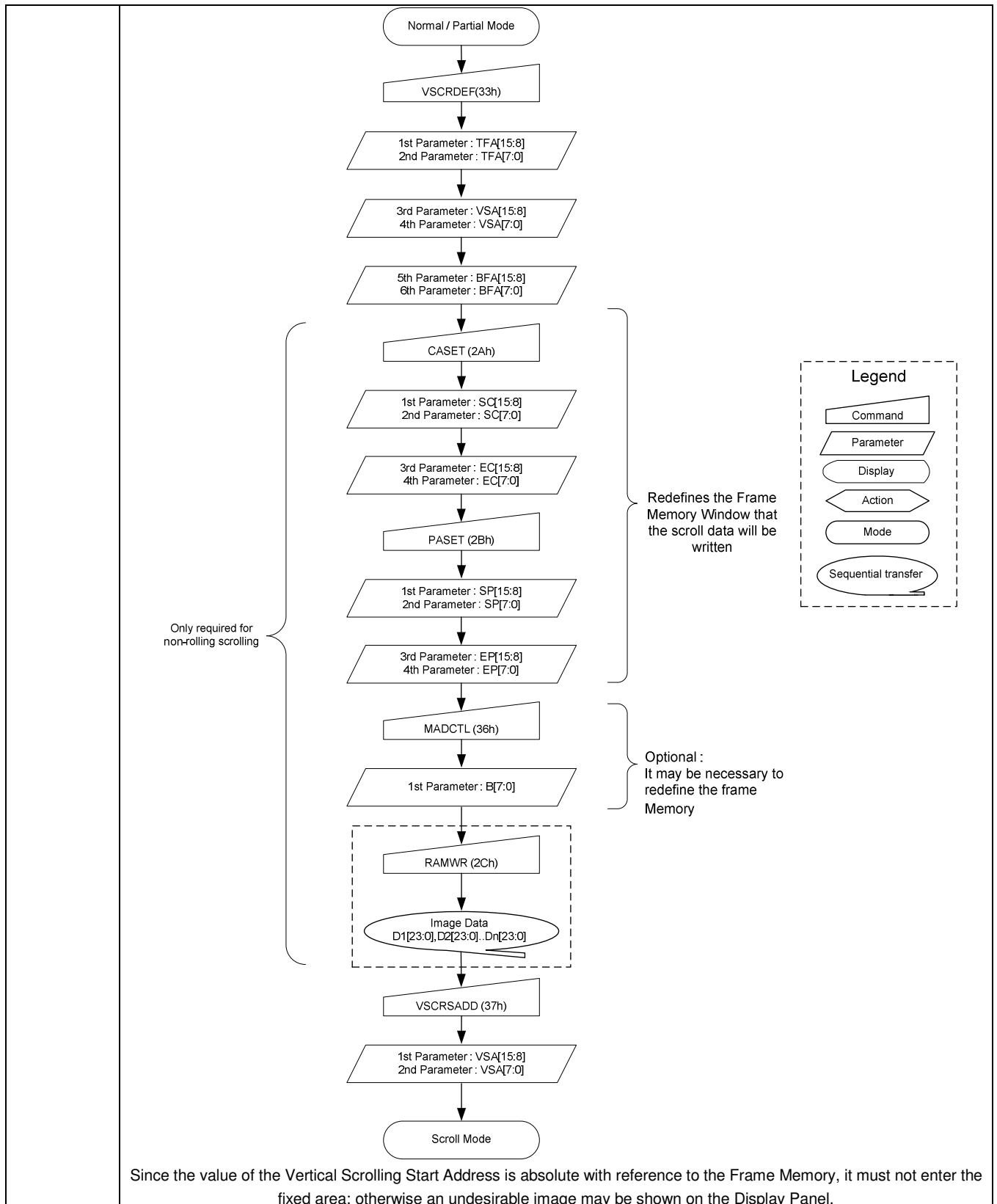
The 1st & 2nd parameter, TFA[8:0], describes the Top Fixed Area in number of lines from the bottom of the frame memory. The bottom of the frame memory and bottom of the display device are aligned.

The 3rd & 4th parameter, VSA[8:0], describes the height of the Vertical Scrolling Area in number of lines of frame memory from the Vertical Scrolling Start Address. The first line of the Vertical Scrolling Area starts immediately after the top most line of the Top Fixed Area. The last line of the Vertical Scrolling Area ends immediately before the bottom most line of the Bottom Fixed Area.

The 5th & 6th parameter, BFA[8:0], describes the Bottom Fixed Area in number of lines from the top of the frame memory. The top of the frame memory and top of the display device are aligned.

TFA, VSA and BFA refer to the Frame Memory Line Pointer.

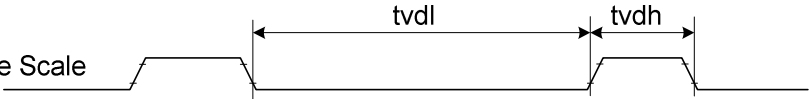
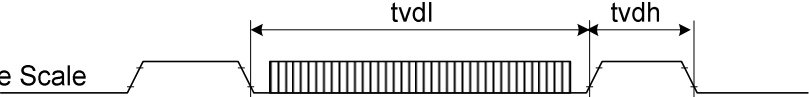
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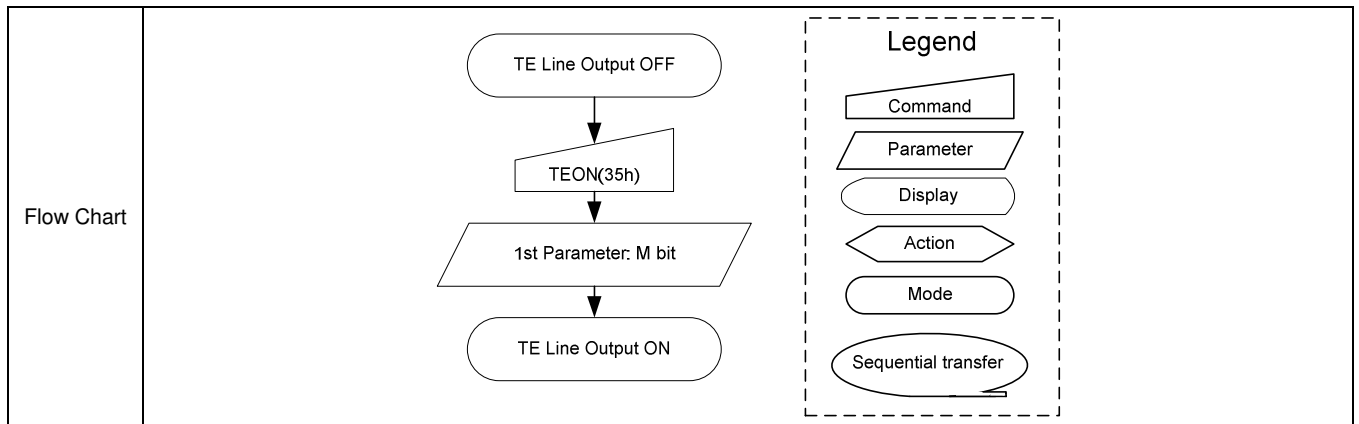


8.2.26. Tearing Effect Line OFF (34h)

34h	TEOFF (Tearing Effect Line OFF)																								
	D/CX	RDX	WRX	D[15:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX												
Command	0	1	↑	XXXXXXXX	0	0	1	1	0	1	0	0	34h												
Parameter	No parameter																								
Description	This command turns off ILI9486's Tearing Effect output signal on the TE signal line.																								
Restriction	This command has no effect when the Tearing Effect output is already off.																								
Register Availability	<table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table>													Status	Availability	Normal Mode On, Idle Mode Off, Sleep Out	Yes	Normal Mode On, Idle Mode On, Sleep Out	Yes	Partial Mode On, Idle Mode Off, Sleep Out	Yes	Partial Mode On, Idle Mode On, Sleep Out	Yes	Sleep In	Yes
Status	Availability																								
Normal Mode On, Idle Mode Off, Sleep Out	Yes																								
Normal Mode On, Idle Mode On, Sleep Out	Yes																								
Partial Mode On, Idle Mode Off, Sleep Out	Yes																								
Partial Mode On, Idle Mode On, Sleep Out	Yes																								
Sleep In	Yes																								
Default	<table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>OFF</td></tr><tr><td>SW Reset</td><td>OFF</td></tr><tr><td>HW Reset</td><td>OFF</td></tr></table>													Status	Default Value	Power On Sequence	OFF	SW Reset	OFF	HW Reset	OFF				
Status	Default Value																								
Power On Sequence	OFF																								
SW Reset	OFF																								
HW Reset	OFF																								
Flow Chart	TE Output ON or OFF ↓ TEOFF (34h) ↓ TE Output OFF Legend Command Parameter Display Action Mode Sequential transfer																								

8.2.27. Tearing Effect Line ON (35h)

35h	TEON (Tearing Effect Line ON)																									
	D/CX	RDX	WRX	D[15:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX													
Command	0	1	↑	XXXXXXXX	0	0	1	1	0	1	0	1	35h													
Parameter	1	1	↑	XXXXXXXX	X	X	X	X	X	X	X	M	XX													
Parameter	No parameter																									
Description	This command is used to turn ON the Tearing Effect output signal from the TE signal line. This output is not affected by changing MADCTL bit B4. The Tearing Effect Line On has one parameter which describes the mode of the Tearing Effect Output Line. (X=Don't Care). When M=0: The Tearing Effect Output line consists of V-Blanking information only: Vertical Time Scale  When M=1: The Tearing Effect Output Line consists of both V-Blanking and H-Blanking information: Vertical Time Scale  Note: During Sleep In Mode with Tearing Effect Line On, Tearing Effect Output pin will be active Low. X = don't care.																									
	Restriction	This command has no effect when the Tearing Effect output is already off.																								
	Register Availability	<table><thead><tr><th>Status</th><th>Availability</th></tr></thead><tbody><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></tbody></table>													Status	Availability	Normal Mode On, Idle Mode Off, Sleep Out	Yes	Normal Mode On, Idle Mode On, Sleep Out	Yes	Partial Mode On, Idle Mode Off, Sleep Out	Yes	Partial Mode On, Idle Mode On, Sleep Out	Yes	Sleep In	Yes
	Status	Availability																								
Normal Mode On, Idle Mode Off, Sleep Out	Yes																									
Normal Mode On, Idle Mode On, Sleep Out	Yes																									
Partial Mode On, Idle Mode Off, Sleep Out	Yes																									
Partial Mode On, Idle Mode On, Sleep Out	Yes																									
Sleep In	Yes																									
Default	<table><thead><tr><th>Status</th><th>Default Value</th></tr></thead><tbody><tr><td>Power On Sequence</td><td>OFF</td></tr><tr><td>SW Reset</td><td>OFF</td></tr><tr><td>HW Reset</td><td>OFF</td></tr></tbody></table>													Status	Default Value	Power On Sequence	OFF	SW Reset	OFF	HW Reset	OFF					
Status	Default Value																									
Power On Sequence	OFF																									
SW Reset	OFF																									
HW Reset	OFF																									



8.2.28. Memory Access Control (36h)

36h	MADCTL (Memory Access Control)												
	D/CX	RDX	WRX	D[15:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XXXXXXXX	0	0	1	1	0	1	1	0	36h
Parameter	1	1	↑	XXXXXXXX	MY	MX	MV	ML	BGR	MH	X	X	XX

This command defines read/write scanning direction of frame memory.

This command makes no change on the other driver status.

Bit	Symbol	Name	Description
D7	MY	Row Address Order	These 3 bits control MPU to memory write/read direction.
D6	MX	Column Address Order	
D5	MV	Row / Column Exchange	
D4	ML	Vertical Refresh Order	LCD vertical refresh direction control.
D3	BGR	RGB-BGR Order	Color selector switch control (0=RGB color filter panel, 1=BGR color filter panel)
D2	MH	Horizontal Refresh ORDER	LCD horizontal refreshing direction control.
D1	X	Reserved	Reserved
D0	X	Reserved	Reserved

X = don't care.

MV(Vertical refresh order bit)="0"

MV(Vertical refresh order bit)="1"

ML(Vertical refresh order bit)="0"

ML(Vertical refresh order bit)="1"

BGR(RGB-BGR Order control bit)="0"

BGR(RGB-BGR Order control bit)="1"

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	MH(Horizontal refresh order control bit)="0" MH(Horizontal refresh order control bit)="1" Note: Top-Left (0,0) means a physical memory location.												
Restriction													
Register Availability	<table border="1"> <thead> <tr> <th>Status</th><th>Availability</th></tr> </thead> <tbody> <tr> <td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr> <tr> <td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr> <tr> <td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr> <tr> <td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr> <tr> <td>Sleep In</td><td>Yes</td></tr> </tbody> </table>	Status	Availability	Normal Mode On, Idle Mode Off, Sleep Out	Yes	Normal Mode On, Idle Mode On, Sleep Out	Yes	Partial Mode On, Idle Mode Off, Sleep Out	Yes	Partial Mode On, Idle Mode On, Sleep Out	Yes	Sleep In	Yes
Status	Availability												
Normal Mode On, Idle Mode Off, Sleep Out	Yes												
Normal Mode On, Idle Mode On, Sleep Out	Yes												
Partial Mode On, Idle Mode Off, Sleep Out	Yes												
Partial Mode On, Idle Mode On, Sleep Out	Yes												
Sleep In	Yes												
Default	<table border="1"> <thead> <tr> <th>Status</th><th>Default Value</th></tr> </thead> <tbody> <tr> <td>Power On Sequence</td><td>00h</td></tr> <tr> <td>SW Reset</td><td>No change</td></tr> <tr> <td>HW Reset</td><td>00h</td></tr> </tbody> </table>	Status	Default Value	Power On Sequence	00h	SW Reset	No change	HW Reset	00h				
Status	Default Value												
Power On Sequence	00h												
SW Reset	No change												
HW Reset	00h												
Flow Chart	MADCTR(36h) ↓ 1st Parameter: MY, MX, MV, ML, RGB, MH Legend - Command - Parameter - Display - Action - Mode - Sequential transfer												

8.2.29. Vertical Scrolling Start Address (37h)

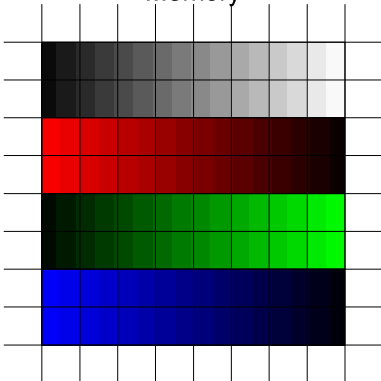
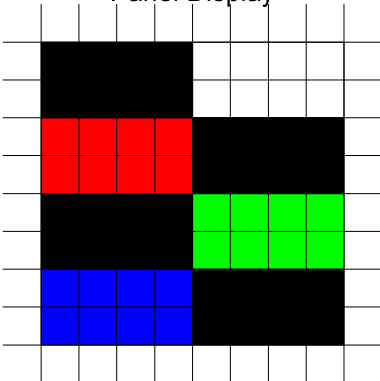
37h	VSCRSADD (Vertical Scrolling Start Address)																						
	D/CX	RDX	WRX	D[15:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX										
Command	0	1	↑	XXXXXXXX	0	0	1	1	0	1	1	1	37h										
1 st Parameter	1	1	↑	XXXXXXXX	VSP[15:8]								XX										
2 nd Parameter	1	1	↑	XXXXXXXX	VSP[7:0]								XX										
Description	This command is used together with Vertical Scrolling Definition (33h). These two commands describe the scrolling area and the scrolling mode. The Vertical Scrolling Start Address command has one parameter which describes the address of the line in the Frame Memory that will be written as the first line after the last line of the Top Fixed Area on the display as illustrated below:- on the display as illustrated below:- When MADCTL B4=0 Example: When Top Fixed Area = Bottom Fixed Area = 00, Vertical Scrolling Area = 480 and VSP='3'. Frame Memory (0, 0) → Line Pointer VSP[15:0] → (0, 479) → Pointer B4=0 <table><tr><td>0</td></tr><tr><td>1</td></tr><tr><td>2</td></tr><tr><td>3</td></tr><tr><td>4</td></tr><tr><td>..</td></tr><tr><td>..</td></tr><tr><td>477</td></tr><tr><td>478</td></tr><tr><td>479</td></tr></table> Display													0	1	2	3	4	..	..	477	478	479
	0																						
	1																						
	2																						
	3																						
	4																						
	..																						
	..																						
	477																						
	478																						
479																							
When MADCTL B4=1 Example: When Top Fixed Area = Bottom Fixed Area = 00, Vertical Scrolling Area = 480 and VSP='3'. Frame Memory (0, 0) → Line Pointer VSP[15:0] → (0, 479) → Pointer B4=1 <table><tr><td>479</td></tr><tr><td>478</td></tr><tr><td>477</td></tr><tr><td>..</td></tr><tr><td>..</td></tr><tr><td>4</td></tr><tr><td>3</td></tr><tr><td>2</td></tr><tr><td>1</td></tr><tr><td>0</td></tr></table> Display													479	478	477	..	..	4	3	2	1	0	
479																							
478																							
477																							
..																							
..																							
4																							
3																							
2																							
1																							
0																							
Notes: (1) When new Pointer position and Picture Data are sent, the result on the display will happen at the next Panel Scan to avoid tearing effect. VSP refers to the Frame Memory line Pointer. X = Don't care																							

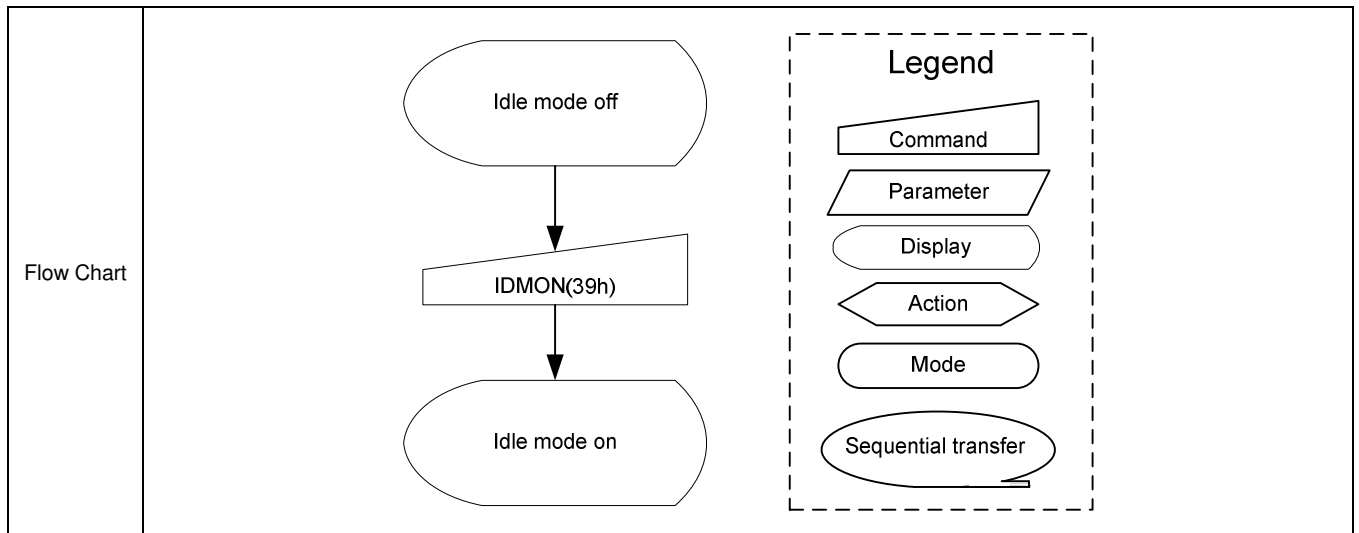
Restriction	Since the value of the Vertical Scrolling Start Address is absolute (with reference to the Frame Memory), it must not enter the fixed area (defined by Vertical Scrolling Definition (33h) – otherwise undesirable image will be displayed on the Panel.												
Register Availability	<table> <tr> <th>Status</th><th>Availability</th></tr> <tr> <td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr> <tr> <td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr> <tr> <td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr> <tr> <td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr> <tr> <td>Sleep In</td><td>Yes</td></tr> </table>	Status	Availability	Normal Mode On, Idle Mode Off, Sleep Out	Yes	Normal Mode On, Idle Mode On, Sleep Out	Yes	Partial Mode On, Idle Mode Off, Sleep Out	Yes	Partial Mode On, Idle Mode On, Sleep Out	Yes	Sleep In	Yes
Status	Availability												
Normal Mode On, Idle Mode Off, Sleep Out	Yes												
Normal Mode On, Idle Mode On, Sleep Out	Yes												
Partial Mode On, Idle Mode Off, Sleep Out	Yes												
Partial Mode On, Idle Mode On, Sleep Out	Yes												
Sleep In	Yes												
Default	<table> <tr> <th>Status</th><th>Default Value</th></tr> <tr> <td>Power On Sequence</td><td>00h</td></tr> <tr> <td>SW Reset</td><td>00h</td></tr> <tr> <td>HW Reset</td><td>00h</td></tr> </table>	Status	Default Value	Power On Sequence	00h	SW Reset	00h	HW Reset	00h				
Status	Default Value												
Power On Sequence	00h												
SW Reset	00h												
HW Reset	00h												
Flow Chart	See Vertical Scrolling Definition (33h) description.												

8.2.30. Idle Mode OFF (38h)

38h	IDMOFF (Idle Mode OFF)																								
	D/CX	RDX	WRX	D[15:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX												
Command	0	1	↑	XXXXXXXX	0	0	1	1	1	0	0	0	38h												
Parameter	No parameter																								
Description	This command causes ILI9486 to exit Idle mode. In Idle OFF mode, display panel can display maximum 262,144 colors.																								
Restriction	This command has no effect when ILI9486 is not in Idle mode.																								
Register Availability	<table><thead><tr><th>Status</th><th>Availability</th></tr></thead><tbody><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></tbody></table>													Status	Availability	Normal Mode On, Idle Mode Off, Sleep Out	Yes	Normal Mode On, Idle Mode On, Sleep Out	Yes	Partial Mode On, Idle Mode Off, Sleep Out	Yes	Partial Mode On, Idle Mode On, Sleep Out	Yes	Sleep In	Yes
Status	Availability																								
Normal Mode On, Idle Mode Off, Sleep Out	Yes																								
Normal Mode On, Idle Mode On, Sleep Out	Yes																								
Partial Mode On, Idle Mode Off, Sleep Out	Yes																								
Partial Mode On, Idle Mode On, Sleep Out	Yes																								
Sleep In	Yes																								
Default	<table><thead><tr><th>Status</th><th>Default Value</th></tr></thead><tbody><tr><td>Power On Sequence</td><td>Idle Mode Off</td></tr><tr><td>SW Reset</td><td>Idle Mode Off</td></tr><tr><td>HW Reset</td><td>Idle Mode Off</td></tr></tbody></table>													Status	Default Value	Power On Sequence	Idle Mode Off	SW Reset	Idle Mode Off	HW Reset	Idle Mode Off				
Status	Default Value																								
Power On Sequence	Idle Mode Off																								
SW Reset	Idle Mode Off																								
HW Reset	Idle Mode Off																								
Flow Chart	Idle Mode ON IDMOFF (38h) Idle Mode OFF Legend Command Parameter Display Action Mode Sequential transfer																								

8.2.31. Idle Mode ON (39h)

39h	IDMON (Idle Mode ON)																																																																																																																																																																																																			
	D/CX	RDX	WRX	D[15:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX																																																																																																																																																																																							
Command	0	1	↑	XXXXXXXX	0	0	1	1	1	0	0	1	39h																																																																																																																																																																																							
Parameter	No parameter																																																																																																																																																																																																			
Description	This command is used to enter into Idle mode on. In the idle on mode, color expression is reduced. The primary and the secondary colors using MSB of each R, G and B in the Frame Memory, 8 color depth data is displayed. Memory  → Panel Display  <table><tr><th colspan="12">Memory Contents vs Display Color</th></tr><tr><th></th><th>R₅</th><th>R₄</th><th>R₃</th><th>R₂</th><th>R₁</th><th>R₀</th><th>G₅</th><th>G₄</th><th>G₃</th><th>G₂</th><th>G₁</th><th>G₀</th><th>B₅</th><th>B₄</th><th>B₃</th><th>B₂</th><th>B₁</th><th>B₀</th></tr><tr><td>Black</td><td>0</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td><td>0</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td><td>0</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td></tr><tr><td>Blue</td><td>0</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td><td>0</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td><td>1</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td></tr><tr><td>Red</td><td>1</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td><td>0</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td><td>0</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td></tr><tr><td>Magenta</td><td>1</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td><td>0</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td><td>1</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td></tr><tr><td>Green</td><td>0</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td><td>1</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td><td>0</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td></tr><tr><td>Cyan</td><td>0</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td><td>1</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td><td>1</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td></tr><tr><td>Yellow</td><td>1</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td><td>1</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td><td>0</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td></tr><tr><td>White</td><td>1</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td><td>1</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td><td>1</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td></tr></table> X = don't care.													Memory Contents vs Display Color													R ₅	R ₄	R ₃	R ₂	R ₁	R ₀	G ₅	G ₄	G ₃	G ₂	G ₁	G ₀	B ₅	B ₄	B ₃	B ₂	B ₁	B ₀	Black	0	X	X	X	X	X	0	X	X	X	X	X	0	X	X	X	X	X	Blue	0	X	X	X	X	X	0	X	X	X	X	X	1	X	X	X	X	X	Red	1	X	X	X	X	X	0	X	X	X	X	X	0	X	X	X	X	X	Magenta	1	X	X	X	X	X	0	X	X	X	X	X	1	X	X	X	X	X	Green	0	X	X	X	X	X	1	X	X	X	X	X	0	X	X	X	X	X	Cyan	0	X	X	X	X	X	1	X	X	X	X	X	1	X	X	X	X	X	Yellow	1	X	X	X	X	X	1	X	X	X	X	X	0	X	X	X	X	X	White	1	X	X	X	X	X	1	X	X	X	X	X	1	X	X	X	X	X
	Memory Contents vs Display Color																																																																																																																																																																																																			
		R ₅	R ₄	R ₃	R ₂	R ₁	R ₀	G ₅	G ₄	G ₃	G ₂	G ₁	G ₀	B ₅	B ₄	B ₃	B ₂	B ₁	B ₀																																																																																																																																																																																	
	Black	0	X	X	X	X	X	0	X	X	X	X	X	0	X	X	X	X	X																																																																																																																																																																																	
Blue	0	X	X	X	X	X	0	X	X	X	X	X	1	X	X	X	X	X																																																																																																																																																																																		
Red	1	X	X	X	X	X	0	X	X	X	X	X	0	X	X	X	X	X																																																																																																																																																																																		
Magenta	1	X	X	X	X	X	0	X	X	X	X	X	1	X	X	X	X	X																																																																																																																																																																																		
Green	0	X	X	X	X	X	1	X	X	X	X	X	0	X	X	X	X	X																																																																																																																																																																																		
Cyan	0	X	X	X	X	X	1	X	X	X	X	X	1	X	X	X	X	X																																																																																																																																																																																		
Yellow	1	X	X	X	X	X	1	X	X	X	X	X	0	X	X	X	X	X																																																																																																																																																																																		
White	1	X	X	X	X	X	1	X	X	X	X	X	1	X	X	X	X	X																																																																																																																																																																																		
Restriction	This command has no effect when module is already in idle off mode.																																																																																																																																																																																																			
Register Availability	<table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table>													Status	Availability	Normal Mode On, Idle Mode Off, Sleep Out	Yes	Normal Mode On, Idle Mode On, Sleep Out	Yes	Partial Mode On, Idle Mode Off, Sleep Out	Yes	Partial Mode On, Idle Mode On, Sleep Out	Yes	Sleep In	Yes																																																																																																																																																																											
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Partial Mode On, Idle Mode Off, Sleep Out	Yes																																																																																																																																																																																																			
Partial Mode On, Idle Mode On, Sleep Out	Yes																																																																																																																																																																																																			
Sleep In	Yes																																																																																																																																																																																																			
Default	<table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>Idle mode OFF</td></tr><tr><td>SW Reset</td><td>Idle mode OFF</td></tr><tr><td>HW Reset</td><td>Idle mode OFF</td></tr></table>													Status	Default Value	Power On Sequence	Idle mode OFF	SW Reset	Idle mode OFF	HW Reset	Idle mode OFF																																																																																																																																																																															
Status	Default Value																																																																																																																																																																																																			
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SW Reset	Idle mode OFF																																																																																																																																																																																																			
HW Reset	Idle mode OFF																																																																																																																																																																																																			

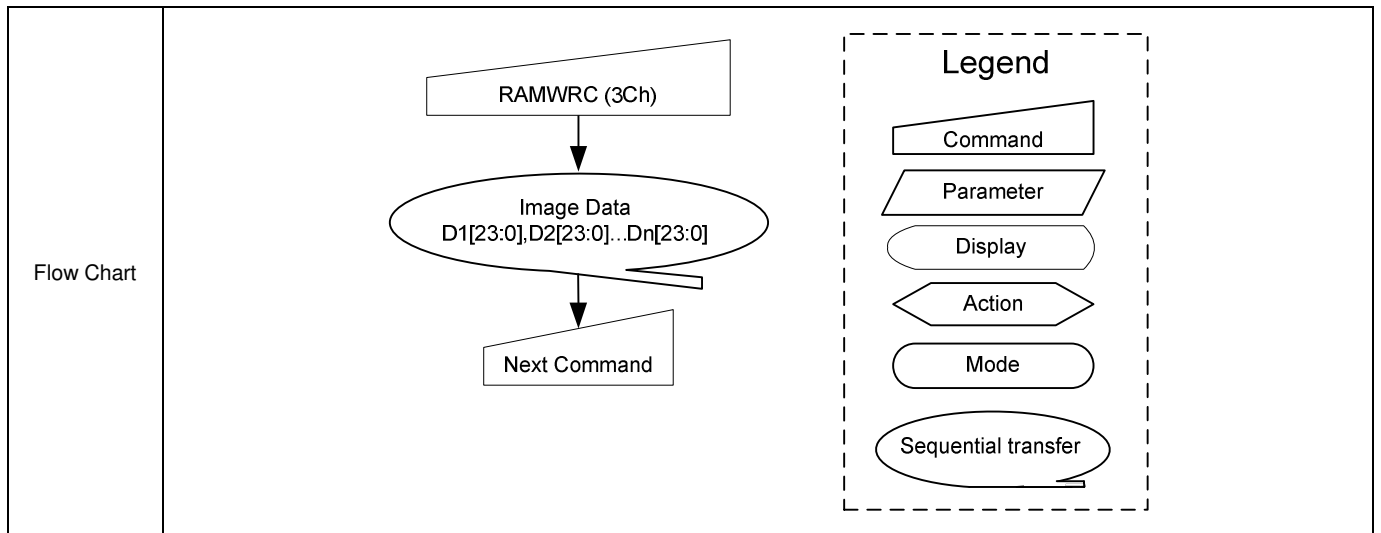


8.2.32. Interface Pixel Format (3Ah)

3Ah	COLMOD (Interface Pixel Format)																																																																																													
	D/CX	RDX	WRX	D[15:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX																																																																																	
Command	0	1	↑	XXXXXXXX	0	0	1	1	1	0	1	0	3Ah																																																																																	
Parameter	1	1	↑	XXXXXXXX	DPI[3:0]				X	DBI[2:0]			XX																																																																																	
Description	This command sets the pixel format for the RGB image data used by the interface. DPI[3:0] is the pixel format select of RGB interface and DBI[2:0] is the pixel format of CPU interface. If a particular interface, either RGB interface or CPU interface, is not used then the corresponding bits in the parameter are ignored. The pixel format are shown in the table below. <table><tr><th colspan="4">DPI[3:0]</th><th>RGB Interface Format</th><th colspan="3">DBI[2:0]</th><th>CPU Interface Format</th></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td><td>Reserved</td><td>0</td><td>0</td><td>0</td><td>Reserved</td></tr><tr><td>0</td><td>0</td><td>0</td><td>1</td><td>Reserved</td><td>0</td><td>0</td><td>1</td><td>Reserved</td></tr><tr><td>0</td><td>0</td><td>1</td><td>0</td><td>Reserved</td><td>0</td><td>1</td><td>0</td><td>Reserved</td></tr><tr><td>0</td><td>0</td><td>1</td><td>1</td><td>Reserved</td><td>0</td><td>1</td><td>1</td><td>Reserved</td></tr><tr><td>0</td><td>1</td><td>0</td><td>0</td><td>Reserved</td><td>1</td><td>0</td><td>0</td><td>Reserved</td></tr><tr><td>0</td><td>1</td><td>0</td><td>1</td><td>16 bits / pixel</td><td>1</td><td>0</td><td>1</td><td>16 bits / pixel</td></tr><tr><td>0</td><td>1</td><td>1</td><td>0</td><td>18 bits / pixel</td><td>1</td><td>1</td><td>0</td><td>18 bits / pixel</td></tr><tr><td>0</td><td>1</td><td>1</td><td>1</td><td>Reserved</td><td>1</td><td>1</td><td>1</td><td>Reserved</td></tr></table> X = don't care													DPI[3:0]				RGB Interface Format	DBI[2:0]			CPU Interface Format	0	0	0	0	Reserved	0	0	0	Reserved	0	0	0	1	Reserved	0	0	1	Reserved	0	0	1	0	Reserved	0	1	0	Reserved	0	0	1	1	Reserved	0	1	1	Reserved	0	1	0	0	Reserved	1	0	0	Reserved	0	1	0	1	16 bits / pixel	1	0	1	16 bits / pixel	0	1	1	0	18 bits / pixel	1	1	0	18 bits / pixel	0	1	1	1	Reserved	1	1	1	Reserved
	DPI[3:0]				RGB Interface Format	DBI[2:0]			CPU Interface Format																																																																																					
	0	0	0	0	Reserved	0	0	0	Reserved																																																																																					
	0	0	0	1	Reserved	0	0	1	Reserved																																																																																					
	0	0	1	0	Reserved	0	1	0	Reserved																																																																																					
	0	0	1	1	Reserved	0	1	1	Reserved																																																																																					
	0	1	0	0	Reserved	1	0	0	Reserved																																																																																					
	0	1	0	1	16 bits / pixel	1	0	1	16 bits / pixel																																																																																					
	0	1	1	0	18 bits / pixel	1	1	0	18 bits / pixel																																																																																					
	0	1	1	1	Reserved	1	1	1	Reserved																																																																																					
Restriction																																																																																														
Register Availability	<table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table>													Status	Availability	Normal Mode On, Idle Mode Off, Sleep Out	Yes	Normal Mode On, Idle Mode On, Sleep Out	Yes	Partial Mode On, Idle Mode Off, Sleep Out	Yes	Partial Mode On, Idle Mode On, Sleep Out	Yes	Sleep In	Yes																																																																					
Status	Availability																																																																																													
Normal Mode On, Idle Mode Off, Sleep Out	Yes																																																																																													
Normal Mode On, Idle Mode On, Sleep Out	Yes																																																																																													
Partial Mode On, Idle Mode Off, Sleep Out	Yes																																																																																													
Partial Mode On, Idle Mode On, Sleep Out	Yes																																																																																													
Sleep In	Yes																																																																																													
Default	<table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>06h</td></tr><tr><td>SW Reset</td><td>06h</td></tr><tr><td>HW Reset</td><td>06h</td></tr></table>													Status	Default Value	Power On Sequence	06h	SW Reset	06h	HW Reset	06h																																																																									
Status	Default Value																																																																																													
Power On Sequence	06h																																																																																													
SW Reset	06h																																																																																													
HW Reset	06h																																																																																													
Flow Chart	n-bit/Pixel Mode ↓ COLMOD (3Ah) ↓ 1st parameter: D[2:0]=" XXX" ↓ New m-bit/Pixel Mode Legend Command Parameter Display Action Mode Sequential transfer																																																																																													

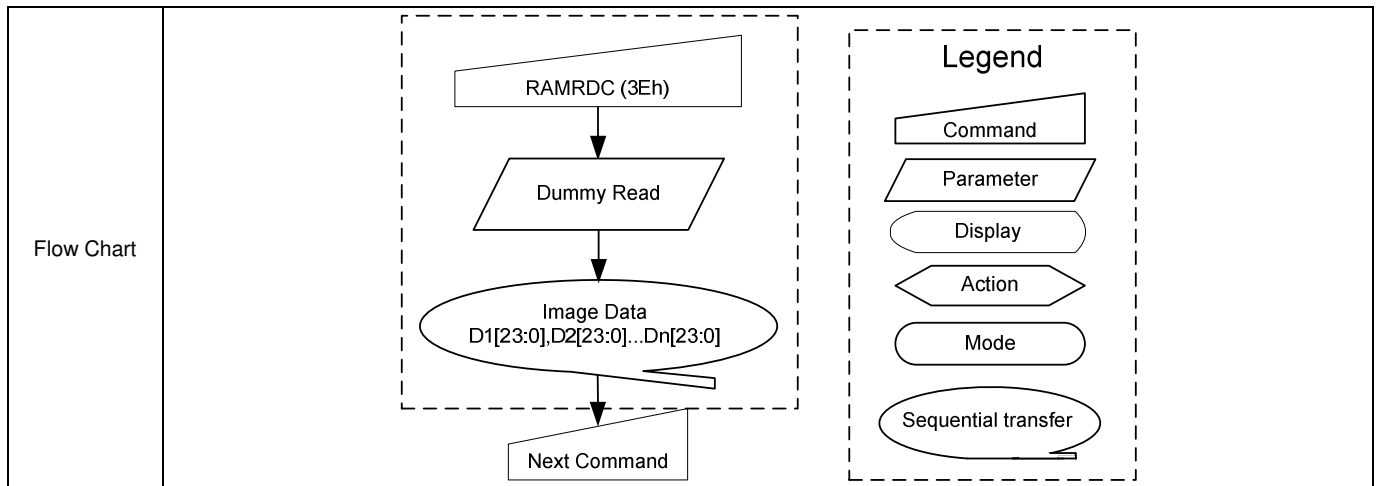
8.2.33. Memory Write Continue (3Ch)

3Ch	RAMWRC (Memory Write Continue)																								
	D/CX	RDX	WRX	D[15:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX												
Command	0	1	↑	XXXXXXXX	0	0	1	1	1	1	0	0	3Ch												
1 st Parameter	1	1	↑	D1[15:0]									XX												
:	1	1	↑	Dx[15:0]									XX												
N th Parameter	1	1	↑	Dn[15:0]									XX												
Description	This command is used to transfer data from MCU to frame memory, if there is wanted to continue memory write after “Memory Write (2Ch)” command.																								
	This command makes no change to the other driver status.																								
	When this command is accepted, the column register and the page register are not reset to the Start Column/Start Page positions as it has been done on “Memory Write (2Ch)” command.																								
	Then D[15:0] is stored in frame memory and the column register and the page register incremented as table below: Column and Page Counter Control.																								
	Condition						Column counter			Page Counter															
	When RAMWR/RAMRD command is accepted						Return to “Start Column”			Return to “Start Page”															
	Complete Pixel Read/Write action						Increment by 1			No change															
	The Column counter value is large than “End Column”						Return to “Start Column”			Increment by 1															
	The Page counter value is large than “End Page”						Return to “Start Column”			Return to “Start Page”															
	Sending any other command can stop frame Write.																								
Restriction	X = don't care.																								
	There is no restriction on length of parameters. No access in the frame memory in Sleep In mode.																								
Register Availability	<table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table>													Status	Availability	Normal Mode On, Idle Mode Off, Sleep Out	Yes	Normal Mode On, Idle Mode On, Sleep Out	Yes	Partial Mode On, Idle Mode Off, Sleep Out	Yes	Partial Mode On, Idle Mode On, Sleep Out	Yes	Sleep In	Yes
Status	Availability																								
Normal Mode On, Idle Mode Off, Sleep Out	Yes																								
Normal Mode On, Idle Mode On, Sleep Out	Yes																								
Partial Mode On, Idle Mode Off, Sleep Out	Yes																								
Partial Mode On, Idle Mode On, Sleep Out	Yes																								
Sleep In	Yes																								
Default	<table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>Contents of memory is set randomly</td></tr><tr><td>SW Reset</td><td>Contents of memory is set randomly</td></tr><tr><td>HW Reset</td><td>Contents of memory is set randomly</td></tr></table>													Status	Default Value	Power On Sequence	Contents of memory is set randomly	SW Reset	Contents of memory is set randomly	HW Reset	Contents of memory is set randomly				
Status	Default Value																								
Power On Sequence	Contents of memory is set randomly																								
SW Reset	Contents of memory is set randomly																								
HW Reset	Contents of memory is set randomly																								

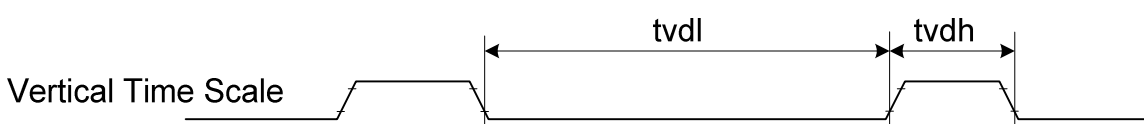


8.2.34. Memory Read Continue (3Eh)

3Eh	RAMRDRC (Memory Read Continue)												
	D/CX	RDX	WRX	D[15:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XXXXXXXX	0	0	1	1	1	1	1	0	3Eh
1 st Parameter	1	↑	1	XXXXXXXX	X	X	X	X	X	X	X	X	XX
2 nd Parameter	1	↑	1	D1[15:0]									XX
:	1	↑	1	Dx[15:0]									XX
N th Parameter	1	↑	1	Dn[15:0]									XX
Description	This command is used to transfer data from frame memory to MCU, if there is wanted to continue memory read after “Memory Read (2Eh)” command.												
	This command makes no change to the other driver status.												
	When this command is accepted, the column register and the page register are not reset to the Start Column/Start Page positions as it has been done on “Memory Read (2Eh)” command.												
	Then D[15:0] is read back from the frame memory and the column register and the page register incremented as table below:												
	Column and Page Counter Control.												
	Condition						Column counter			Page Counter			
	When RAMWR/RAMRD command is accepted						Return to “Start Column”			Return to “Start Page”			
	Complete Pixel Read/Write action						Increment by 1			No change			
	The Column counter value is large than “End Column”						Return to “Start Column”			Increment by 1			
	The Page counter value is large than “End Page”						Return to “Start Column”			Return to “Start Page”			
Frame Read can be stopped by sending any other command.													
X = can be ‘0’ or ‘1’													
Restriction	There is no restriction on length of parameters.												
	No access in the frame memory in Sleep In mode.												
Register Availability													
	Status								Availability				
	Normal Mode On, Idle Mode Off, Sleep Out								Yes				
	Normal Mode On, Idle Mode On, Sleep Out								Yes				
	Partial Mode On, Idle Mode Off, Sleep Out								Yes				
	Partial Mode On, Idle Mode On, Sleep Out								Yes				
Default													
	Status						Default Value						
	Power On Sequence						Contents of memory is set randomly						
	SW Reset						Contents of memory is set randomly						
	HW Reset						Contents of memory is set randomly						



8.2.35. Write Tear Scan Line (44h)

44h	TESLWR (Write Tear Scan Line)																								
	D/CX	RDX	WRX	D[15:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX												
Command	0	1	↑	XXXXXXXX	0	1	0	0	0	1	0	0	44h												
1 st Parameter	1	1	↑	XXXXXXXX	N[15:8]								XX												
2 nd Parameter	1	1	↑	XXXXXXXX	N[7:0]								XX												
Description	This command turns on the display Tearing Effect output signal on the TE signal line when the display reaches line N. The TE signal is not affected by changing Memory Access Control bit B4. The Tearing Effect Line On has one parameter that describes the Tearing Effect Output Line mode. The Tearing Effect Output line consists of V-Blanking information only. Vertical Time Scale  Note that Set Tear Scan Line with N = 0 is equivalent to Tearing Effect Line ON with M = 0. The Tearing Effect Output line shall be active low when ILI9486 is in Sleep mode.																								
Restriction	This command has no effect when Tearing Effect output is already ON.																								
Register Availability	<table><thead><tr><th>Status</th><th>Availability</th></tr></thead><tbody><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></tbody></table>													Status	Availability	Normal Mode On, Idle Mode Off, Sleep Out	Yes	Normal Mode On, Idle Mode On, Sleep Out	Yes	Partial Mode On, Idle Mode Off, Sleep Out	Yes	Partial Mode On, Idle Mode On, Sleep Out	Yes	Sleep In	Yes
Status	Availability																								
Normal Mode On, Idle Mode Off, Sleep Out	Yes																								
Normal Mode On, Idle Mode On, Sleep Out	Yes																								
Partial Mode On, Idle Mode Off, Sleep Out	Yes																								
Partial Mode On, Idle Mode On, Sleep Out	Yes																								
Sleep In	Yes																								
Default	<table><thead><tr><th>Status</th><th>Default Value</th></tr></thead><tbody><tr><td>Power On Sequence</td><td>00h</td></tr><tr><td>SW Reset</td><td>No change</td></tr><tr><td>HW Reset</td><td>00h</td></tr></tbody></table>													Status	Default Value	Power On Sequence	00h	SW Reset	No change	HW Reset	00h				
Status	Default Value																								
Power On Sequence	00h																								
SW Reset	No change																								
HW Reset	00h																								
Flow Chart	TE Output ON or OFF ↓ TESLWR (44h) ↓ 1st Parameter : N[15:8] ↓ 2nd Parameter : N[7:0] ↓ TE Line Output ON Legend Command Parameter Display Action Mode Sequential transfer																								

8.2.36. Read Scan Line (45h)

45h	TESLRD (Read Tear Scan Line)																								
	D/CX	RDX	WRX	D[15:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX												
Command	0	1	↑	XXXXXXXX	0	1	0	0	0	1	0	1	45h												
1 st Parameter	1	↑	1	XXXXXXXX	X	X	X	X	X	X	X	X	XX												
2 nd Parameter	1	↑	1	XXXXXXXX	N[15:8]								XX												
3 rd Parameter	1	↑	1	XXXXXXXX	N[7:0]								XX												
Description	The display returns the current scan line, N, used to update the display device. The total number of scan lines on a display device is defined as VSYNC + VBP + VACT + VFP. The first scan line is defined as the first line of V-Sync and is denoted as Line 0. When in Sleep Mode, the value returned by Read Scan Line command is undefined.																								
Restriction	None																								
Register Availability	<table><thead><tr><th>Status</th><th>Availability</th></tr></thead><tbody><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></tbody></table>													Status	Availability	Normal Mode On, Idle Mode Off, Sleep Out	Yes	Normal Mode On, Idle Mode On, Sleep Out	Yes	Partial Mode On, Idle Mode Off, Sleep Out	Yes	Partial Mode On, Idle Mode On, Sleep Out	Yes	Sleep In	Yes
Status	Availability																								
Normal Mode On, Idle Mode Off, Sleep Out	Yes																								
Normal Mode On, Idle Mode On, Sleep Out	Yes																								
Partial Mode On, Idle Mode Off, Sleep Out	Yes																								
Partial Mode On, Idle Mode On, Sleep Out	Yes																								
Sleep In	Yes																								
Default	<table><thead><tr><th>Status</th><th>Default Value</th></tr></thead><tbody><tr><td>Power On Sequence</td><td>00h</td></tr><tr><td>SW Reset</td><td>No change</td></tr><tr><td>HW Reset</td><td>00h</td></tr></tbody></table>													Status	Default Value	Power On Sequence	00h	SW Reset	No change	HW Reset	00h				
Status	Default Value																								
Power On Sequence	00h																								
SW Reset	No change																								
HW Reset	00h																								
Flow Chart	TESLRD (45h) Dummy Read 2nd Parameter : N[15:8] 3rd Parameter : N[7:0] Host ILI9486 Legend Command Parameter Display Action Mode Sequential transfer																								

8.2.37. Write Display Brightness Value (51h)

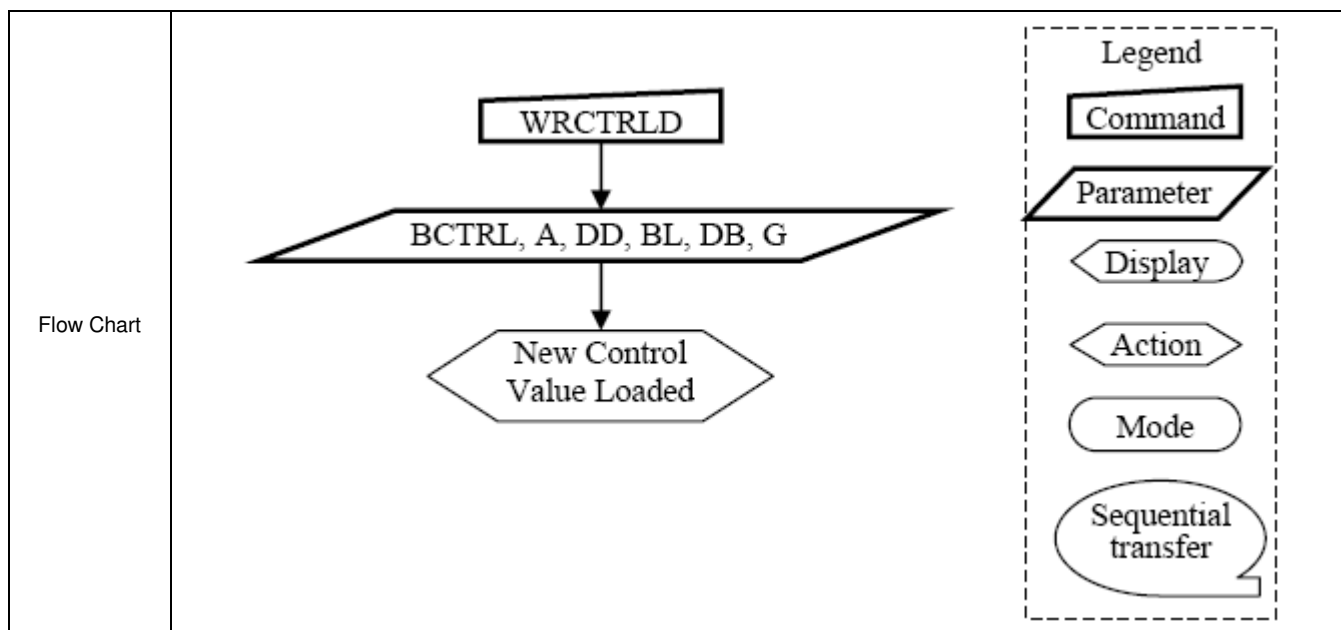
51h	WRDISBV (Write Display Brightness)																								
	D/CX	RDX	WRX	D[15:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX												
Command	0	1	↑	XXXXXXXX	0	1	0	1	0	0	0	1	51h												
1 st Parameter	1	1	↑	XXXXXXXX	DBV[7:0]								XX												
Description	This command is used to adjust the brightness value of the display. DBV[7:0]: 8 bit, for display brightness of manual brightness setting and CABC in ILI9486. There is a PWM output signal, PWM_OUT pin, to control the LED driver IC in order to control display brightness. In principle relationship is that 00h value means the lowest brightness and FFh value means the highest brightness.																								
Restriction																									
Register Availability	<table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode ON, Idle Mode OFF, Sleep OUT</td><td>Yes</td></tr><tr><td>Normal Mode ON, Idle Mode ON, Sleep OUT</td><td>Yes</td></tr><tr><td>Partial Mode ON, Idle Mode OFF, Sleep OUT</td><td>Yes</td></tr><tr><td>Partial Mode ON, Idle Mode ON, Sleep OUT</td><td>Yes</td></tr><tr><td>Sleep IN</td><td>Yes</td></tr></table>													Status	Availability	Normal Mode ON, Idle Mode OFF, Sleep OUT	Yes	Normal Mode ON, Idle Mode ON, Sleep OUT	Yes	Partial Mode ON, Idle Mode OFF, Sleep OUT	Yes	Partial Mode ON, Idle Mode ON, Sleep OUT	Yes	Sleep IN	Yes
Status	Availability																								
Normal Mode ON, Idle Mode OFF, Sleep OUT	Yes																								
Normal Mode ON, Idle Mode ON, Sleep OUT	Yes																								
Partial Mode ON, Idle Mode OFF, Sleep OUT	Yes																								
Partial Mode ON, Idle Mode ON, Sleep OUT	Yes																								
Sleep IN	Yes																								
Default	<table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power ON Sequence</td><td>00h</td></tr><tr><td>H/W Reset</td><td>00h</td></tr></table>													Status	Default Value	Power ON Sequence	00h	H/W Reset	00h						
Status	Default Value																								
Power ON Sequence	00h																								
H/W Reset	00h																								
Flow Chart	WRDISBV ↓ DBV[7..0] ↓ New Display Luminance Value Loaded Legend Command Parameter Display Action Mode Sequential transfer																								

8.2.38. Read Display Brightness Value (52h)

52h	RDDISBV (Read Display Brightness Value)																								
	D/CX	RDX	WRX	D[15:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX												
Command	0	1	↑	XXXXXXXX	0	1	0	1	0	0	1	0	52h												
1 st Parameter	1	↑	1	XXXXXXXX	X	X	X	X	X	X	X	X	XX												
2 nd Parameter	1	↑	1	XXXXXXXX	DBV[7:0]							XX													
Description	This command is used to return the brightness value of the display. DBV[7:0] is reset when display is in sleep-in mode. DBV[7:0] is '0' when bit BCTRL of “Write CTRL Display (53h)” command is '0'. DBV[7:0] is manual set brightness specified with “Write CTRL Display (53h)” command when BCTRL bit is '1'. When bit BCTRL of “Write CTRL Display (53h)” command is '1' and C1/C0 bit of “Write Content Adaptive Brightness Control (55h)” command are '0', DBV[7:0] output is the brightness value specified with “ Write Display Brightness (51h)” command.																								
Restriction	ILI9486 is sending 2nd parameter value on the data lines if the MCU wants to read more than one parameter (= more than 2 RDX cycle) on parallel MCU interface. Only 2nd parameter is sent on DSI (The 1st parameter is not sent).																								
Register Availability	<table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode ON, Idle Mode OFF, Sleep OUT</td><td>Yes</td></tr><tr><td>Normal Mode ON, Idle Mode ON, Sleep OUT</td><td>Yes</td></tr><tr><td>Partial Mode ON, Idle Mode OFF, Sleep OUT</td><td>Yes</td></tr><tr><td>Partial Mode ON, Idle Mode ON, Sleep OUT</td><td>Yes</td></tr><tr><td>Sleep IN</td><td>Yes</td></tr></table>													Status	Availability	Normal Mode ON, Idle Mode OFF, Sleep OUT	Yes	Normal Mode ON, Idle Mode ON, Sleep OUT	Yes	Partial Mode ON, Idle Mode OFF, Sleep OUT	Yes	Partial Mode ON, Idle Mode ON, Sleep OUT	Yes	Sleep IN	Yes
Status	Availability																								
Normal Mode ON, Idle Mode OFF, Sleep OUT	Yes																								
Normal Mode ON, Idle Mode ON, Sleep OUT	Yes																								
Partial Mode ON, Idle Mode OFF, Sleep OUT	Yes																								
Partial Mode ON, Idle Mode ON, Sleep OUT	Yes																								
Sleep IN	Yes																								
Default	<table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power ON Sequence</td><td>00h</td></tr><tr><td>H/W Reset</td><td>00h</td></tr></table>													Status	Default Value	Power ON Sequence	00h	H/W Reset	00h						
Status	Default Value																								
Power ON Sequence	00h																								
H/W Reset	00h																								
	Serial I/F Mode (P/SX = Low) Read RDDISBV ↓ Send 2nd Parameter Parallel I/F Mode (P/SX = High) Read RDDISBV ↓ Dummy Read ↓ Send 2nd Parameter Host Display Legend Command Parameter Display Action Mode Sequential transfer																								

8.2.39. Write CTRL Display Value (53h)

53h	WRCTRLD (Write Control Display)																								
	D/CX	RDX	WRX	D[15:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX												
Command	0	1	↑	XXXXXXXX	0	1	0	1	0	0	1	1	53h												
1 st Parameter	1	1	↑	XXXXXXXX	X	X	BCTRL	X	DD	BL	X	X	XX												
Description	This command is used to control display brightness.																								
	BCTRL : Brightness Control Block On/Off, This bit is always used to switch brightness for display.																								
	<table><tr><th>BCTRL</th><th>Description</th></tr><tr><td>0</td><td>Brightness Control Block OFF (DBV[7:0]=00h)</td></tr><tr><td>1</td><td>Brightness Control Block ON (DBV[7:0] is active)</td></tr></table>													BCTRL	Description	0	Brightness Control Block OFF (DBV[7:0]=00h)	1	Brightness Control Block ON (DBV[7:0] is active)						
	BCTRL	Description																							
	0	Brightness Control Block OFF (DBV[7:0]=00h)																							
	1	Brightness Control Block ON (DBV[7:0] is active)																							
	DD : Display Dimming Control. This function is only for manual brightness setting.																								
	<table><tr><th>DD</th><th>Description</th></tr><tr><td>0</td><td>Display Dimming OFF</td></tr><tr><td>1</td><td>Display Dimming ON</td></tr></table>													DD	Description	0	Display Dimming OFF	1	Display Dimming ON						
	DD	Description																							
	0	Display Dimming OFF																							
1	Display Dimming ON																								
BL : Backlight Control On/Off																									
<table><tr><th>BL</th><th>Description</th></tr><tr><td>0</td><td>Backlight Control OFF</td></tr><tr><td>1</td><td>Backlight Control ON</td></tr></table>													BL	Description	0	Backlight Control OFF	1	Backlight Control ON							
BL	Description																								
0	Backlight Control OFF																								
1	Backlight Control ON																								
Dimming function is adapted to the brightness registers for display when bit BCTRL is changed at DD=1, e.g. BCTRL: 0 -> 1 or 1-> 0.																									
When BL bit change from “On” to “Off”, backlight is turned off without gradual dimming, even if dimming-on (DD=1) are selected.																									
X = Don't care																									
Restriction																									
Register Availability	<table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode ON, Idle Mode OFF, Sleep OUT</td><td>Yes</td></tr><tr><td>Normal Mode ON, Idle Mode ON, Sleep OUT</td><td>Yes</td></tr><tr><td>Partial Mode ON, Idle Mode OFF, Sleep OUT</td><td>Yes</td></tr><tr><td>Partial Mode ON, Idle Mode ON, Sleep OUT</td><td>Yes</td></tr><tr><td>Sleep IN</td><td>Yes</td></tr></table>													Status	Availability	Normal Mode ON, Idle Mode OFF, Sleep OUT	Yes	Normal Mode ON, Idle Mode ON, Sleep OUT	Yes	Partial Mode ON, Idle Mode OFF, Sleep OUT	Yes	Partial Mode ON, Idle Mode ON, Sleep OUT	Yes	Sleep IN	Yes
Status	Availability																								
Normal Mode ON, Idle Mode OFF, Sleep OUT	Yes																								
Normal Mode ON, Idle Mode ON, Sleep OUT	Yes																								
Partial Mode ON, Idle Mode OFF, Sleep OUT	Yes																								
Partial Mode ON, Idle Mode ON, Sleep OUT	Yes																								
Sleep IN	Yes																								
Default	<table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power ON Sequence</td><td>00h</td></tr><tr><td>H/W Reset</td><td>00h</td></tr></table>													Status	Default Value	Power ON Sequence	00h	H/W Reset	00h						
Status	Default Value																								
Power ON Sequence	00h																								
H/W Reset	00h																								



8.2.40. Read CTRL Display Value (54h)

54h	RDCTRLD (Read Control Display Value)																								
	D/CX	RDX	WRX	D[15:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX												
Command	0	1	↑	XXXXXXXX	0	1	0	1	0	1	0	0	54h												
1 st Parameter	1	↑	1	XXXXXXXX	X	X	X	X	X	X	X	X	XX												
2 nd Parameter	1	↑	1	XXXXXXXX	X	X	BCTRL	X	DD	BL	X	X	XX												
Description	This command is used to control display brightness.																								
	BCTRL: Brightness Control Block On/Off, This bit is always used to switch brightness for display.																								
	<table><tr><th>BCTRL</th><th>Description</th></tr><tr><td>0</td><td>Brightness Control Block OFF (DBV[7:0]=00h)</td></tr><tr><td>1</td><td>Brightness Control Block ON (DBV[7:0] is active)</td></tr></table>													BCTRL	Description	0	Brightness Control Block OFF (DBV[7:0]=00h)	1	Brightness Control Block ON (DBV[7:0] is active)						
	BCTRL	Description																							
	0	Brightness Control Block OFF (DBV[7:0]=00h)																							
	1	Brightness Control Block ON (DBV[7:0] is active)																							
DD: Display Dimming Control. This function is only for manual brightness setting.																									
<table><tr><th>DD</th><th>Description</th></tr><tr><td>0</td><td>Display Dimming OFF</td></tr><tr><td>1</td><td>Display Dimming ON</td></tr></table>													DD	Description	0	Display Dimming OFF	1	Display Dimming ON							
DD	Description																								
0	Display Dimming OFF																								
1	Display Dimming ON																								
BL: Backlight Control On/Off																									
<table><tr><th>BL</th><th>Description</th></tr><tr><td>0</td><td>Backlight Control OFF</td></tr><tr><td>1</td><td>Backlight Control ON</td></tr></table>													BL	Description	0	Backlight Control OFF	1	Backlight Control ON							
BL	Description																								
0	Backlight Control OFF																								
1	Backlight Control ON																								
X = Don't care																									
Restriction																									
Register Availability	<table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode ON, Idle Mode OFF, Sleep OUT</td><td>Yes</td></tr><tr><td>Normal Mode ON, Idle Mode ON, Sleep OUT</td><td>Yes</td></tr><tr><td>Partial Mode ON, Idle Mode OFF, Sleep OUT</td><td>Yes</td></tr><tr><td>Partial Mode ON, Idle Mode ON, Sleep OUT</td><td>Yes</td></tr><tr><td>Sleep IN</td><td>Yes</td></tr></table>													Status	Availability	Normal Mode ON, Idle Mode OFF, Sleep OUT	Yes	Normal Mode ON, Idle Mode ON, Sleep OUT	Yes	Partial Mode ON, Idle Mode OFF, Sleep OUT	Yes	Partial Mode ON, Idle Mode ON, Sleep OUT	Yes	Sleep IN	Yes
Status	Availability																								
Normal Mode ON, Idle Mode OFF, Sleep OUT	Yes																								
Normal Mode ON, Idle Mode ON, Sleep OUT	Yes																								
Partial Mode ON, Idle Mode OFF, Sleep OUT	Yes																								
Partial Mode ON, Idle Mode ON, Sleep OUT	Yes																								
Sleep IN	Yes																								
Default	<table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power ON Sequence</td><td>00h</td></tr><tr><td>H/W Reset</td><td>00h</td></tr></table>													Status	Default Value	Power ON Sequence	00h	H/W Reset	00h						
Status	Default Value																								
Power ON Sequence	00h																								
H/W Reset	00h																								
Flow Chart	Serial I/F Mode (P/SX = Low) Read RDCTRLD Send 2nd Parameter Parallel I/F Mode (P/SX = High) Read RDCTRLD Dummy Read Send 2nd Parameter Host Display Legend Command Parameter Display Action Mode Sequential transfer																								

8.2.41. Write Content Adaptive Brightness Control Value (55h)

55h	WRCABC (Write Content Adaptive Brightness Control)																														
	D/CX	RDX	WRX	D[15:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX																		
Command	0	1	↑	XXXXXXXX	0	1	0	1	0	1	0	1	55h																		
1 st Parameter	1	1	↑	XXXXXXXX	X	X	X	X	X	X	C[1:0]		XX																		
Description	This command is used to set parameters for image content based adaptive brightness control functionality.																														
	There is possible to use 4 different modes for content adaptive image functionality, which are defined on a table below.																														
	<table><tr><td colspan="2">C[1:0]</td><td>Description</td></tr><tr><td>0</td><td>0</td><td>CABC OFF</td></tr><tr><td>0</td><td>1</td><td>User Interface Image</td></tr><tr><td>1</td><td>0</td><td>Still Picture</td></tr><tr><td>1</td><td>1</td><td>Moving Image</td></tr></table>													C[1:0]		Description	0	0	CABC OFF	0	1	User Interface Image	1	0	Still Picture	1	1	Moving Image			
	C[1:0]		Description																												
	0	0	CABC OFF																												
0	1	User Interface Image																													
1	0	Still Picture																													
1	1	Moving Image																													
X = Don't care																															
Restriction																															
Register Availability	<table><tr><td colspan="2">Status</td><td>Availability</td></tr><tr><td colspan="2">Normal Mode ON, Idle Mode OFF, Sleep OUT</td><td>Yes</td></tr><tr><td colspan="2">Normal Mode ON, Idle Mode ON, Sleep OUT</td><td>Yes</td></tr><tr><td colspan="2">Partial Mode ON, Idle Mode OFF, Sleep OUT</td><td>Yes</td></tr><tr><td colspan="2">Partial Mode ON, Idle Mode ON, Sleep OUT</td><td>Yes</td></tr><tr><td colspan="2">Sleep IN</td><td>Yes</td></tr></table>													Status		Availability	Normal Mode ON, Idle Mode OFF, Sleep OUT		Yes	Normal Mode ON, Idle Mode ON, Sleep OUT		Yes	Partial Mode ON, Idle Mode OFF, Sleep OUT		Yes	Partial Mode ON, Idle Mode ON, Sleep OUT		Yes	Sleep IN		Yes
	Status		Availability																												
	Normal Mode ON, Idle Mode OFF, Sleep OUT		Yes																												
	Normal Mode ON, Idle Mode ON, Sleep OUT		Yes																												
	Partial Mode ON, Idle Mode OFF, Sleep OUT		Yes																												
Partial Mode ON, Idle Mode ON, Sleep OUT		Yes																													
Sleep IN		Yes																													
Default	<table><tr><td>Status</td><td>Default Value</td></tr><tr><td>Power ON Sequence</td><td>00h</td></tr><tr><td>H/W Reset</td><td>00h</td></tr></table>													Status	Default Value	Power ON Sequence	00h	H/W Reset	00h												
	Status	Default Value																													
	Power ON Sequence	00h																													
H/W Reset	00h																														
Flow Chart	WRCABC ↓ 1st parameter: C[1:0] ↓ New Adaptive Image Mode Legend Command Parameter Display Action Mode Sequential transfer																														

8.2.42. Read Content Adaptive Brightness Control Value (56h)

56h	RDCABC (Read Content Adaptive Brightness Control)																								
	D/CX	RDX	WRX	D[15:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX												
Command	0	1	↑	XXXXXXXX	0	1	0	1	0	1	1	0	56h												
1 st Parameter	1	↑	1	XXXXXXXX	X	X	X	X	X	X	X	X	XX												
2 nd Parameter	1	↑	1	XXXXXXXX	X	X	X	X	X	X	C[1:0]		XX												
Description	This command is used to read the settings for image content based adaptive brightness control functionality. There is possible to use 4 different modes for content adaptive image functionality which are defined on the table below. <table><tr><th>C[1:0]</th><th>Description</th></tr><tr><td>0 0</td><td>CABC OFF</td></tr><tr><td>0 1</td><td>User Interface Image</td></tr><tr><td>1 0</td><td>Still Picture</td></tr><tr><td>1 1</td><td>Moving Image</td></tr></table> X = Don't care													C[1:0]	Description	0 0	CABC OFF	0 1	User Interface Image	1 0	Still Picture	1 1	Moving Image		
C[1:0]	Description																								
0 0	CABC OFF																								
0 1	User Interface Image																								
1 0	Still Picture																								
1 1	Moving Image																								
Restriction																									
Register Availability	<table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode ON, Idle Mode OFF, Sleep OUT</td><td>Yes</td></tr><tr><td>Normal Mode ON, Idle Mode ON, Sleep OUT</td><td>Yes</td></tr><tr><td>Partial Mode ON, Idle Mode OFF, Sleep OUT</td><td>Yes</td></tr><tr><td>Partial Mode ON, Idle Mode ON, Sleep OUT</td><td>Yes</td></tr><tr><td>Sleep IN</td><td>Yes</td></tr></table>													Status	Availability	Normal Mode ON, Idle Mode OFF, Sleep OUT	Yes	Normal Mode ON, Idle Mode ON, Sleep OUT	Yes	Partial Mode ON, Idle Mode OFF, Sleep OUT	Yes	Partial Mode ON, Idle Mode ON, Sleep OUT	Yes	Sleep IN	Yes
Status	Availability																								
Normal Mode ON, Idle Mode OFF, Sleep OUT	Yes																								
Normal Mode ON, Idle Mode ON, Sleep OUT	Yes																								
Partial Mode ON, Idle Mode OFF, Sleep OUT	Yes																								
Partial Mode ON, Idle Mode ON, Sleep OUT	Yes																								
Sleep IN	Yes																								
Default	<table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power ON Sequence</td><td>00h</td></tr><tr><td>H/W Reset</td><td>00h</td></tr></table>													Status	Default Value	Power ON Sequence	00h	H/W Reset	00h						
Status	Default Value																								
Power ON Sequence	00h																								
H/W Reset	00h																								
Flow Chart	Serial I/F Mode (P/SX = Low) Read RDCABC Send 2nd Parameter Parallel I/F Mode (P/SX = High) Read RDCABC Dummy Read Send 2nd Parameter Host Display Legend Command Parameter Display Action Mode Sequential transfer																								

8.2.43. Write CABC Minimum Brightness (5Eh)

5Eh	WRCABCMB (Write CABC Minimum Brightness)																								
	D/CX	RDX	WRX	D[15:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX												
Command	0	1	↑	XXXXXXXX	0	1	0	1	1	1	1	0	5Eh												
1 st Parameter	1	1	↑	XXXXXXXX	CMB[7:0]								XX												
Description	This command is used to set the minimum brightness value of the display for CABC function. CMB[7:0]: CABC minimum brightness control, this parameter is used to avoid too much brightness reduction. When CABC is active, CABC can not reduce the display brightness to less than CABC minimum brightness setting. Image processing function is worked as normal, even if the brightness can not be changed. This function does not affect to the other function, manual brightness setting. Manual brightness can be set the display brightness to less than CABC minimum brightness. Smooth transition and dimming function can be worked as normal. When display brightness is turned off (BCTRL=0 of “Write CTRL Display (53h)”), CABC minimum brightness setting is ignored. In principle relationship is that 00h value means the lowest brightness for CABC and FFh value means the highest brightness for CABC.																								
Restriction																									
Register Availability	<table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode ON, Idle Mode OFF, Sleep OUT</td><td>Yes</td></tr><tr><td>Normal Mode ON, Idle Mode ON, Sleep OUT</td><td>Yes</td></tr><tr><td>Partial Mode ON, Idle Mode OFF, Sleep OUT</td><td>Yes</td></tr><tr><td>Partial Mode ON, Idle Mode ON, Sleep OUT</td><td>Yes</td></tr><tr><td>Sleep IN</td><td>Yes</td></tr></table>													Status	Availability	Normal Mode ON, Idle Mode OFF, Sleep OUT	Yes	Normal Mode ON, Idle Mode ON, Sleep OUT	Yes	Partial Mode ON, Idle Mode OFF, Sleep OUT	Yes	Partial Mode ON, Idle Mode ON, Sleep OUT	Yes	Sleep IN	Yes
Status	Availability																								
Normal Mode ON, Idle Mode OFF, Sleep OUT	Yes																								
Normal Mode ON, Idle Mode ON, Sleep OUT	Yes																								
Partial Mode ON, Idle Mode OFF, Sleep OUT	Yes																								
Partial Mode ON, Idle Mode ON, Sleep OUT	Yes																								
Sleep IN	Yes																								
Default	<table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power ON Sequence</td><td>00h</td></tr><tr><td>H/W Reset</td><td>00h</td></tr></table>													Status	Default Value	Power ON Sequence	00h	H/W Reset	00h						
Status	Default Value																								
Power ON Sequence	00h																								
H/W Reset	00h																								
Flow Chart	WRCABCMB CMB[7..0] New Display Luminance Value Loaded Legend Command Parameter Display Action Mode Sequential transfer																								

8.2.44. Read CABC Minimum Brightness (5Fh)

5Fh	RDCABCMB (Read CABC Minimum Brightness)																								
	D/CX	RDX	WRX	D[15:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX												
Command	0	1	↑	XXXXXXXX	0	1	0	1	1	1	1	1	5Fh												
1 st Parameter	1	↑	1	XXXXXXXX	X	X	X	X	X	X	X	X	XX												
2 nd Parameter	1	↑	1	XXXXXXXX	CMB[7:0]								XX												
Description	This command returns the minimum brightness value of CABC function. In principle the relationship is that 00h value means the lowest brightness and FFh value means the highest brightness. CMB[7:0] is CABC minimum brightness specified with “Write CABC minimum brightness (5Eh)” command.																								
Restriction	ILI9486 is sending 2nd parameter value on the data lines if the MCU wants to read more than one parameter (= more than 2 RDX cycle) on parallel MCU interface. Only 2nd parameter is sent on DSI (The 1st parameter is not sent).																								
Register Availability	<table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode ON, Idle Mode OFF, Sleep OUT</td><td>Yes</td></tr><tr><td>Normal Mode ON, Idle Mode ON, Sleep OUT</td><td>Yes</td></tr><tr><td>Partial Mode ON, Idle Mode OFF, Sleep OUT</td><td>Yes</td></tr><tr><td>Partial Mode ON, Idle Mode ON, Sleep OUT</td><td>Yes</td></tr><tr><td>Sleep IN</td><td>Yes</td></tr></table>													Status	Availability	Normal Mode ON, Idle Mode OFF, Sleep OUT	Yes	Normal Mode ON, Idle Mode ON, Sleep OUT	Yes	Partial Mode ON, Idle Mode OFF, Sleep OUT	Yes	Partial Mode ON, Idle Mode ON, Sleep OUT	Yes	Sleep IN	Yes
Status	Availability																								
Normal Mode ON, Idle Mode OFF, Sleep OUT	Yes																								
Normal Mode ON, Idle Mode ON, Sleep OUT	Yes																								
Partial Mode ON, Idle Mode OFF, Sleep OUT	Yes																								
Partial Mode ON, Idle Mode ON, Sleep OUT	Yes																								
Sleep IN	Yes																								
Default	<table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power ON Sequence</td><td>00h</td></tr><tr><td>H/W Reset</td><td>00h</td></tr></table>													Status	Default Value	Power ON Sequence	00h	H/W Reset	00h						
Status	Default Value																								
Power ON Sequence	00h																								
H/W Reset	00h																								
Flow Chart	Serial I/F Mode (P/SX = Low) Read RDCABCMB Send 2nd Parameter Parallel I/F Mode (P/SX = High) Read RDCABCMB Dummy Read Send 2nd Parameter Host Display Legend Command Parameter Display Action Mode Sequential transfer																								

8.2.45. Read First Checksum (AAh)

AAh	RDFCS (Read First Checksum)																								
	D/CX	RDX	WRX	D[15:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX												
Command	0	1	↑	XXXXXXXX	1	0	1	0	1	0	1	0	AAh												
1 st Parameter	1	↑	1	XXXXXXXX	X	X	X	X	X	X	X	X	XX												
2 nd Parameter	1	↑	1	XXXXXXXX	FCS[7:0]								XX												
Description	This command returns the first checksum what has been calculated from User's area registers and the frame memory after the write access to those registers and/or frame memory has been done. X = can be '0' or '1'																								
Restriction	It will be necessary to wait 150ms after there is the last write access on User area registers before there can read this checksum value. ILI9486 is sending 2nd parameter value on the data lines if the MCU wants to read more than one parameter (= more than 2 RDX cycle) on parallel MCU interface. Only 2nd parameter is sent on DSI (The 1st parameter is not sent).																								
Register Availability	<table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode ON, Idle Mode OFF, Sleep OUT</td><td>Yes</td></tr><tr><td>Normal Mode ON, Idle Mode ON, Sleep OUT</td><td>Yes</td></tr><tr><td>Partial Mode ON, Idle Mode OFF, Sleep OUT</td><td>Yes</td></tr><tr><td>Partial Mode ON, Idle Mode ON, Sleep OUT</td><td>Yes</td></tr><tr><td>Sleep IN</td><td>Yes</td></tr></table>													Status	Availability	Normal Mode ON, Idle Mode OFF, Sleep OUT	Yes	Normal Mode ON, Idle Mode ON, Sleep OUT	Yes	Partial Mode ON, Idle Mode OFF, Sleep OUT	Yes	Partial Mode ON, Idle Mode ON, Sleep OUT	Yes	Sleep IN	Yes
Status	Availability																								
Normal Mode ON, Idle Mode OFF, Sleep OUT	Yes																								
Normal Mode ON, Idle Mode ON, Sleep OUT	Yes																								
Partial Mode ON, Idle Mode OFF, Sleep OUT	Yes																								
Partial Mode ON, Idle Mode ON, Sleep OUT	Yes																								
Sleep IN	Yes																								
Default	<table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power ON Sequence</td><td>00h</td></tr><tr><td>H/W Reset</td><td>00h</td></tr></table>													Status	Default Value	Power ON Sequence	00h	H/W Reset	00h						
Status	Default Value																								
Power ON Sequence	00h																								
H/W Reset	00h																								
Flow Chart	RDFCS ↓ Send 1st Parameter ↓ Send FCS[7:0] Host Display Legend Command Parameter Display Action Mode Sequential transfer																								

8.2.46. Read Continue Checksum (AFh)

AFh	RDCFCS (Read Continue Checksum)																								
	D/CX	RDX	WRX	D[15:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX												
Command	0	1	↑	XXXXXXXX	1	0	1	0	1	1	1	1	AFh												
1 st Parameter	1	↑	1	XXXXXXXX	X	X	X	X	X	X	X	X	XX												
2 nd Parameter	1	↑	1	XXXXXXXX	CCS[7:0]								XX												
Description	This command returns the continue checksum what has been calculated continuously after the first checksum has calculated from User's area registers and the frame memory after the write access to those registers and/or frame memory has been done. X = can be '0' or '1'																								
Restriction	It will be necessary to wait 300ms after there is the last write access on User area registers before there can read this checksum value in the first time. ILI9486 is sending 2nd parameter value on the data lines if the MCU wants to read more than one parameter (= more than 2 RDX cycle) on parallel MCU interface. Only 2nd parameter is sent on DSI (The 1st parameter is not sent).																								
Register Availability	<table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode ON, Idle Mode OFF, Sleep OUT</td><td>Yes</td></tr><tr><td>Normal Mode ON, Idle Mode ON, Sleep OUT</td><td>Yes</td></tr><tr><td>Partial Mode ON, Idle Mode OFF, Sleep OUT</td><td>Yes</td></tr><tr><td>Partial Mode ON, Idle Mode ON, Sleep OUT</td><td>Yes</td></tr><tr><td>Sleep IN</td><td>Yes</td></tr></table>													Status	Availability	Normal Mode ON, Idle Mode OFF, Sleep OUT	Yes	Normal Mode ON, Idle Mode ON, Sleep OUT	Yes	Partial Mode ON, Idle Mode OFF, Sleep OUT	Yes	Partial Mode ON, Idle Mode ON, Sleep OUT	Yes	Sleep IN	Yes
Status	Availability																								
Normal Mode ON, Idle Mode OFF, Sleep OUT	Yes																								
Normal Mode ON, Idle Mode ON, Sleep OUT	Yes																								
Partial Mode ON, Idle Mode OFF, Sleep OUT	Yes																								
Partial Mode ON, Idle Mode ON, Sleep OUT	Yes																								
Sleep IN	Yes																								
Default	<table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power ON Sequence</td><td>00h</td></tr><tr><td>H/W Reset</td><td>00h</td></tr></table>													Status	Default Value	Power ON Sequence	00h	H/W Reset	00h						
Status	Default Value																								
Power ON Sequence	00h																								
H/W Reset	00h																								
Flow Chart	RDCCS ↓ Send 1st Parameter ↓ Send CCS[7:0] Host Display Legend Command Parameter Display Action Mode Sequential transfer																								

8.2.47. Read ID1 (DAh)

DAh	RDID1 (Read ID1)																								
	D/CX	RDX	WRX	D[15:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX												
Command	0	1	↑	XXXXXXXX	1	1	0	1	1	0	1	0	DAh												
1 st parameter	1	↑	1	XXXXXXXX	X	X	X	X	X	X	X	X	XX												
2 nd parameter	1	↑	1	XXXXXXXX	ID1[7:0]								XX												
Description	This read byte identifies the LCD module's manufacturer ID and it is specified by User The 1st parameter is dummy data. The 2nd parameter is LCD module's manufacturer ID. X = Don't care																								
Restriction	ILI9486 is sending 2nd parameter value on the data lines if the MCU wants to read more than one parameter (= more than 2 RDX cycle) on parallel MCU interface. Only 2nd parameter is sent on DSI (The 1st parameter is not sent).																								
Register Availability	<table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table>													Status	Availability	Normal Mode On, Idle Mode Off, Sleep Out	Yes	Normal Mode On, Idle Mode On, Sleep Out	Yes	Partial Mode On, Idle Mode Off, Sleep Out	Yes	Partial Mode On, Idle Mode On, Sleep Out	Yes	Sleep In	Yes
Status	Availability																								
Normal Mode On, Idle Mode Off, Sleep Out	Yes																								
Normal Mode On, Idle Mode On, Sleep Out	Yes																								
Partial Mode On, Idle Mode Off, Sleep Out	Yes																								
Partial Mode On, Idle Mode On, Sleep Out	Yes																								
Sleep In	Yes																								
Default	<table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>XXh</td></tr><tr><td>HW Reset</td><td>XXh</td></tr></table>													Status	Default Value	Power On Sequence	XXh	HW Reset	XXh						
Status	Default Value																								
Power On Sequence	XXh																								
HW Reset	XXh																								
Flow Chart	RDID1(DAh) Host Driver 1st Parameter: Dummy Read 2nd Parameter: Send ID1[7:0] Legend Command Parameter Display Action Mode Sequential transfer																								

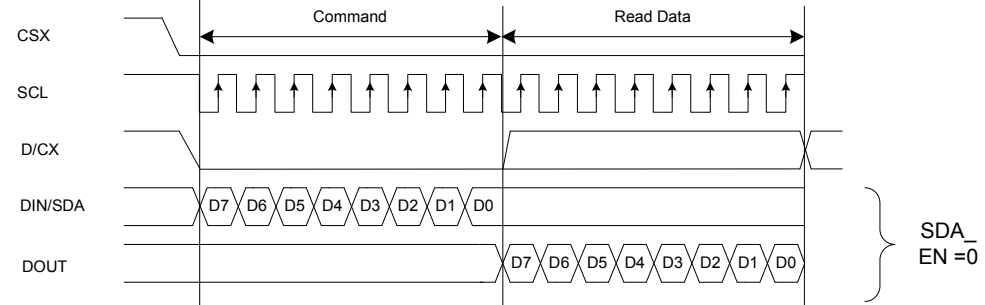
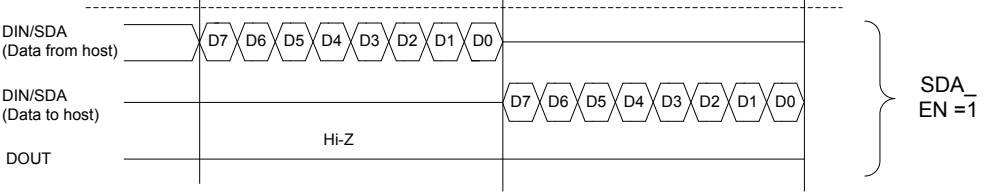
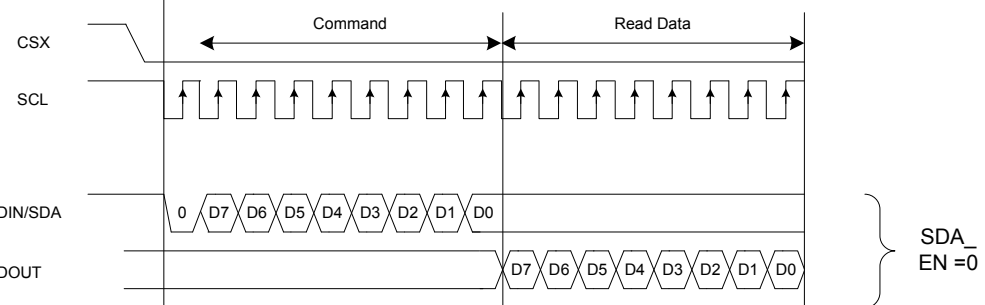
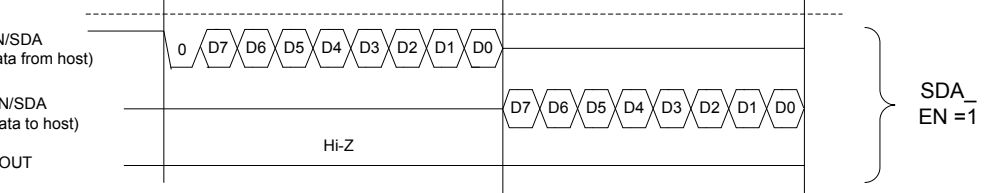
8.2.48. Read ID2 (DBh)

DBh	RDID2 (Read ID2)																								
	D/CX	RDX	WRX	D[15:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX												
Command	0	1	↑	XXXXXXXX	1	1	0	1	1	0	1	1	DBh												
1 st parameter	1	↑	1	XXXXXXXX	X	X	X	X	X	X	X	X	XX												
2 nd arameter	1	↑	1	XXXXXXXX	1	ID2[6:0]							XX												
Description	This read byte is used to track the LCD module/driver version. It is defined by display supplier (with User’s agreement) and changes each time a revision is made to the display, material or construction specifications. The 1st parameter is dummy data. The 2nd parameter is LCD module/driver version ID and the ID parameter range is from 80h to FFh. The ID2 can be programmed by OTP function. X = Don’t care																								
Restriction	ILI9486 is sending 2nd parameter value on the data lines if the MCU wants to read more than one parameter (= more than 2 RDX cycle) on parallel MCU interface. Only 2nd parameter is sent on DSI (The 1st parameter is not sent).																								
Register Availability	<table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table>													Status	Availability	Normal Mode On, Idle Mode Off, Sleep Out	Yes	Normal Mode On, Idle Mode On, Sleep Out	Yes	Partial Mode On, Idle Mode Off, Sleep Out	Yes	Partial Mode On, Idle Mode On, Sleep Out	Yes	Sleep In	Yes
Status	Availability																								
Normal Mode On, Idle Mode Off, Sleep Out	Yes																								
Normal Mode On, Idle Mode On, Sleep Out	Yes																								
Partial Mode On, Idle Mode Off, Sleep Out	Yes																								
Partial Mode On, Idle Mode On, Sleep Out	Yes																								
Sleep In	Yes																								
Default	<table><tr><th>Status</th><th>Default Value (Before OTP program)</th><th>Default Value (After OTP program)</th></tr><tr><td>Power On Sequence</td><td>80h</td><td>OTP value</td></tr><tr><td>SW Reset</td><td>80h</td><td>OTP value</td></tr><tr><td>HW Reset</td><td>80h</td><td>OTP value</td></tr></table>													Status	Default Value (Before OTP program)	Default Value (After OTP program)	Power On Sequence	80h	OTP value	SW Reset	80h	OTP value	HW Reset	80h	OTP value
Status	Default Value (Before OTP program)	Default Value (After OTP program)																							
Power On Sequence	80h	OTP value																							
SW Reset	80h	OTP value																							
HW Reset	80h	OTP value																							
Flow Chart	RDID2(DBh) ↓ 1st Parameter: Dummy Read ↓ 2nd Parameter: Send ID2[7:0] Host Driver Legend Command Parameter Display Action Mode Sequential transfer																								

8.2.49. Read ID3 (DCh)

DCh	RDID3 (Read ID3)																								
	D/CX	RDX	WRX	D[15:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX												
Command	0	1	↑	XXXXXXXX	1	1	0	1	1	1	0	0	DCh												
1 st parameter	1	↑	1	XXXXXXXX	X	X	X	X	X	X	X	X	XX												
2 nd parameter	1	↑	1	XXXXXXXX	ID3[7:0]								XX												
Description	This read byte identifies the LCD module/driver and It is specified by User. The 1st parameter is dummy data. The 2nd parameter is LCD module/driver ID. The ID3 can be programmed by OTP function. X = Don't care																								
Restriction	ILI9486 is sending 2nd parameter value on the data lines if the MCU wants to read more than one parameter (= more than 2 RDX cycle) on parallel MCU interface. Only 2nd parameter is sent on DSI (The 1st parameter is not sent).																								
Register Availability	<table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table>													Status	Availability	Normal Mode On, Idle Mode Off, Sleep Out	Yes	Normal Mode On, Idle Mode On, Sleep Out	Yes	Partial Mode On, Idle Mode Off, Sleep Out	Yes	Partial Mode On, Idle Mode On, Sleep Out	Yes	Sleep In	Yes
Status	Availability																								
Normal Mode On, Idle Mode Off, Sleep Out	Yes																								
Normal Mode On, Idle Mode On, Sleep Out	Yes																								
Partial Mode On, Idle Mode Off, Sleep Out	Yes																								
Partial Mode On, Idle Mode On, Sleep Out	Yes																								
Sleep In	Yes																								
Default	<table><tr><th>Status</th><th>Default Value (Before OTP program)</th><th>Default Value (After OTP program)</th></tr><tr><td>Power On Sequence</td><td>00h</td><td>OTP value</td></tr><tr><td>SW Reset</td><td>00h</td><td>OTP value</td></tr><tr><td>HW Reset</td><td>00h</td><td>OTP value</td></tr></table>													Status	Default Value (Before OTP program)	Default Value (After OTP program)	Power On Sequence	00h	OTP value	SW Reset	00h	OTP value	HW Reset	00h	OTP value
Status	Default Value (Before OTP program)	Default Value (After OTP program)																							
Power On Sequence	00h	OTP value																							
SW Reset	00h	OTP value																							
HW Reset	00h	OTP value																							
Flow Chart	RDID3(DCh) ↓ 1st Parameter: Dummy Read ↓ 2nd Parameter: Send ID3[7:0] Host Driver Legend Command Parameter Display Action Mode Sequential transfer																								

8.2.50. Interface Mode Control (B0h)

B0h	IFMODE (Interface Mode Control)												
	D/CX	RDX	WRX	D[15:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XXXXXXXX	1	0	1	1	0	0	0	0	B0h
1 st Parameter	1	1	↑	XXXXXXXX	SDA_EN	0	0	0	VSPL	HSPL	DPL	EPL	XX
Description	Sets the operation status of the display interface. The setting becomes effective as soon as the command is received.												
	EPL : DE polarity ("0"= High enable for RGB interface, "1"=Low enable for RGB interface)												
	DPL : PCLK polarity set ("0"=data fetched at the rising time, "1"=data fetched at the falling time)												
	HSPL : HSYNC polarity ("0"=Low level sync clock, "1"=High level sync clock)												
	VSPL : VSYNC polarity ("0"= Low level sync clock, "1"= High level sync clock)												
	SDA_EN : 3/4 wire serial interface selection												
	SDA_EN = "0", DIN and DOUT pins are used for 3/4 wire serial interface.												
	SDA_EN = "1", DIN/SDA pin is used for 3/4 wire serial interface and DOUT pin is not used.												
													
													
													
													
Restriction													

Register Availability	<table><tr><td colspan="4">Status</td><td colspan="2">Availability</td></tr><tr><td colspan="4">Normal Mode ON, Idle Mode OFF, Sleep OUT</td><td colspan="2">Yes</td></tr><tr><td colspan="4">Normal Mode ON, Idle Mode ON, Sleep OUT</td><td colspan="2">Yes</td></tr><tr><td colspan="4">Partial Mode ON, Idle Mode OFF, Sleep OUT</td><td colspan="2">Yes</td></tr><tr><td colspan="4">Partial Mode ON, Idle Mode ON, Sleep OUT</td><td colspan="2">Yes</td></tr><tr><td colspan="4">Sleep IN</td><td colspan="2">Yes</td></tr></table>						Status				Availability		Normal Mode ON, Idle Mode OFF, Sleep OUT				Yes		Normal Mode ON, Idle Mode ON, Sleep OUT				Yes		Partial Mode ON, Idle Mode OFF, Sleep OUT				Yes		Partial Mode ON, Idle Mode ON, Sleep OUT				Yes		Sleep IN				Yes	
	Status				Availability																																					
	Normal Mode ON, Idle Mode OFF, Sleep OUT				Yes																																					
	Normal Mode ON, Idle Mode ON, Sleep OUT				Yes																																					
	Partial Mode ON, Idle Mode OFF, Sleep OUT				Yes																																					
	Partial Mode ON, Idle Mode ON, Sleep OUT				Yes																																					
Sleep IN				Yes																																						
Default	<table><tr><td rowspan="2">Status</td><td colspan="5">Default Value</td></tr><tr><td>SDA_EN</td><td>EPL</td><td>DPL</td><td>HSPL</td><td>VSPL</td></tr><tr><td>Power ON Sequence</td><td>0b</td><td>0b</td><td>0b</td><td>0b</td><td>0b</td></tr><tr><td>H/W Reset</td><td>0b</td><td>0b</td><td>0b</td><td>0b</td><td>0b</td></tr></table>						Status	Default Value					SDA_EN	EPL	DPL	HSPL	VSPL	Power ON Sequence	0b	0b	0b	0b	0b	H/W Reset	0b	0b	0b	0b	0b													
	Status	Default Value																																								
		SDA_EN	EPL	DPL	HSPL	VSPL																																				
	Power ON Sequence	0b	0b	0b	0b	0b																																				
H/W Reset	0b	0b	0b	0b	0b																																					

8.2.51. Frame Rate Control (In Normal Mode/Full Colors) (B1h)

B1h		FRMCTR1 (Frame Rate Control (In Normal Mode / Full colors))																																																																																																																																																																																																																					
	D/CX	RDX	WRX	D[15:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX																																																																																																																																																																																																										
Command	0	1	↑	XXXXXXXX	1	0	1	1	0	0	0	1	B1h																																																																																																																																																																																																										
1 st Parameter	1	1	↑	XXXXXXXX	FRS[3:0]				0	0	DIVA[1:0]		XX																																																																																																																																																																																																										
2 nd parameter	1	1	↑	XXXXXXXX	0	0	0	RTNA[4:0]					XX																																																																																																																																																																																																										
Description	FRS[3:0]: Sets the frame frequency of full color normal mode.																																																																																																																																																																																																																						
	<table><tr><th colspan="4">FRS[3:0]</th><th>Frame rate(Hz)</th></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td><td>28</td></tr><tr><td>0</td><td>0</td><td>0</td><td>1</td><td>30</td></tr><tr><td>0</td><td>0</td><td>1</td><td>0</td><td>32</td></tr><tr><td>0</td><td>0</td><td>1</td><td>1</td><td>34</td></tr><tr><td>0</td><td>1</td><td>0</td><td>0</td><td>36</td></tr><tr><td>0</td><td>1</td><td>0</td><td>1</td><td>39</td></tr><tr><td>0</td><td>1</td><td>1</td><td>0</td><td>42</td></tr><tr><td>0</td><td>1</td><td>1</td><td>1</td><td>46</td></tr><tr><td>1</td><td>0</td><td>0</td><td>0</td><td>50</td></tr><tr><td>1</td><td>0</td><td>0</td><td>1</td><td>56</td></tr><tr><td>1</td><td>0</td><td>1</td><td>0</td><td>62</td></tr><tr><td>1</td><td>0</td><td>1</td><td>1</td><td>70</td></tr><tr><td>1</td><td>1</td><td>0</td><td>0</td><td>81</td></tr><tr><td>1</td><td>1</td><td>0</td><td>1</td><td>96</td></tr><tr><td>1</td><td>1</td><td>1</td><td>0</td><td>117</td></tr><tr><td>1</td><td>1</td><td>1</td><td>1</td><td>117</td></tr></table>													FRS[3:0]				Frame rate(Hz)	0	0	0	0	28	0	0	0	1	30	0	0	1	0	32	0	0	1	1	34	0	1	0	0	36	0	1	0	1	39	0	1	1	0	42	0	1	1	1	46	1	0	0	0	50	1	0	0	1	56	1	0	1	0	62	1	0	1	1	70	1	1	0	0	81	1	1	0	1	96	1	1	1	0	117	1	1	1	1	117																																																																																																																					
	FRS[3:0]				Frame rate(Hz)																																																																																																																																																																																																																		
	0	0	0	0	28																																																																																																																																																																																																																		
	0	0	0	1	30																																																																																																																																																																																																																		
	0	0	1	0	32																																																																																																																																																																																																																		
	0	0	1	1	34																																																																																																																																																																																																																		
	0	1	0	0	36																																																																																																																																																																																																																		
	0	1	0	1	39																																																																																																																																																																																																																		
	0	1	1	0	42																																																																																																																																																																																																																		
	0	1	1	1	46																																																																																																																																																																																																																		
	1	0	0	0	50																																																																																																																																																																																																																		
	1	0	0	1	56																																																																																																																																																																																																																		
	1	0	1	0	62																																																																																																																																																																																																																		
	1	0	1	1	70																																																																																																																																																																																																																		
	1	1	0	0	81																																																																																																																																																																																																																		
	1	1	0	1	96																																																																																																																																																																																																																		
	1	1	1	0	117																																																																																																																																																																																																																		
	1	1	1	1	117																																																																																																																																																																																																																		
	DIVA [1:0] : division ratio for internal clocks when Normal mode.																																																																																																																																																																																																																						
	<table><tr><th colspan="2">DIVA[1:0]</th><th>Division Ratio</th></tr><tr><td>0</td><td>0</td><td>fosc</td></tr><tr><td>0</td><td>1</td><td>fosc / 2</td></tr><tr><td>1</td><td>0</td><td>fosc / 4</td></tr><tr><td>1</td><td>1</td><td>fosc / 8</td></tr></table>													DIVA[1:0]		Division Ratio	0	0	fosc	0	1	fosc / 2	1	0	fosc / 4	1	1	fosc / 8																																																																																																																																																																																											
	DIVA[1:0]		Division Ratio																																																																																																																																																																																																																				
	0	0	fosc																																																																																																																																																																																																																				
	0	1	fosc / 2																																																																																																																																																																																																																				
	1	0	fosc / 4																																																																																																																																																																																																																				
1	1	fosc / 8																																																																																																																																																																																																																					
RTNA [4:0] : RTNA[4:0] is used to set 1H (line) period of Normal mode at CPU interface.																																																																																																																																																																																																																							
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Register Availability	<table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode ON, Idle Mode OFF, Sleep OUT</td><td>Yes</td></tr><tr><td>Normal Mode ON, Idle Mode ON, Sleep OUT</td><td>Yes</td></tr><tr><td>Partial Mode ON, Idle Mode OFF, Sleep OUT</td><td>Yes</td></tr><tr><td>Partial Mode ON, Idle Mode ON, Sleep OUT</td><td>Yes</td></tr><tr><td>Sleep IN</td><td>Yes</td></tr></table>	Status	Availability	Normal Mode ON, Idle Mode OFF, Sleep OUT	Yes	Normal Mode ON, Idle Mode ON, Sleep OUT	Yes	Partial Mode ON, Idle Mode OFF, Sleep OUT	Yes	Partial Mode ON, Idle Mode ON, Sleep OUT	Yes	Sleep IN	Yes			
Status	Availability															
Normal Mode ON, Idle Mode OFF, Sleep OUT	Yes															
Normal Mode ON, Idle Mode ON, Sleep OUT	Yes															
Partial Mode ON, Idle Mode OFF, Sleep OUT	Yes															
Partial Mode ON, Idle Mode ON, Sleep OUT	Yes															
Sleep IN	Yes															
Default	<table><tr><th rowspan="2">Status</th><th colspan="3">Default Value</th></tr><tr><th>FRS [3:0]</th><th>DIVA[1:0]</th><th>RTNA[4:0]</th></tr><tr><td>Power ON Sequence</td><td>4'b1011</td><td>2'b00</td><td>5'b10001</td></tr><tr><td>H/W Reset</td><td>4'b1011</td><td>2'b00</td><td>5'b10001</td></tr></table>	Status	Default Value			FRS [3:0]	DIVA[1:0]	RTNA[4:0]	Power ON Sequence	4'b1011	2'b00	5'b10001	H/W Reset	4'b1011	2'b00	5'b10001
Status	Default Value															
	FRS [3:0]	DIVA[1:0]	RTNA[4:0]													
Power ON Sequence	4'b1011	2'b00	5'b10001													
H/W Reset	4'b1011	2'b00	5'b10001													

8.2.52. Frame Rate Control (In Idle Mode/8 colors) (B2h)

_B2h				FRMCTR2 (Frame Rate Control (In Idle Mode / 8 colors))																																																																																	
	D/CX	RDX	WRX	D[15:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX																																																																								
Command	0	1	↑	XXXXXXXX	1	0	1	1	0	0	1	0	B2h																																																																								
1 st Parameter	1	1	↑	XXXXXXXX	0	0	0	0	0	0	DIVB[1:0]		XX																																																																								
2 nd parameter	1	1	↑	XXXXXXXX	0	0	0	RTNB[4:0]					XX																																																																								
Description	Sets the division ratio for internal clocks of Idle mode at CPU interface.																																																																																				
	DIVB [1:0] : division ratio for internal clocks when Idle mode.																																																																																				
	<table><tr><th colspan="2">DIVB[1:0]</th><th>Division Ratio</th></tr><tr><td>0</td><td>0</td><td>fosc</td></tr><tr><td>0</td><td>1</td><td>fosc / 2</td></tr><tr><td>1</td><td>0</td><td>fosc / 4</td></tr><tr><td>1</td><td>1</td><td>fosc / 8</td></tr></table>													DIVB[1:0]		Division Ratio	0	0	fosc	0	1	fosc / 2	1	0	fosc / 4	1	1	fosc / 8																																																									
	DIVB[1:0]		Division Ratio																																																																																		
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	0	1	fosc / 2																																																																																		
	1	0	fosc / 4																																																																																		
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	RTNB [4:0] : RTNB[4:0] is used to set 1H (line) period of Idle mode at CPU interface.																																																																																				
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<table><tr><th colspan="5">RTNB[4:0]</th><th>Clock per Line</th></tr><tr><td>0</td><td>1</td><td>0</td><td>1</td><td>1</td><td>Setting prohibited</td></tr><tr><td>0</td><td>1</td><td>1</td><td>0</td><td>0</td><td>Setting prohibited</td></tr><tr><td>0</td><td>1</td><td>1</td><td>0</td><td>1</td><td>Setting prohibited</td></tr><tr><td>0</td><td>1</td><td>1</td><td>1</td><td>0</td><td>Setting prohibited</td></tr><tr><td>0</td><td>1</td><td>1</td><td>1</td><td>1</td><td>Setting prohibited</td></tr><tr><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>16 clocks</td></tr><tr><td>1</td><td>0</td><td>0</td><td>0</td><td>1</td><td>17 clocks</td></tr><tr><td>1</td><td>0</td><td>0</td><td>1</td><td>0</td><td>18 clocks</td></tr><tr><td>1</td><td>0</td><td>0</td><td>1</td><td>1</td><td>19 clocks</td></tr><tr><td>1</td><td>0</td><td>1</td><td>0</td><td>0</td><td>20 clocks</td></tr><tr><td>1</td><td>0</td><td>1</td><td>0</td><td>1</td><td>21 clocks</td></tr></table>													RTNB[4:0]					Clock per Line	0	1	0	1	1	Setting prohibited	0	1	1	0	0	Setting prohibited	0	1	1	0	1	Setting prohibited	0	1	1	1	0	Setting prohibited	0	1	1	1	1	Setting prohibited	1	0	0	0	0	16 clocks	1	0	0	0	1	17 clocks	1	0	0	1	0	18 clocks	1	0	0	1	1	19 clocks	1	0	1	0	0	20 clocks	1	0	1	0	1	21 clocks	
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Status	Availability																																																																																				
Normal Mode ON, Idle Mode OFF, Sleep OUT	Yes																																																																																				
Normal Mode ON, Idle Mode ON, Sleep OUT	Yes																																																																																				
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Default	<table><tr><th rowspan="2">Status</th><th colspan="2">Default Value</th></tr><tr><th>DIVB[1:0]</th><th>RTNB[4:0]</th></tr><tr><td>Power ON Sequence</td><td>2'b00</td><td>5'b10001</td></tr><tr><td>H/W Reset</td><td>2'b00</td><td>5'b10001</td></tr></table>													Status	Default Value		DIVB[1:0]	RTNB[4:0]	Power ON Sequence	2'b00	5'b10001	H/W Reset	2'b00	5'b10001																																																													
Status	Default Value																																																																																				
	DIVB[1:0]	RTNB[4:0]																																																																																			
Power ON Sequence	2'b00	5'b10001																																																																																			
H/W Reset	2'b00	5'b10001																																																																																			

8.2.53. Frame Rate control (In Partial Mode/Full Colors) (B3h)

_B3h		FRMCTR3 (Frame Rate Control (In Partial Mode / Full colors))																																																																																			
	D/CX	RDX	WRX	D[15:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX																																																																								
Command	0	1	↑	XXXXXXXX	1	0	1	1	0	0	1	1	B3h																																																																								
1 st Parameter	1	1	↑	XXXXXXXX	0	0	0	0	0	0	DIVC[1:0]		XX																																																																								
2 nd parameter	1	1	↑	XXXXXXXX	0	0	0	RTNC[4:0]					XX																																																																								
Description	Sets the division ratio for internal clocks of Partial mode (Idle mode off) at CPU interface.																																																																																				
	DIVC [1:0] : division ratio for internal clocks when Partial mode.																																																																																				
	<table><tr><th colspan="2">DIVC[1:0]</th><th>Division Ratio</th></tr><tr><td>0</td><td>0</td><td>fosc</td></tr><tr><td>0</td><td>1</td><td>fosc / 2</td></tr><tr><td>1</td><td>0</td><td>fosc / 4</td></tr><tr><td>1</td><td>1</td><td>fosc / 8</td></tr></table>													DIVC[1:0]		Division Ratio	0	0	fosc	0	1	fosc / 2	1	0	fosc / 4	1	1	fosc / 8																																																									
	DIVC[1:0]		Division Ratio																																																																																		
	0	0	fosc																																																																																		
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	1	0	fosc / 4																																																																																		
	1	1	fosc / 8																																																																																		
	RTNC [4:0] : RTNC[4:0] is used to set 1H (line) period of Partial mode at CPU interface.																																																																																				
	<table><tr><th colspan="5">RTNC[4:0]</th><th>Clock per Line</th></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>Setting prohibited</td></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>Setting prohibited</td></tr><tr><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td><td>Setting prohibited</td></tr><tr><td>0</td><td>0</td><td>0</td><td>1</td><td>1</td><td>Setting prohibited</td></tr><tr><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td><td>Setting prohibited</td></tr><tr><td>0</td><td>0</td><td>1</td><td>0</td><td>1</td><td>Setting prohibited</td></tr><tr><td>0</td><td>0</td><td>1</td><td>1</td><td>0</td><td>Setting prohibited</td></tr><tr><td>0</td><td>0</td><td>1</td><td>1</td><td>1</td><td>Setting prohibited</td></tr><tr><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td><td>Setting prohibited</td></tr><tr><td>0</td><td>1</td><td>0</td><td>0</td><td>1</td><td>Setting prohibited</td></tr><tr><td>0</td><td>1</td><td>0</td><td>1</td><td>0</td><td>Setting prohibited</td></tr></table>													RTNC[4:0]					Clock per Line	0	0	0	0	0	Setting prohibited	0	0	0	0	1	Setting prohibited	0	0	0	1	0	Setting prohibited	0	0	0	1	1	Setting prohibited	0	0	1	0	0	Setting prohibited	0	0	1	0	1	Setting prohibited	0	0	1	1	0	Setting prohibited	0	0	1	1	1	Setting prohibited	0	1	0	0	0	Setting prohibited	0	1	0	0	1	Setting prohibited	0	1	0	1	0	Setting prohibited
	RTNC[4:0]					Clock per Line																																																																															
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<table><tr><th colspan="5">RTNC[4:0]</th><th>Clock per Line</th></tr><tr><td>0</td><td>1</td><td>0</td><td>1</td><td>1</td><td>Setting prohibited</td></tr><tr><td>0</td><td>1</td><td>1</td><td>0</td><td>0</td><td>Setting prohibited</td></tr><tr><td>0</td><td>1</td><td>1</td><td>0</td><td>1</td><td>Setting prohibited</td></tr><tr><td>0</td><td>1</td><td>1</td><td>1</td><td>0</td><td>Setting prohibited</td></tr><tr><td>0</td><td>1</td><td>1</td><td>1</td><td>1</td><td>Setting prohibited</td></tr><tr><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>16 clocks</td></tr><tr><td>1</td><td>0</td><td>0</td><td>0</td><td>1</td><td>17 clocks</td></tr><tr><td>1</td><td>0</td><td>0</td><td>1</td><td>0</td><td>18 clocks</td></tr><tr><td>1</td><td>0</td><td>0</td><td>1</td><td>1</td><td>19 clocks</td></tr><tr><td>1</td><td>0</td><td>1</td><td>0</td><td>0</td><td>20 clocks</td></tr><tr><td>1</td><td>0</td><td>1</td><td>0</td><td>1</td><td>21 clocks</td></tr></table>													RTNC[4:0]					Clock per Line	0	1	0	1	1	Setting prohibited	0	1	1	0	0	Setting prohibited	0	1	1	0	1	Setting prohibited	0	1	1	1	0	Setting prohibited	0	1	1	1	1	Setting prohibited	1	0	0	0	0	16 clocks	1	0	0	0	1	17 clocks	1	0	0	1	0	18 clocks	1	0	0	1	1	19 clocks	1	0	1	0	0	20 clocks	1	0	1	0	1	21 clocks	
RTNC[4:0]					Clock per Line																																																																																
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1	0	1	0	1	21 clocks																																																																																
<table><tr><th colspan="5">RTNC[4:0]</th><th>Clock per Line</th></tr><tr><td>1</td><td>0</td><td>1</td><td>1</td><td>0</td><td>22 clocks</td></tr><tr><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td><td>23 clocks</td></tr><tr><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>24 clocks</td></tr><tr><td>1</td><td>0</td><td>0</td><td>0</td><td>1</td><td>25 clocks</td></tr><tr><td>1</td><td>0</td><td>0</td><td>1</td><td>0</td><td>26 clocks</td></tr><tr><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td><td>27 clocks</td></tr><tr><td>1</td><td>1</td><td>0</td><td>0</td><td>0</td><td>28 clocks</td></tr><tr><td>1</td><td>1</td><td>0</td><td>0</td><td>1</td><td>29 clocks</td></tr><tr><td>1</td><td>1</td><td>0</td><td>1</td><td>0</td><td>30 clocks</td></tr><tr><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>31 clocks</td></tr></table>													RTNC[4:0]					Clock per Line	1	0	1	1	0	22 clocks	1	0	1	1	1	23 clocks	1	0	0	0	0	24 clocks	1	0	0	0	1	25 clocks	1	0	0	1	0	26 clocks	1	0	1	1	1	27 clocks	1	1	0	0	0	28 clocks	1	1	0	0	1	29 clocks	1	1	0	1	0	30 clocks	1	1	1	1	1	31 clocks							
RTNC[4:0]					Clock per Line																																																																																
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Restriction																																																																																					
Register Availability	<table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode ON, Idle Mode OFF, Sleep OUT</td><td>Yes</td></tr><tr><td>Normal Mode ON, Idle Mode ON, Sleep OUT</td><td>Yes</td></tr><tr><td>Partial Mode ON, Idle Mode OFF, Sleep OUT</td><td>Yes</td></tr><tr><td>Partial Mode ON, Idle Mode ON, Sleep OUT</td><td>Yes</td></tr><tr><td>Sleep IN</td><td>Yes</td></tr></table>													Status	Availability	Normal Mode ON, Idle Mode OFF, Sleep OUT	Yes	Normal Mode ON, Idle Mode ON, Sleep OUT	Yes	Partial Mode ON, Idle Mode OFF, Sleep OUT	Yes	Partial Mode ON, Idle Mode ON, Sleep OUT	Yes	Sleep IN	Yes																																																												
Status	Availability																																																																																				
Normal Mode ON, Idle Mode OFF, Sleep OUT	Yes																																																																																				
Normal Mode ON, Idle Mode ON, Sleep OUT	Yes																																																																																				
Partial Mode ON, Idle Mode OFF, Sleep OUT	Yes																																																																																				
Partial Mode ON, Idle Mode ON, Sleep OUT	Yes																																																																																				
Sleep IN	Yes																																																																																				
Default	<table><tr><th rowspan="2">Status</th><th colspan="2">Default Value</th></tr><tr><th>DIVC[1:0]</th><th>RTNC[4:0]</th></tr><tr><td>Power ON Sequence</td><td>2'b00</td><td>5'b10001</td></tr><tr><td>H/W Reset</td><td>2'b00</td><td>5'b10001</td></tr></table>													Status	Default Value		DIVC[1:0]	RTNC[4:0]	Power ON Sequence	2'b00	5'b10001	H/W Reset	2'b00	5'b10001																																																													
Status	Default Value																																																																																				
	DIVC[1:0]	RTNC[4:0]																																																																																			
Power ON Sequence	2'b00	5'b10001																																																																																			
H/W Reset	2'b00	5'b10001																																																																																			

8.2.54. Display Inversion Control (B4h)

_B4h				INVTR (Display Inversion Control)																																																																																																																																																																																																																																									
	D/CX	RDX	WRX	D[15:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX																																																																																																																																																																																																																																
Command	0	1	↑	XXXXXXXX	1	0	1	1	0	1	0	0	B4h																																																																																																																																																																																																																																
1 st Parameter	1	1	↑	XXXXXXXX	0	0	0	ZINV	0	0	DINV[1:0]		XX																																																																																																																																																																																																																																
Description	ZINV : Set Z-inversion mode 0 : Disable Z-inversion 1 : Enable Z-inversion mode DINV[1:0] : Set the inversion mode																																																																																																																																																																																																																																												
	<table><tr><th>DINV [1:0]</th><th colspan="12">Dot inversion mode</th></tr><tr><td rowspan="5">2'b00</td><td rowspan="5">Column inversion</td><td colspan="6">1st frame</td><td colspan="6">2nd frame</td></tr><tr><td>1 line</td><td>+</td><td>-</td><td>+</td><td>-</td><td>+</td><td>-</td><td>1 line</td><td>-</td><td>+</td><td>-</td><td>+</td><td>-</td></tr><tr><td>2 line</td><td>+</td><td>-</td><td>+</td><td>-</td><td>+</td><td>-</td><td>2 line</td><td>-</td><td>+</td><td>-</td><td>+</td><td>-</td></tr><tr><td>3 line</td><td>+</td><td>-</td><td>+</td><td>-</td><td>+</td><td>-</td><td>3 line</td><td>-</td><td>+</td><td>-</td><td>+</td><td>-</td></tr><tr><td>4 line</td><td>+</td><td>-</td><td>+</td><td>-</td><td>+</td><td>-</td><td>4 line</td><td>-</td><td>+</td><td>-</td><td>+</td><td>-</td></tr><tr><td rowspan="5">2'b01</td><td rowspan="5">1-dot inversion</td><td colspan="6">1st frame</td><td colspan="6">2nd frame</td></tr><tr><td>1 line</td><td>+</td><td>-</td><td>+</td><td>-</td><td>+</td><td>-</td><td>1 line</td><td>-</td><td>+</td><td>-</td><td>+</td><td>-</td></tr><tr><td>2 line</td><td>-</td><td>+</td><td>-</td><td>+</td><td>-</td><td>+</td><td>2 line</td><td>+</td><td>-</td><td>+</td><td>-</td><td>+</td></tr><tr><td>3 line</td><td>+</td><td>-</td><td>+</td><td>-</td><td>+</td><td>-</td><td>3 line</td><td>-</td><td>+</td><td>-</td><td>+</td><td>-</td></tr><tr><td>4 line</td><td>-</td><td>+</td><td>-</td><td>+</td><td>-</td><td>+</td><td>4 line</td><td>+</td><td>-</td><td>+</td><td>-</td><td>+</td></tr><tr><td rowspan="5">2'b10</td><td rowspan="5">2-dot inversion</td><td colspan="6">1st frame</td><td colspan="6">2nd frame</td></tr><tr><td>1 line</td><td>+</td><td>-</td><td>+</td><td>-</td><td>+</td><td>-</td><td>1 line</td><td>-</td><td>+</td><td>-</td><td>+</td><td>-</td></tr><tr><td>2 line</td><td>+</td><td>-</td><td>+</td><td>-</td><td>+</td><td>-</td><td>2 line</td><td>-</td><td>+</td><td>-</td><td>+</td><td>-</td></tr><tr><td>3 line</td><td>-</td><td>+</td><td>-</td><td>+</td><td>-</td><td>+</td><td>3 line</td><td>+</td><td>-</td><td>+</td><td>-</td><td>+</td></tr><tr><td>4 line</td><td>-</td><td>+</td><td>-</td><td>+</td><td>-</td><td>+</td><td>4 line</td><td>+</td><td>-</td><td>+</td><td>-</td><td>+</td></tr><tr><td>2'b11</td><td colspan="12">Setting prohibited</td></tr></table>													DINV [1:0]	Dot inversion mode												2'b00	Column inversion	1st frame						2nd frame						1 line	+	-	+	-	+	-	1 line	-	+	-	+	-	2 line	+	-	+	-	+	-	2 line	-	+	-	+	-	3 line	+	-	+	-	+	-	3 line	-	+	-	+	-	4 line	+	-	+	-	+	-	4 line	-	+	-	+	-	2'b01	1-dot inversion	1st frame						2nd frame						1 line	+	-	+	-	+	-	1 line	-	+	-	+	-	2 line	-	+	-	+	-	+	2 line	+	-	+	-	+	3 line	+	-	+	-	+	-	3 line	-	+	-	+	-	4 line	-	+	-	+	-	+	4 line	+	-	+	-	+	2'b10	2-dot inversion	1st frame						2nd frame						1 line	+	-	+	-	+	-	1 line	-	+	-	+	-	2 line	+	-	+	-	+	-	2 line	-	+	-	+	-	3 line	-	+	-	+	-	+	3 line	+	-	+	-	+	4 line	-	+	-	+	-	+	4 line	+	-	+	-	+	2'b11	Setting prohibited											
	DINV [1:0]	Dot inversion mode																																																																																																																																																																																																																																											
	2'b00	Column inversion	1st frame						2nd frame																																																																																																																																																																																																																																				
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			4 line	+	-	+	-	+	-	4 line	-	+	-	+	-																																																																																																																																																																																																																														
	2'b01	1-dot inversion	1st frame						2nd frame																																																																																																																																																																																																																																				
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4 line			-	+	-	+	-	+	4 line	+	-	+	-	+																																																																																																																																																																																																																															
2'b10	2-dot inversion	1st frame						2nd frame																																																																																																																																																																																																																																					
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Normal Mode ON, Idle Mode OFF, Sleep OUT	Yes																																																																																																																																																																																																																																												
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Partial Mode ON, Idle Mode OFF, Sleep OUT	Yes																																																																																																																																																																																																																																												
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Sleep IN	Yes																																																																																																																																																																																																																																												
Default	<table><tr><th rowspan="2">Status</th><th colspan="2">Default Value</th></tr><tr><th>ZINV</th><th>DINV[1:0]</th></tr><tr><td>Power ON Sequence</td><td>1'b0</td><td>2'b00</td></tr><tr><td>H/W Reset</td><td>1'b0</td><td>2'b00</td></tr></table>													Status	Default Value		ZINV	DINV[1:0]	Power ON Sequence	1'b0	2'b00	H/W Reset	1'b0	2'b00																																																																																																																																																																																																																					
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Power ON Sequence	1'b0	2'b00																																																																																																																																																																																																																																											
H/W Reset	1'b0	2'b00																																																																																																																																																																																																																																											

8.2.55. Blanking Porph Control (B5h)

_B5h				PRCTR (Blanking Porph)																																																					
	D/CX	RDX	WRX	D[15:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX																																												
Command	0	1	↑	XXXXXXXX	1	0	1	1	0	1	0	1	B5h																																												
1 st parameter	1	1	↑	XXXXXXXX	VFP[7:0]								XX																																												
2 nd parameter	1	1	↑	XXXXXXXX	VBP[7:0]								XX																																												
3 nd parameter	1	1	↑	XXXXXXXX	0	0	0	HFP[4:0]					XX																																												
4 nd parameter	1	1	↑	XXXXXXXX	HBP[7:0]								XX																																												
Description	VFP [7:0] / VBP [7:0]: The FP [7:0] and BP [7:0] bits specify the line number of vertical front and back porch period respectively.																																																								
	<table><tr><th>FP[7:0]</th><th>Number of lines of front porch</th></tr><tr><td>00000000</td><td>Setting prohibited</td></tr><tr><td>00000001</td><td>Setting prohibited</td></tr><tr><td>00000010</td><td>2</td></tr><tr><td>00000011</td><td>3</td></tr><tr><td>:</td><td>:</td></tr><tr><td>:</td><td>:</td></tr><tr><td>11111100</td><td>252</td></tr><tr><td>11111101</td><td>253</td></tr><tr><td>11111110</td><td>254</td></tr><tr><td>11111111</td><td>255</td></tr></table>				FP[7:0]	Number of lines of front porch	00000000	Setting prohibited	00000001	Setting prohibited	00000010	2	00000011	3	:	:	:	:	11111100	252	11111101	253	11111110	254	11111111	255	<table><tr><th>BP[7:0]</th><th>Number of lines of back porch</th></tr><tr><td>00000000</td><td>Setting prohibited</td></tr><tr><td>00000001</td><td>Setting prohibited</td></tr><tr><td>00000010</td><td>2</td></tr><tr><td>00000011</td><td>3</td></tr><tr><td>:</td><td>:</td></tr><tr><td>:</td><td>:</td></tr><tr><td>11111100</td><td>252</td></tr><tr><td>11111101</td><td>253</td></tr><tr><td>11111110</td><td>254</td></tr><tr><td>11111111</td><td>254</td></tr></table>									BP[7:0]	Number of lines of back porch	00000000	Setting prohibited	00000001	Setting prohibited	00000010	2	00000011	3	:	:	:	:	11111100	252	11111101	253	11111110	254	11111111	254
	FP[7:0]	Number of lines of front porch																																																							
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	11111101	253																																																							
	11111110	254																																																							
	11111111	254																																																							
	HFP [4:0]: The HFP [4:0] bits specify the dotclk number of horizontal front porch period.																																																								
<table><tr><th>HFP[4:0]</th><th>Number of dotclk of front porch</th></tr><tr><td>00000</td><td>Setting prohibited</td></tr><tr><td>00001</td><td>Setting prohibited</td></tr><tr><td>00010</td><td>2</td></tr><tr><td>00011</td><td>3</td></tr><tr><td>:</td><td>:</td></tr><tr><td>:</td><td>:</td></tr><tr><td>11100</td><td>28</td></tr><tr><td>11101</td><td>29</td></tr><tr><td>11110</td><td>30</td></tr><tr><td>11111</td><td>31</td></tr></table>				HFP[4:0]	Number of dotclk of front porch	00000	Setting prohibited	00001	Setting prohibited	00010	2	00011	3	:	:	:	:	11100	28	11101	29	11110	30	11111	31																																
HFP[4:0]	Number of dotclk of front porch																																																								
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<table><tr><th>HBP[7:0]</th><th>Number of dotclk of front porch</th></tr><tr><td>00000000</td><td>Setting prohibited</td></tr><tr><td>00000001</td><td>Setting prohibited</td></tr><tr><td>00000010</td><td>2</td></tr><tr><td>00000011</td><td>3</td></tr><tr><td>:</td><td>:</td></tr><tr><td>:</td><td>:</td></tr><tr><td>11111100</td><td>252</td></tr><tr><td>11111101</td><td>253</td></tr><tr><td>11111110</td><td>254</td></tr><tr><td>11111111</td><td>255</td></tr></table>				HBP[7:0]	Number of dotclk of front porch	00000000	Setting prohibited	00000001	Setting prohibited	00000010	2	00000011	3	:	:	:	:	11111100	252	11111101	253	11111110	254	11111111	255																																
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11111110	254																																																								
11111111	255																																																								
Restriction																																																									

Register Availability				
Default				

8.2.56. Display Function Control (B6h)

B6h	DISCTRL (Display Function Control)																															
	D/CX	RDX	WRX	D[15:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX																			
Command	0	1	↑	XXXXXXXX	1	0	1	1	0	1	1	0	B6h																			
1 st parameter	1	1	↑	XXXXXXXX	BYPASS	0	RM	DM	PTG[1:0]		PT[1:0]		XX																			
2 nd parameter	1	1	↑	XXXXXXXX	0	GS	SS	SM	ISC[3:0]				XX																			
3 rd Parameter	1	1	↑	XXXXXXXX	0	0	NL[5:0]						XX																			
Description	DM: Select the display operation mode. <table><tr><th>DM</th><th>Interface Mode</th></tr><tr><td>0</td><td>Internal system clock</td></tr><tr><td>1</td><td>RGB interface</td></tr></table>													DM	Interface Mode	0	Internal system clock	1	RGB interface													
	DM	Interface Mode																														
	0	Internal system clock																														
	1	RGB interface																														
	RM: Select the interface to access the GRAM. When RM='0', the driver will write display data to GRAM via system interface and the driver will write display data to GRAM via RGB interface when RM='1'. <table><tr><th>RM</th><th>Interface for RAM access</th></tr><tr><td>0</td><td>System interface</td></tr><tr><td>1</td><td>RGB interface</td></tr></table>													RM	Interface for RAM access	0	System interface	1	RGB interface													
	RM	Interface for RAM access																														
	0	System interface																														
	1	RGB interface																														
	BYPASS: Select the display data path whether memory or direct to shift register when RGB interface is used. <table><tr><th>BYPASS</th><th>Display data path</th></tr><tr><td>0</td><td>Memory</td></tr><tr><td>1</td><td>Direct to shift register</td></tr></table>													BYPASS	Display data path	0	Memory	1	Direct to shift register													
	BYPASS	Display data path																														
	0	Memory																														
	1	Direct to shift register																														
Note: RGB input signal, when set to bypass mode the Hsync low ≥ 3, HBP ≥ 3, HFP ≥ 10.																																
PTG [1:0]: Set the scan mode in non-display area. <table><tr><th>PTG1</th><th>PTG0</th><th>Gate outputs in non-display area</th><th>Source outputs in non-display area</th></tr><tr><td>0</td><td>0</td><td>Normal scan</td><td>Set with the PT[2:0] bits</td></tr><tr><td>0</td><td>1</td><td>Setting prohibited</td><td>---</td></tr><tr><td>1</td><td>0</td><td>Interval scan</td><td>Set with the PT[2:0] bits</td></tr><tr><td>1</td><td>1</td><td>Setting prohibited</td><td>---</td></tr></table>													PTG1	PTG0	Gate outputs in non-display area	Source outputs in non-display area	0	0	Normal scan	Set with the PT[2:0] bits	0	1	Setting prohibited	---	1	0	Interval scan	Set with the PT[2:0] bits	1	1	Setting prohibited	---
PTG1	PTG0	Gate outputs in non-display area	Source outputs in non-display area																													
0	0	Normal scan	Set with the PT[2:0] bits																													
0	1	Setting prohibited	---																													
1	0	Interval scan	Set with the PT[2:0] bits																													
1	1	Setting prohibited	---																													
PT [1:0]: Determine source/VCOM output in a non-display area in the partial display mode. <table><tr><th colspan="2">PT[1:0]</th><th>Source output on non-display area</th></tr><tr><td>0</td><td>0</td><td>V63</td></tr><tr><td>0</td><td>1</td><td>V0</td></tr><tr><td>1</td><td>0</td><td>AGND</td></tr><tr><td>1</td><td>1</td><td>Hi-Z</td></tr></table>													PT[1:0]		Source output on non-display area	0	0	V63	0	1	V0	1	0	AGND	1	1	Hi-Z					
PT[1:0]		Source output on non-display area																														
0	0	V63																														
0	1	V0																														
1	0	AGND																														
1	1	Hi-Z																														
SS: Select the shift direction of outputs from the source driver. <table><tr><th>SS</th><th>Source Output Scan Direction</th></tr><tr><td>0</td><td>S1 → S960</td></tr><tr><td>1</td><td>S960 → S1</td></tr></table>													SS	Source Output Scan Direction	0	S1 → S960	1	S960 → S1														
SS	Source Output Scan Direction																															
0	S1 → S960																															
1	S960 → S1																															
In addition to the shift direction, the settings for both SS and BGR bits are required to change the assignment of R, G, B dots to the source driver pins.																																
To assign R, G, B dots to the source driver pins from S1 to S960, set SS = 0.																																

To assign R, G, B dots to the source driver pins from S960 to S1, set SS = 1.

ISC[3:0]: Set the scan cycle when PTG selects interval scan in non-display area drive period. The scan cycle is defined by n frame periods, where n is an odd number from 3 to 31. The polarity of liquid crystal drive voltage from the gate driver is inverted in the same timing as the interval scan cycle.

ISC[3:0]	Scan cycle	(f _{FRAME})=60Hz
4'h0	Setting inhibited	—
4'h1	3 frames	50ms
4'h2	5 frames	84ms
4'h3	7 frames	117ms
4'h4	9 frames	150ms
4'h5	11 frames	184ms
4'h6	13 frames	217ms
4'h7	15 frames	251ms
4'h8	17 frames	284ms
4'h9	19 frames	317ms
4'hA	21 frames	351ms
4'hB	23 frames	384ms
4'hC	25 frames	418ms
4'hD	27 frames	451ms
4'hE	29 frames	484ms
4'hF	31 frames	518ms

GS: Sets the direction of scan by the gate driver.

GS	Gate Output Scan Direction
0	G1 → G480
1	G480 → G1

SM: Sets the gate driver pin arrangement in combination with the GS bit (R60h) to select the optimal scan mode for the module.

SM	GS	Scan Direction	Gate Output Sequence
0	0		G1, G2, G3, G4, ..., G476 G477, G478, G479, G480

			G1 G3 G477 G479 TFT Panel G2 G4 G378 G480 Odd-number Even-number G479 to G1 G480 to G2 IC	G480, G479, G478, ..., G9 G7, G5, G4, G3, G2, G1
		G1 G479 TFT Panel G2 G480 Odd-number Even-number G1 to G479 G2 to G480 IC	G1, G3, G5, G7, ...,G471 G473, G475, G477, G479 G2, G4, G6, G8, ...,G472 G474, G476, G478, G480	
		G1 G479 TFT Panel G2 G480 Odd-number Even-number G479 to G1 G480 to G2 IC	G480, G478, G476, ...,G14 G12, G10, G8, G6, G4, G2 G479, G477, G475,...,G13 G11, G9, G7, G5, G3, G1	

NL [5:0]: Sets the number of lines to drive the LCD at an interval of 8 lines. The GRAM address mapping is not affected by the number of lines set by NL[5:0]. The number of lines must be the same or more than the number of lines necessary for the size of the liquid crystal panel.

NL[5:0]	LCD Drive Line
6'h00 ~ 6'h3B	8 * (NL5:0)+1 lines
Others	Setting inhibited

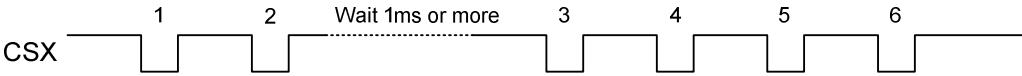
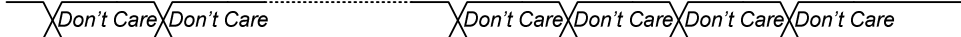
Restriction													
Register Availability	<table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode ON, Idle Mode OFF, Sleep OUT</td><td>Yes</td></tr><tr><td>Normal Mode ON, Idle Mode ON, Sleep OUT</td><td>Yes</td></tr><tr><td>Partial Mode ON, Idle Mode OFF, Sleep OUT</td><td>Yes</td></tr><tr><td>Partial Mode ON, Idle Mode ON, Sleep OUT</td><td>Yes</td></tr><tr><td>Sleep IN</td><td>Yes</td></tr></table>	Status	Availability	Normal Mode ON, Idle Mode OFF, Sleep OUT	Yes	Normal Mode ON, Idle Mode ON, Sleep OUT	Yes	Partial Mode ON, Idle Mode OFF, Sleep OUT	Yes	Partial Mode ON, Idle Mode ON, Sleep OUT	Yes	Sleep IN	Yes
Status	Availability												
Normal Mode ON, Idle Mode OFF, Sleep OUT	Yes												
Normal Mode ON, Idle Mode ON, Sleep OUT	Yes												
Partial Mode ON, Idle Mode OFF, Sleep OUT	Yes												
Partial Mode ON, Idle Mode ON, Sleep OUT	Yes												
Sleep IN	Yes												

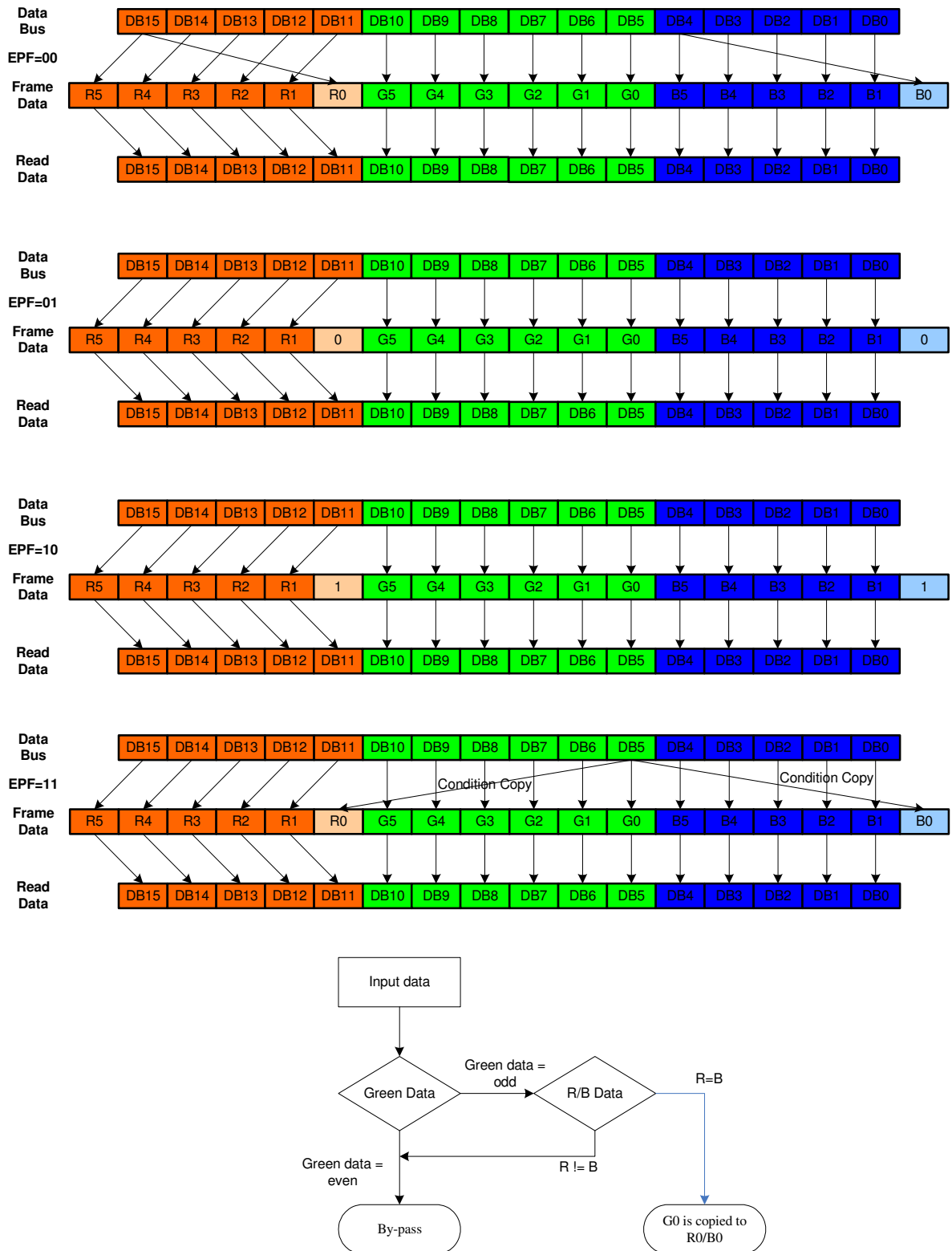
Default

Status	Default Value						
	PTG[1:0]	PT[1:0]	GS	SS	SM	ISC[3:0]	NL[5:0]
Power ON Sequence	2'b00	2'b00	1'b0	1'b0	1'b0	4'b0010	6'b111011
H/W Reset	2'b00	2'b00	1'b0	1'b0	1'b0	4'b0010	6'b111011

Status	Default Value		
	RM	DM	BYPASS
Power ON Sequence	1'b0	1'b0	1'b1
H/W Reset	1'b0	1'b0	1'b1

8.2.57. Entry Mode Set (B7h)

B7h	ETMOD (Entry Mode Set)																										
	D/CX	RDX	WRX	D[15:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX														
Command	0	1	↑	XXXXXXXX	1	0	1	1	0	1	1	1	B7h														
Parameter	1	1	↑	XXXXXXXX	EPF[1:0]		0	0	DSTB	GON	DTE	GAS	XX														
Description	DSTB: The ILI9486 driver enters the Deep Standby Mode when DSTB is set to high ("1"). In Deep Standby mode, both internal logic power and SRAM power are turn off, the display data stored in the Frame Memory and the instructions are not saved. Rewrite Frame Memory content and instructions after the Deep Standby Mode is exited.																										
	<i>Note: ILI9486 provides two ways to exit the Deep Standby Mode:</i>																										
	<i>(1) Exit Deep Standby Mode by pull down CSX to low ("0") 6 times.</i>																										
	<i>(2) Input a RESX pulse with effective low level duration to start up the inside logic regulator and makes a transition to the initial state.</i>																										
																											
	WRX "High"																										
	RDX "High"																										
	D/CX "Low" or "High"																										
	D[17:0] 																										
	GAS: Low voltage detection control.																										
<table border="1"><thead><tr><th>GAS</th><th>Low voltage detection</th></tr></thead><tbody><tr><td>0</td><td>Enable</td></tr><tr><td>1</td><td>Disable</td></tr></tbody></table>													GAS	Low voltage detection	0	Enable	1	Disable									
GAS	Low voltage detection																										
0	Enable																										
1	Disable																										
GON/DTE: Set the output level of gate driver G1 ~ G320 as follows																											
<table border="1"><thead><tr><th>GON</th><th>DTE</th><th>G1~G320 Gate Output</th></tr></thead><tbody><tr><td>0</td><td>0</td><td>VGH</td></tr><tr><td>0</td><td>1</td><td>VGH</td></tr><tr><td>1</td><td>0</td><td>VGL</td></tr><tr><td>1</td><td>1</td><td>Normal display</td></tr></tbody></table>													GON	DTE	G1~G320 Gate Output	0	0	VGH	0	1	VGH	1	0	VGL	1	1	Normal display
GON	DTE	G1~G320 Gate Output																									
0	0	VGH																									
0	1	VGH																									
1	0	VGL																									
1	1	Normal display																									
EPF[1:0] Set the data format when 16bbp (R,G,B) to 18 bbp (r, g, b) is stored in the internal GRAM																											



Restriction

Register Availability	<table><tr><th>Status</th><th colspan="5">Availability</th></tr><tr><td>Normal Mode ON, Idle Mode OFF, Sleep OUT</td><td colspan="5">Yes</td></tr><tr><td>Normal Mode ON, Idle Mode ON, Sleep OUT</td><td colspan="5">Yes</td></tr><tr><td>Partial Mode ON, Idle Mode OFF, Sleep OUT</td><td colspan="5">Yes</td></tr><tr><td>Partial Mode ON, Idle Mode ON, Sleep OUT</td><td colspan="5">Yes</td></tr><tr><td>Sleep IN</td><td colspan="5">Yes</td></tr></table>						Status	Availability					Normal Mode ON, Idle Mode OFF, Sleep OUT	Yes					Normal Mode ON, Idle Mode ON, Sleep OUT	Yes					Partial Mode ON, Idle Mode OFF, Sleep OUT	Yes					Partial Mode ON, Idle Mode ON, Sleep OUT	Yes					Sleep IN	Yes				
	Status	Availability																																								
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Sleep IN	Yes																																									
Default	<table><tr><th rowspan="2">Status</th><th colspan="5">Default Value</th></tr><tr><th>EPF[1:0]</th><th>DSTB</th><th>GON</th><th>DTE</th><th>GAS</th></tr><tr><td>Power ON Sequence</td><td>2'b00</td><td>1'b0</td><td>1'b1</td><td>1'b1</td><td>1'b0</td></tr><tr><td>H/W Reset</td><td>2b'00</td><td>1'b0</td><td>1'b1</td><td>1'b1</td><td>1'b0</td></tr></table>						Status	Default Value					EPF[1:0]	DSTB	GON	DTE	GAS	Power ON Sequence	2'b00	1'b0	1'b1	1'b1	1'b0	H/W Reset	2b'00	1'b0	1'b1	1'b1	1'b0													
	Status	Default Value																																								
		EPF[1:0]	DSTB	GON	DTE	GAS																																				
	Power ON Sequence	2'b00	1'b0	1'b1	1'b1	1'b0																																				
H/W Reset	2b'00	1'b0	1'b1	1'b1	1'b0																																					

8.2.58. Power Control 1 (C0h)

C0h	PWCTRL 1 (Power Control 1)																																																																															
	D/CX	RDX	WRX	D[15:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX																																																																			
Command	0	1	↑	XXXXXXXX	1	1	0	0	0	0	0	0	C0h																																																																			
1 st Parameter	1	1	↑	XXXXXXXX	0	0	0	VRH1[4:0]					XX																																																																			
2 nd Parameter	1	1	↑	XXXXXXXX	0	0	0	VRH2[4:0]					XX																																																																			
Description	VRH1[4:0]: Sets the VREG1OUT voltage for positive gamma																																																																															
	<table><tr><th>VRH1[4:0]</th><th>VREG1OUT</th><th>VRH1[4:0]</th><th>VREG1OUT</th></tr><tr><td>5'h00</td><td>Halt (Vreg1out =Hiz)</td><td>5'h10</td><td>1.25 x 3.65 = 4.5625</td></tr><tr><td>5'h01</td><td>1.25 x 2.90 = 3.6250</td><td>5'h11</td><td>1.25 x 3.70 = 4.6250</td></tr><tr><td>5'h02</td><td>1.25 x 2.95 = 3.6875</td><td>5'h12</td><td>1.25 x 3.75 = 4.6875</td></tr><tr><td>5'h03</td><td>1.25 x 3.00 = 3.7500</td><td>5'h13</td><td>1.25 x 3.80 = 4.7500</td></tr><tr><td>5'h04</td><td>1.25 x 3.05 = 3.8125</td><td>5'h14</td><td>1.25 x 3.85 = 4.8125</td></tr><tr><td>5'h05</td><td>1.25 x 3.10 = 3.8750</td><td>5'h15</td><td>1.25 x 3.90 = 4.8750</td></tr><tr><td>5'h06</td><td>1.25 x 3.15 = 3.9375</td><td>5'h16</td><td>1.25 x 3.95 = 4.9375</td></tr><tr><td>5'h07</td><td>1.25 x 3.20 = 4.0000</td><td>5'h17</td><td>1.25 x 4.00 = 5.0000</td></tr><tr><td>5'h08</td><td>1.25 x 3.25 = 4.0625</td><td>5'h18</td><td>1.25 x 4.05 = 5.0625</td></tr><tr><td>5'h09</td><td>1.25 x 3.30 = 4.1250</td><td>5'h19</td><td>1.25 x 4.10 = 5.1250</td></tr><tr><td>5'h0A</td><td>1.25 x 3.35 = 4.1875</td><td>5'h1A</td><td>1.25 x 4.15 = 5.1875</td></tr><tr><td>5'h0B</td><td>1.25 x 3.40 = 4.2500</td><td>5'h1B</td><td>1.25 x 4.20 = 5.2500</td></tr><tr><td>5'h0C</td><td>1.25 x 3.45 = 4.3125</td><td>5'h1C</td><td>1.25 x 4.25 = 5.3125</td></tr><tr><td>5'h0D</td><td>1.25 x 3.50 = 4.3750</td><td>5'h1D</td><td>1.25 x 4.30 = 5.3750</td></tr><tr><td>5'h0E</td><td>1.25 x 3.55 = 4.4375</td><td>5'h1E</td><td>1.25 x 4.35 = 5.4375</td></tr><tr><td>5'h0F</td><td>1.25 x 3.60 = 4.5000</td><td>5'h1F</td><td>1.25 x 4.40 = 5.5000</td></tr></table>													VRH1[4:0]	VREG1OUT	VRH1[4:0]	VREG1OUT	5'h00	Halt (Vreg1out =Hiz)	5'h10	1.25 x 3.65 = 4.5625	5'h01	1.25 x 2.90 = 3.6250	5'h11	1.25 x 3.70 = 4.6250	5'h02	1.25 x 2.95 = 3.6875	5'h12	1.25 x 3.75 = 4.6875	5'h03	1.25 x 3.00 = 3.7500	5'h13	1.25 x 3.80 = 4.7500	5'h04	1.25 x 3.05 = 3.8125	5'h14	1.25 x 3.85 = 4.8125	5'h05	1.25 x 3.10 = 3.8750	5'h15	1.25 x 3.90 = 4.8750	5'h06	1.25 x 3.15 = 3.9375	5'h16	1.25 x 3.95 = 4.9375	5'h07	1.25 x 3.20 = 4.0000	5'h17	1.25 x 4.00 = 5.0000	5'h08	1.25 x 3.25 = 4.0625	5'h18	1.25 x 4.05 = 5.0625	5'h09	1.25 x 3.30 = 4.1250	5'h19	1.25 x 4.10 = 5.1250	5'h0A	1.25 x 3.35 = 4.1875	5'h1A	1.25 x 4.15 = 5.1875	5'h0B	1.25 x 3.40 = 4.2500	5'h1B	1.25 x 4.20 = 5.2500	5'h0C	1.25 x 3.45 = 4.3125	5'h1C	1.25 x 4.25 = 5.3125	5'h0D	1.25 x 3.50 = 4.3750	5'h1D	1.25 x 4.30 = 5.3750	5'h0E	1.25 x 3.55 = 4.4375	5'h1E	1.25 x 4.35 = 5.4375	5'h0F	1.25 x 3.60 = 4.5000	5'h1F
VRH1[4:0]	VREG1OUT	VRH1[4:0]	VREG1OUT																																																																													
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5'h0F	1.25 x 3.60 = 4.5000	5'h1F	1.25 x 4.40 = 5.5000																																																																													
Description	VRH2[4:0]: Sets the VREG2OUT voltage for negative gamma																																																																															
	<table><tr><th>VRH2[4:0]</th><th>VREG2OUT</th><th>VRH2[4:0]</th><th>VREG2OUT</th></tr><tr><td>5'h00</td><td>Halt (Vreg2out =Hiz)</td><td>5'h10</td><td>-1.25 x 3.65 = -4.5625</td></tr><tr><td>5'h01</td><td>-1.25 x 2.90 = -3.6250</td><td>5'h11</td><td>-1.25 x 3.70 = -4.6250</td></tr><tr><td>5'h02</td><td>-1.25 x 2.95 = -3.6875</td><td>5'h12</td><td>-1.25 x 3.75 = -4.6875</td></tr><tr><td>5'h03</td><td>-1.25 x 3.00 = -3.7500</td><td>5'h13</td><td>-1.25 x 3.80 = -4.7500</td></tr><tr><td>5'h04</td><td>-1.25 x 3.05 = -3.8125</td><td>5'h14</td><td>-1.25 x 3.85 = -4.8125</td></tr><tr><td>5'h05</td><td>-1.25 x 3.10 = -3.8750</td><td>5'h15</td><td>-1.25 x 3.90 = -4.8750</td></tr><tr><td>5'h06</td><td>-1.25 x 3.15 = -3.9375</td><td>5'h16</td><td>-1.25 x 3.95 = -4.9375</td></tr><tr><td>5'h07</td><td>-1.25 x 3.20 = -4.0000</td><td>5'h17</td><td>-1.25 x 4.00 = -5.0000</td></tr><tr><td>5'h08</td><td>-1.25 x 3.25 = -4.0625</td><td>5'h18</td><td>-1.25 x 4.05 = -5.0625</td></tr><tr><td>5'h09</td><td>-1.25 x 3.30 = -4.1250</td><td>5'h19</td><td>-1.25 x 4.10 = -5.1250</td></tr><tr><td>5'h0A</td><td>-1.25 x 3.35 = -4.1875</td><td>5'h1A</td><td>-1.25 x 4.15 = -5.1875</td></tr><tr><td>5'h0B</td><td>-1.25 x 3.40 = -4.2500</td><td>5'h1B</td><td>-1.25 x 4.20 = -5.2500</td></tr><tr><td>5'h0C</td><td>-1.25 x 3.45 = -4.3125</td><td>5'h1C</td><td>-1.25 x 4.25 = -5.3125</td></tr><tr><td>5'h0D</td><td>-1.25 x 3.50 = -4.3750</td><td>5'h1D</td><td>-1.25 x 4.30 = -5.3750</td></tr><tr><td>5'h0E</td><td>-1.25 x 3.55 = -4.4375</td><td>5'h1E</td><td>-1.25 x 4.35 = -5.4375</td></tr><tr><td>5'h0F</td><td>-1.25 x 3.60 = -4.5000</td><td>5'h1F</td><td>-1.25 x 4.40 = -5.5000</td></tr></table>													VRH2[4:0]	VREG2OUT	VRH2[4:0]	VREG2OUT	5'h00	Halt (Vreg2out =Hiz)	5'h10	-1.25 x 3.65 = -4.5625	5'h01	-1.25 x 2.90 = -3.6250	5'h11	-1.25 x 3.70 = -4.6250	5'h02	-1.25 x 2.95 = -3.6875	5'h12	-1.25 x 3.75 = -4.6875	5'h03	-1.25 x 3.00 = -3.7500	5'h13	-1.25 x 3.80 = -4.7500	5'h04	-1.25 x 3.05 = -3.8125	5'h14	-1.25 x 3.85 = -4.8125	5'h05	-1.25 x 3.10 = -3.8750	5'h15	-1.25 x 3.90 = -4.8750	5'h06	-1.25 x 3.15 = -3.9375	5'h16	-1.25 x 3.95 = -4.9375	5'h07	-1.25 x 3.20 = -4.0000	5'h17	-1.25 x 4.00 = -5.0000	5'h08	-1.25 x 3.25 = -4.0625	5'h18	-1.25 x 4.05 = -5.0625	5'h09	-1.25 x 3.30 = -4.1250	5'h19	-1.25 x 4.10 = -5.1250	5'h0A	-1.25 x 3.35 = -4.1875	5'h1A	-1.25 x 4.15 = -5.1875	5'h0B	-1.25 x 3.40 = -4.2500	5'h1B	-1.25 x 4.20 = -5.2500	5'h0C	-1.25 x 3.45 = -4.3125	5'h1C	-1.25 x 4.25 = -5.3125	5'h0D	-1.25 x 3.50 = -4.3750	5'h1D	-1.25 x 4.30 = -5.3750	5'h0E	-1.25 x 3.55 = -4.4375	5'h1E	-1.25 x 4.35 = -5.4375	5'h0F	-1.25 x 3.60 = -4.5000	5'h1F
VRH2[4:0]	VREG2OUT	VRH2[4:0]	VREG2OUT																																																																													
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5'h0E	-1.25 x 3.55 = -4.4375	5'h1E	-1.25 x 4.35 = -5.4375																																																																													
5'h0F	-1.25 x 3.60 = -4.5000	5'h1F	-1.25 x 4.40 = -5.5000																																																																													
Restriction																																																																																

Register Availability	<table><tr><th>Status</th><th colspan="2">Availability</th></tr><tr><td>Normal Mode ON, Idle Mode OFF, Sleep OUT</td><td colspan="2">Yes</td></tr><tr><td>Normal Mode ON, Idle Mode ON, Sleep OUT</td><td colspan="2">Yes</td></tr><tr><td>Partial Mode ON, Idle Mode OFF, Sleep OUT</td><td colspan="2">Yes</td></tr><tr><td>Partial Mode ON, Idle Mode ON, Sleep OUT</td><td colspan="2">Yes</td></tr><tr><td>Sleep IN</td><td colspan="2">Yes</td></tr></table>			Status	Availability		Normal Mode ON, Idle Mode OFF, Sleep OUT	Yes		Normal Mode ON, Idle Mode ON, Sleep OUT	Yes		Partial Mode ON, Idle Mode OFF, Sleep OUT	Yes		Partial Mode ON, Idle Mode ON, Sleep OUT	Yes		Sleep IN	Yes	
	Status	Availability																			
	Normal Mode ON, Idle Mode OFF, Sleep OUT	Yes																			
	Normal Mode ON, Idle Mode ON, Sleep OUT	Yes																			
	Partial Mode ON, Idle Mode OFF, Sleep OUT	Yes																			
	Partial Mode ON, Idle Mode ON, Sleep OUT	Yes																			
Sleep IN	Yes																				
Default	<table><tr><th>Status</th><th>VRH1</th><th>VRH2</th></tr><tr><td>Power ON Sequence</td><td>5'b01110</td><td>5'b01110</td></tr><tr><td>H/W Reset</td><td>5'b01110</td><td>5'b01110</td></tr></table>			Status	VRH1	VRH2	Power ON Sequence	5'b01110	5'b01110	H/W Reset	5'b01110	5'b01110									
	Status	VRH1	VRH2																		
	Power ON Sequence	5'b01110	5'b01110																		
H/W Reset	5'b01110	5'b01110																			

8.2.59. Power Control 2 (C1h)

C1h	PWCTRL 2 (Power Control 2)																																																								
	D/CX	RDX	WRX	D[15:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX																																												
Command	0	1	↑	XXXXXXXX	1	1	0	0	0	0	0	1	C1h																																												
1 st parameter	1	1	↑	XXXXXXXX	0	SAP[2:0]			0	BT[2:0]			XX																																												
2 nd parameter	1	1	↑	XXXXXXXX	0	0	0	0	0	VC[2:0]			XX																																												
Description	BT [2:0]: Sets the factor used in the step-up circuits. Select the optimal step-up factor for the operating voltage. To reduce power consumption, set a smaller factor.																																																								
	<table><tr><th>BT[2:0]</th><th>DDVDH</th><th>DDVDL</th><th>VCL</th><th>VGH</th><th>VGL</th></tr><tr><td>4'h0</td><td rowspan="8">Vci1 x 2</td><td rowspan="8">-(VCI1-VCL)</td><td rowspan="8">- Vci1</td><td rowspan="3">Vci1 x 6</td><td>- Vci1 x 5</td></tr><tr><td>4'h1</td><td>- Vci1 x 4</td></tr><tr><td>4'h2</td><td>- Vci1 x 3</td></tr><tr><td>4'h3</td><td rowspan="3">Vci1 x 5</td><td>- Vci1 x 5</td></tr><tr><td>4'h4</td><td>- Vci1 x 4</td></tr><tr><td>4'h5</td><td>- Vci1 x 3</td></tr><tr><td>4'h6</td><td rowspan="2">Vci1 x 4</td><td>- Vci1 x4</td></tr><tr><td>4'h7</td><td>- Vci1 x3</td></tr></table>													BT[2:0]	DDVDH	DDVDL	VCL	VGH	VGL	4'h0	Vci1 x 2	-(VCI1-VCL)	- Vci1	Vci1 x 6	- Vci1 x 5	4'h1	- Vci1 x 4	4'h2	- Vci1 x 3	4'h3	Vci1 x 5	- Vci1 x 5	4'h4	- Vci1 x 4	4'h5	- Vci1 x 3	4'h6	Vci1 x 4	- Vci1 x4	4'h7	- Vci1 x3																
	BT[2:0]	DDVDH	DDVDL	VCL	VGH	VGL																																																			
	4'h0	Vci1 x 2	-(VCI1-VCL)	- Vci1	Vci1 x 6	- Vci1 x 5																																																			
	4'h1					- Vci1 x 4																																																			
	4'h2					- Vci1 x 3																																																			
	4'h3				Vci1 x 5	- Vci1 x 5																																																			
	4'h4					- Vci1 x 4																																																			
	4'h5					- Vci1 x 3																																																			
	4'h6				Vci1 x 4	- Vci1 x4																																																			
4'h7	- Vci1 x3																																																								
<i>Note: To prevent the device damage, please keep VGH – DDVDH < 8V condition.</i>																																																									
SAP [2:0]: It is used to adjust the constant current in the operational amplifier circuit in the LCD power supply circuit. Larger constant current enhances the drivability of the LCD, but it also increases the current consumption. Adjust the constant current taking the trade-off between the display quality and the current consumption into account.																																																									
<table><tr><th colspan="3">SAP[2:0]</th><th>Description</th><th>Gamma bias control</th></tr><tr><td>0</td><td>0</td><td>0</td><td>Small</td><td>0.71 x I</td></tr><tr><td>0</td><td>0</td><td>1</td><td>Small</td><td>0.71 x I</td></tr><tr><td>0</td><td>1</td><td>0</td><td>Small</td><td>0.71 x I</td></tr><tr><td>0</td><td>1</td><td>1</td><td>Small</td><td>0.71 x I</td></tr><tr><td>1</td><td>0</td><td>0</td><td>Medium</td><td>1.00 x I</td></tr><tr><td>1</td><td>0</td><td>1</td><td>Medium to Large</td><td>1.25 x I</td></tr><tr><td>1</td><td>1</td><td>0</td><td>Large</td><td>1.43 x I</td></tr><tr><td>1</td><td>1</td><td>1</td><td>Large</td><td>1.43 x I</td></tr></table>													SAP[2:0]			Description	Gamma bias control	0	0	0	Small	0.71 x I	0	0	1	Small	0.71 x I	0	1	0	Small	0.71 x I	0	1	1	Small	0.71 x I	1	0	0	Medium	1.00 x I	1	0	1	Medium to Large	1.25 x I	1	1	0	Large	1.43 x I	1	1	1	Large	1.43 x I
SAP[2:0]			Description	Gamma bias control																																																					
0	0	0	Small	0.71 x I																																																					
0	0	1	Small	0.71 x I																																																					
0	1	0	Small	0.71 x I																																																					
0	1	1	Small	0.71 x I																																																					
1	0	0	Medium	1.00 x I																																																					
1	0	1	Medium to Large	1.25 x I																																																					
1	1	0	Large	1.43 x I																																																					
1	1	1	Large	1.43 x I																																																					
When starting the charge-pump of LCD in the Power ON stage, make sure that SAP=0, and set the SAP=1, after starting up the LCD power supply circuit.																																																									
VC [2:0]: Sets VCI1 regulator output voltage.																																																									
<table><tr><th>VC[2:0]</th><th>Vci1 voltage</th></tr><tr><td>3'h0</td><td>External VCI</td></tr><tr><td>3'h1</td><td>3.1V</td></tr><tr><td>3'h2</td><td>3.0V</td></tr><tr><td>3'h3</td><td>2.9V</td></tr><tr><td>3'h4</td><td>2.8V</td></tr><tr><td>3'h5</td><td>2.7V</td></tr><tr><td>3'h6</td><td>2.6V</td></tr><tr><td>3'h7</td><td>2.5V</td></tr></table>													VC[2:0]	Vci1 voltage	3'h0	External VCI	3'h1	3.1V	3'h2	3.0V	3'h3	2.9V	3'h4	2.8V	3'h5	2.7V	3'h6	2.6V	3'h7	2.5V																											
VC[2:0]	Vci1 voltage																																																								
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	Status	Availability																										
	Normal Mode ON, Idle Mode OFF, Sleep OUT	Yes																										
	Normal Mode ON, Idle Mode ON, Sleep OUT	Yes																										
	Partial Mode ON, Idle Mode OFF, Sleep OUT	Yes																										
	Partial Mode ON, Idle Mode ON, Sleep OUT	Yes																										
Sleep IN	Yes																											
Default	<table><tr><th rowspan="2">Status</th><th colspan="3">Default Value</th></tr><tr><th>BT[2:0]</th><th>SAP[2:0]</th><th>VC[2:0]</th></tr><tr><td>Power ON Sequence</td><td>3'b000</td><td>3'b110</td><td>3'b000</td></tr><tr><td>H/W Reset</td><td>3'b000</td><td>3'b110</td><td>3'b000</td></tr></table>				Status	Default Value			BT[2:0]	SAP[2:0]	VC[2:0]	Power ON Sequence	3'b000	3'b110	3'b000	H/W Reset	3'b000	3'b110	3'b000									
	Status	Default Value																										
		BT[2:0]	SAP[2:0]	VC[2:0]																								
	Power ON Sequence	3'b000	3'b110	3'b000																								
H/W Reset	3'b000	3'b110	3'b000																									

8.2.60. Power Control 3 (For Normal Mode) (C2h)

C2h	PWCTRL 3 (Power Control 3)																																																																																				
	D/CX	RDX	WRX	D[15:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX																																																																								
Command	0	1	↑	XXXXXXXX	1	1	0	0	0	0	1	0	C2h																																																																								
1 st parameter	1	1	↑	XXXXXXXX	0	DCA1[2:0]			0	DCA0[2:0]			XX																																																																								
Description	DCA0 [2:0]: Selects the operating frequency of the step-up circuit 1/4/5 for Normal mode. The higher step-up operating frequency enhances the drivability of the step-up circuit and the quality of display but increases the current consumption. Adjust the frequency taking the trade-off between the display quality and the current consumption into account. DCA1 [2:0]: Selects the operating frequency of the step-up circuit 2/3 for Normal mode. The higher step-up operating frequency enhances the drivability of the step-up circuit and the quality of display but increases the current consumption. Adjust the frequency taking the trade-off between the display quality and the current consumption into account.																																																																																				
	<table><tr><th colspan="3">DCA0[2:0]</th><th>Step-up cycle for step-up circuit 1/4/5</th></tr><tr><td>0</td><td>0</td><td>0</td><td>1/8 H</td></tr><tr><td>0</td><td>0</td><td>1</td><td>1/4 H</td></tr><tr><td>0</td><td>1</td><td>0</td><td>1/2 H</td></tr><tr><td>0</td><td>1</td><td>1</td><td>1 H</td></tr><tr><td>1</td><td>0</td><td>0</td><td>2 H</td></tr><tr><td>1</td><td>0</td><td>1</td><td>4 H</td></tr><tr><td>1</td><td>1</td><td>0</td><td>8 H</td></tr><tr><td>1</td><td>1</td><td>1</td><td>16 H</td></tr></table>				DCA0[2:0]			Step-up cycle for step-up circuit 1/4/5	0	0	0	1/8 H	0	0	1	1/4 H	0	1	0	1/2 H	0	1	1	1 H	1	0	0	2 H	1	0	1	4 H	1	1	0	8 H	1	1	1	16 H	<table><tr><th colspan="3">DCA1[2:0]</th><th>Step-up cycle for step-up circuit 2/3</th></tr><tr><td>0</td><td>0</td><td>0</td><td>1/2 H</td></tr><tr><td>0</td><td>0</td><td>1</td><td>1 H</td></tr><tr><td>0</td><td>1</td><td>0</td><td>2 H</td></tr><tr><td>0</td><td>1</td><td>1</td><td>4 H</td></tr><tr><td>1</td><td>0</td><td>0</td><td>8 H</td></tr><tr><td>1</td><td>0</td><td>1</td><td>16 H</td></tr><tr><td>1</td><td>1</td><td>0</td><td>32 H</td></tr><tr><td>1</td><td>1</td><td>1</td><td>64 H</td></tr></table>									DCA1[2:0]			Step-up cycle for step-up circuit 2/3	0	0	0	1/2 H	0	0	1	1 H	0	1	0	2 H	0	1	1	4 H	1	0	0	8 H	1	0	1	16 H	1	1	0	32 H	1	1	1	64 H
	DCA0[2:0]			Step-up cycle for step-up circuit 1/4/5																																																																																	
	0	0	0	1/8 H																																																																																	
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	1	0	1	4 H																																																																																	
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DCA1[2:0]			Step-up cycle for step-up circuit 2/3																																																																																		
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0	1	0	2 H																																																																																		
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Default	<table><tr><th rowspan="2">Status</th><th colspan="2">Default Value</th></tr><tr><th>DCA0[2:0]</th><th>DCA1[2:0]</th></tr><tr><td>Power ON Sequence</td><td>3'b011</td><td>3'b011</td></tr><tr><td>H/W Reset</td><td>3'b011</td><td>3'b011</td></tr></table>													Status	Default Value		DCA0[2:0]	DCA1[2:0]	Power ON Sequence	3'b011	3'b011	H/W Reset	3'b011	3'b011																																																													
Status	Default Value																																																																																				
	DCA0[2:0]	DCA1[2:0]																																																																																			
Power ON Sequence	3'b011	3'b011																																																																																			
H/W Reset	3'b011	3'b011																																																																																			

8.2.61. Power Control 4 (For Idle Mode) (C3h)

C3h	PWCTRL 4 (Power Control 4)																																																																																				
	D/CX	RDX	WRX	D[15:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX																																																																								
Command	0	1	↑	XXXXXXXX	1	1	0	0	0	0	1	1	C3h																																																																								
1 st Parameter	1	1	↑	XXXXXXXX	0	DCB1[2:0]			0	DCB0[2:0]			XX																																																																								
Description	DCB0 [2:0]: Selects the operating frequency of the step-up circuit 1/4/5 for Idle mode. The higher step-up operating frequency enhances the drivability of the step-up circuit and the quality of display but increases the current consumption. Adjust the frequency taking the trade-off between the display quality and the current consumption into account. DCB1 [2:0]: Selects the operating frequency of the step-up circuit 2/3 for Idle mode. The higher step-up operating frequency enhances the drivability of the step-up circuit and the quality of display but increases the current consumption. Adjust the frequency taking the trade-off between the display quality and the current consumption into account.																																																																																				
	<table><tr><th colspan="3">DCB0[2:0]</th><th>Step-up cycle for step-up circuit 1/4/5</th></tr><tr><td>0</td><td>0</td><td>0</td><td>1/8 H</td></tr><tr><td>0</td><td>0</td><td>1</td><td>1/4 H</td></tr><tr><td>0</td><td>1</td><td>0</td><td>1/2 H</td></tr><tr><td>0</td><td>1</td><td>1</td><td>1 H</td></tr><tr><td>1</td><td>0</td><td>0</td><td>2 H</td></tr><tr><td>1</td><td>0</td><td>1</td><td>4 H</td></tr><tr><td>1</td><td>1</td><td>0</td><td>8 H</td></tr><tr><td>1</td><td>1</td><td>1</td><td>16 H</td></tr></table>				DCB0[2:0]			Step-up cycle for step-up circuit 1/4/5	0	0	0	1/8 H	0	0	1	1/4 H	0	1	0	1/2 H	0	1	1	1 H	1	0	0	2 H	1	0	1	4 H	1	1	0	8 H	1	1	1	16 H	<table><tr><th colspan="3">DCB1[2:0]</th><th>Step-up cycle for step-up circuit 2/3</th></tr><tr><td>0</td><td>0</td><td>0</td><td>1/2 H</td></tr><tr><td>0</td><td>0</td><td>1</td><td>1 H</td></tr><tr><td>0</td><td>1</td><td>0</td><td>2 H</td></tr><tr><td>0</td><td>1</td><td>1</td><td>4 H</td></tr><tr><td>1</td><td>0</td><td>0</td><td>8 H</td></tr><tr><td>1</td><td>0</td><td>1</td><td>16 H</td></tr><tr><td>1</td><td>1</td><td>0</td><td>32 H</td></tr><tr><td>1</td><td>1</td><td>1</td><td>64 H</td></tr></table>									DCB1[2:0]			Step-up cycle for step-up circuit 2/3	0	0	0	1/2 H	0	0	1	1 H	0	1	0	2 H	0	1	1	4 H	1	0	0	8 H	1	0	1	16 H	1	1	0	32 H	1	1	1	64 H
	DCB0[2:0]			Step-up cycle for step-up circuit 1/4/5																																																																																	
	0	0	0	1/8 H																																																																																	
	0	0	1	1/4 H																																																																																	
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Status	Default Value																																																																																				
	DCB0[2:0]	DCB1[2:0]																																																																																			
Power ON Sequence	3'b011	3'b011																																																																																			
H/W Reset	3'b011	3'b011																																																																																			

8.2.62. Power Control 5 (For Partial Mode) (C4h)

C4h	PWCTRL 5 (Power Control 5)																																																																																				
	D/CX	RDX	WRX	D[15:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX																																																																								
Command	0	1	↑	XXXXXXXX	1	1	0	0	0	1	0	0	C4h																																																																								
1 st Parameter	1	1	↑	XXXXXXXX	0	DCC1[2:0]			0	DCC0[2:0]			XX																																																																								
Description	DCC0 [2:0]: Selects the operating frequency of the step-up circuit 1/4/5 for Partial mode. The higher step-up operating frequency enhances the drivability of the step-up circuit and the quality of display but increases the current consumption. Adjust the frequency taking the trade-off between the display quality and the current consumption into account. DCC1 [2:0]: Selects the operating frequency of the step-up circuit 2/3 for Partial mode. The higher step-up operating frequency enhances the drivability of the step-up circuit and the quality of display but increases the current consumption. Adjust the frequency taking the trade-off between the display quality and the current consumption into account.																																																																																				
	<table><tr><th colspan="3">DCC0[2:0]</th><th>Step-up cycle for step-up circuit 1/4/5</th></tr><tr><td>0</td><td>0</td><td>0</td><td>1/8 H</td></tr><tr><td>0</td><td>0</td><td>1</td><td>1/4 H</td></tr><tr><td>0</td><td>1</td><td>0</td><td>1/2 H</td></tr><tr><td>0</td><td>1</td><td>1</td><td>1 H</td></tr><tr><td>1</td><td>0</td><td>0</td><td>2 H</td></tr><tr><td>1</td><td>0</td><td>1</td><td>4 H</td></tr><tr><td>1</td><td>1</td><td>0</td><td>8 H</td></tr><tr><td>1</td><td>1</td><td>1</td><td>16 H</td></tr></table>				DCC0[2:0]			Step-up cycle for step-up circuit 1/4/5	0	0	0	1/8 H	0	0	1	1/4 H	0	1	0	1/2 H	0	1	1	1 H	1	0	0	2 H	1	0	1	4 H	1	1	0	8 H	1	1	1	16 H	<table><tr><th colspan="3">DCC1[2:0]</th><th>Step-up cycle for step-up circuit 2/3</th></tr><tr><td>0</td><td>0</td><td>0</td><td>1/2 H</td></tr><tr><td>0</td><td>0</td><td>1</td><td>1 H</td></tr><tr><td>0</td><td>1</td><td>0</td><td>2 H</td></tr><tr><td>0</td><td>1</td><td>1</td><td>4 H</td></tr><tr><td>1</td><td>0</td><td>0</td><td>8 H</td></tr><tr><td>1</td><td>0</td><td>1</td><td>16 H</td></tr><tr><td>1</td><td>1</td><td>0</td><td>32 H</td></tr><tr><td>1</td><td>1</td><td>1</td><td>64 H</td></tr></table>									DCC1[2:0]			Step-up cycle for step-up circuit 2/3	0	0	0	1/2 H	0	0	1	1 H	0	1	0	2 H	0	1	1	4 H	1	0	0	8 H	1	0	1	16 H	1	1	0	32 H	1	1	1	64 H
	DCC0[2:0]			Step-up cycle for step-up circuit 1/4/5																																																																																	
	0	0	0	1/8 H																																																																																	
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DCC1[2:0]			Step-up cycle for step-up circuit 2/3																																																																																		
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Status	Default Value																																																																																				
	DCC0[2:0]	DCC1[2:0]																																																																																			
Power ON Sequence	3'b011	3'b011																																																																																			
H/W Reset	3'b011	3'b011																																																																																			

8.2.63. VCOM Control (C5h)

C5h	VMCTRL (VCOM Control)												
	D/CX	RDX	WRX	D[15:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XXXXXXXX	1	1	0	0	0	1	0	1	C5h
1 st Parameter	1	↑	1	XXXXXXXX	0	0	0	0	0	0	0	nVM	XX
2 nd Parameter	1	1	↑	XXXXXXXX	VCM_REG[7:0]								XX
3 rd Parameter	1	1	↑	XXXXXXXX	VCM_REG_EN	0	0	0	0	0	0	0	XX
4 rd Parameter	1	↑	1	XXXXXXXX	VCM_OUT[7:0]								XX
Description	nVM : When the NV memory is programmed, the nVM will be set as '1' automatically. 0 : NV memory is not programmed 1 : NV memory is programmed												
	VCM_REG [7:0] is used to set factor to generate VCOM voltage from the reference voltage VREG2OUT.												
	VCM[7:0]								VCOM				
	0	0	0	0	0	0	0	0	-2				
	0	0	0	0	0	0	0	1	-1.98438				
	0	0	0	0	0	0	1	0	-1.96875				
	0	0	0	0	0	0	1	1	-1.95313				
	0	0	0	0	0	1	0	0	-1.9375				
	0	0	0	0	0	1	0	1	-1.92188				
	0	0	0	0	0	1	1	0	-1.90625				
	0	0	0	0	0	1	1	1	-1.89063				
	0	0	0	0	1	0	0	0	-1.875				
	0	0	0	0	1	0	0	1	-1.85938				
	0	0	0	0	1	0	1	0	-1.84375				
	0	0	0	0	1	0	1	1	-1.82813				
	0	0	0	0	1	1	0	0	-1.8125				
	0	0	0	0	1	1	0	1	-1.79688				
	0	0	0	0	1	1	1	0	-1.78125				
	0	0	0	0	1	1	1	1	-1.76563				
	0	0	0	1	0	0	0	0	-1.75				
	0	0	0	1	0	0	0	1	-1.73438				
	0	0	0	1	0	0	1	0	-1.71875				
	0	0	0	1	0	0	1	1	-1.70313				
	0	0	0	1	0	1	0	0	-1.6875				
	0	0	0	1	0	1	0	1	-1.67188				
	0	0	0	1	0	1	1	0	-1.65625				
	0	0	0	1	0	1	1	1	-1.64063				
	0	0	0	1	1	0	0	0	-1.625				
	0	0	0	1	1	0	0	1	-1.60938				
	0	0	0	1	1	0	1	0	-1.59375				
	0	0	0	1	1	0	1	1	-1.57813				
	0	0	0	1	1	1	0	0	-1.5625				
	0	0	0	1	1	1	0	1	-1.54688				
	0	0	0	1	1	1	1	0	-1.53125				
	0	0	0	1	1	1	1	1	-1.51563				
	0	0	1	0	0	0	0	0	-1.5				
	0	0	1	0	0	0	0	1	-1.48438				
	0	0	1	0	0	0	1	0	-1.46875				
	0	0	1	0	0	0	1	1	-1.45313				
	0	0	1	0	0	1	0	0	-1.4375				
	0	0	1	0	0	1	0	1	-1.42188				
	0	0	1	0	0	1	1	0	-1.40625				
	0	0	1	0	0	1	1	1	-1.39063				
	0	0	1	0	1	0	0	0	-1.375				
	0	0	1	0	1	0	0	1	-1.35938				
	0	0	1	0	1	0	1	0	-1.34375				
	0	0	1	0	1	0	1	1	-1.32813				

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0	0	1	0	1	1	0	0	-1.3125
0	0	1	0	1	1	0	1	-1.29688
0	0	1	0	1	1	1	0	-1.28125
0	0	1	0	1	1	1	1	-1.26563
0	0	1	1	0	0	0	0	-1.25
0	0	1	1	0	0	0	1	-1.23438
0	0	1	1	0	0	1	0	-1.21875
0	0	1	1	0	0	1	1	-1.20313
0	0	1	1	0	1	0	0	-1.1875
0	0	1	1	0	1	0	1	-1.17188
0	0	1	1	0	1	1	0	-1.15625
0	0	1	1	0	1	1	1	-1.14063
0	0	1	1	1	0	0	0	-1.125
0	0	1	1	1	0	0	1	-1.10938
0	0	1	1	1	0	1	0	-1.09375
0	0	1	1	1	0	1	1	-1.07813
0	0	1	1	1	1	0	0	-1.0625
0	0	1	1	1	1	0	1	-1.04688
0	0	1	1	1	1	1	0	-1.03125
0	0	1	1	1	1	1	1	-1.01563
0	1	0	0	0	0	0	0	-1
0	1	0	0	0	0	0	1	-0.98438
0	1	0	0	0	0	1	0	-0.96875
0	1	0	0	0	0	1	1	-0.95313
0	1	0	0	0	1	0	0	-0.9375
0	1	0	0	0	1	0	1	-0.92188
0	1	0	0	0	1	1	0	-0.90625
0	1	0	0	0	1	1	1	-0.89063
0	1	0	0	1	0	0	0	-0.875
0	1	0	0	1	0	0	1	-0.85938
0	1	0	0	1	0	1	0	-0.84375
0	1	0	0	1	0	1	1	-0.82813
0	1	0	0	1	1	0	0	-0.8125
0	1	0	0	1	1	0	1	-0.79688
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0	1	0	0	1	1	1	1	-0.76563
0	1	0	1	0	0	0	0	-0.75
0	1	0	1	0	0	0	1	-0.73438
0	1	0	1	0	0	1	0	-0.71875
0	1	0	1	0	0	1	1	-0.70313
0	1	0	1	0	1	0	0	-0.6875
0	1	0	1	0	1	0	1	-0.67188
0	1	0	1	0	1	1	0	-0.65625
0	1	0	1	0	1	1	1	-0.64063
0	1	0	1	1	0	0	0	-0.625
0	1	0	1	1	0	0	1	-0.60938
0	1	0	1	1	0	1	0	-0.59375
0	1	0	1	1	0	1	1	-0.57813
0	1	0	1	1	1	0	0	-0.5625
0	1	0	1	1	1	0	1	-0.54688
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0	1	0	1	1	1	1	1	-0.51563
0	1	1	0	0	0	0	0	-0.5
0	1	1	0	0	0	0	1	-0.48438
0	1	1	0	0	0	1	0	-0.46875
0	1	1	0	0	0	1	1	-0.45313
0	1	1	0	0	1	0	0	-0.4375
0	1	1	0	0	1	0	1	-0.42188

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	<table><tr><td>0</td><td>1</td><td>1</td><td>0</td><td>0</td><td>1</td><td>1</td><td>0</td><td>-0.40625</td></tr><tr><td>0</td><td>1</td><td>1</td><td>0</td><td>0</td><td>1</td><td>1</td><td>1</td><td>-0.39063</td></tr><tr><td>0</td><td>1</td><td>1</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td><td>-0.375</td></tr><tr><td>0</td><td>1</td><td>1</td><td>0</td><td>1</td><td>0</td><td>0</td><td>1</td><td>-0.35938</td></tr><tr><td>0</td><td>1</td><td>1</td><td>0</td><td>1</td><td>0</td><td>1</td><td>0</td><td>-0.34375</td></tr><tr><td>0</td><td>1</td><td>1</td><td>0</td><td>1</td><td>0</td><td>1</td><td>1</td><td>-0.32813</td></tr><tr><td>0</td><td>1</td><td>1</td><td>0</td><td>1</td><td>1</td><td>0</td><td>0</td><td>-0.3125</td></tr><tr><td>0</td><td>1</td><td>1</td><td>0</td><td>1</td><td>1</td><td>0</td><td>1</td><td>-0.29688</td></tr><tr><td>0</td><td>1</td><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td><td>0</td><td>-0.28125</td></tr><tr><td>0</td><td>1</td><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td><td>1</td><td>-0.26563</td></tr><tr><td>0</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>-0.25</td></tr><tr><td>0</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td><td>0</td><td>1</td><td>-0.23438</td></tr><tr><td>0</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td><td>1</td><td>0</td><td>-0.21875</td></tr><tr><td>0</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td><td>1</td><td>1</td><td>-0.20313</td></tr><tr><td>0</td><td>1</td><td>1</td><td>1</td><td>0</td><td>1</td><td>0</td><td>0</td><td>-0.1875</td></tr><tr><td>0</td><td>1</td><td>1</td><td>1</td><td>0</td><td>1</td><td>0</td><td>1</td><td>-0.17188</td></tr><tr><td>0</td><td>1</td><td>1</td><td>1</td><td>0</td><td>1</td><td>1</td><td>0</td><td>-0.15625</td></tr><tr><td>0</td><td>1</td><td>1</td><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td><td>-0.14063</td></tr><tr><td>0</td><td>1</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td><td>0</td><td>-0.125</td></tr><tr><td>0</td><td>1</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td><td>1</td><td>-0.10938</td></tr><tr><td>0</td><td>1</td><td>1</td><td>1</td><td>1</td><td>0</td><td>1</td><td>0</td><td>-0.09375</td></tr><tr><td>0</td><td>1</td><td>1</td><td>1</td><td>1</td><td>0</td><td>1</td><td>1</td><td>-0.07813</td></tr><tr><td>0</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td><td>-0.0625</td></tr><tr><td>0</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>0</td><td>1</td><td>-0.04688</td></tr><tr><td>0</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>0</td><td>-0.03125</td></tr><tr><td>0</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>-0.01563</td></tr><tr><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td colspan="8">10000001~11111110</td><td>Inhibit</td></tr><tr><td colspan="8">11111111</td><td>Halt</td></tr></table> VCM_REG_EN: Select the Vcom value from VCM_REG [7:0] or NV memory. 0: VCOM value from NV memory. 1: VCOM value from VCM_REG [7:0]. VCM_OUT [7:0]: NV memory programmed value.	0	1	1	0	0	1	1	0	-0.40625	0	1	1	0	0	1	1	1	-0.39063	0	1	1	0	1	0	0	0	-0.375	0	1	1	0	1	0	0	1	-0.35938	0	1	1	0	1	0	1	0	-0.34375	0	1	1	0	1	0	1	1	-0.32813	0	1	1	0	1	1	0	0	-0.3125	0	1	1	0	1	1	0	1	-0.29688	0	1	1	0	1	1	1	0	-0.28125	0	1	1	0	1	1	1	1	-0.26563	0	1	1	1	0	0	0	0	-0.25	0	1	1	1	0	0	0	1	-0.23438	0	1	1	1	0	0	1	0	-0.21875	0	1	1	1	0	0	1	1	-0.20313	0	1	1	1	0	1	0	0	-0.1875	0	1	1	1	0	1	0	1	-0.17188	0	1	1	1	0	1	1	0	-0.15625	0	1	1	1	0	1	1	1	-0.14063	0	1	1	1	1	0	0	0	-0.125	0	1	1	1	1	0	0	1	-0.10938	0	1	1	1	1	0	1	0	-0.09375	0	1	1	1	1	0	1	1	-0.07813	0	1	1	1	1	1	0	0	-0.0625	0	1	1	1	1	1	0	1	-0.04688	0	1	1	1	1	1	1	0	-0.03125	0	1	1	1	1	1	1	1	-0.01563	1	0	0	0	0	0	0	0	0	10000001~11111110								Inhibit	11111111								Halt
0	1	1	0	0	1	1	0	-0.40625																																																																																																																																																																																																																																																														
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0	1	1	0	1	0	0	0	-0.375																																																																																																																																																																																																																																																														
0	1	1	0	1	0	0	1	-0.35938																																																																																																																																																																																																																																																														
0	1	1	0	1	0	1	0	-0.34375																																																																																																																																																																																																																																																														
0	1	1	0	1	0	1	1	-0.32813																																																																																																																																																																																																																																																														
0	1	1	0	1	1	0	0	-0.3125																																																																																																																																																																																																																																																														
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0	1	1	0	1	1	1	1	-0.26563																																																																																																																																																																																																																																																														
0	1	1	1	0	0	0	0	-0.25																																																																																																																																																																																																																																																														
0	1	1	1	0	0	0	1	-0.23438																																																																																																																																																																																																																																																														
0	1	1	1	0	0	1	0	-0.21875																																																																																																																																																																																																																																																														
0	1	1	1	0	0	1	1	-0.20313																																																																																																																																																																																																																																																														
0	1	1	1	0	1	0	0	-0.1875																																																																																																																																																																																																																																																														
0	1	1	1	0	1	0	1	-0.17188																																																																																																																																																																																																																																																														
0	1	1	1	0	1	1	0	-0.15625																																																																																																																																																																																																																																																														
0	1	1	1	0	1	1	1	-0.14063																																																																																																																																																																																																																																																														
0	1	1	1	1	0	0	0	-0.125																																																																																																																																																																																																																																																														
0	1	1	1	1	0	0	1	-0.10938																																																																																																																																																																																																																																																														
0	1	1	1	1	0	1	0	-0.09375																																																																																																																																																																																																																																																														
0	1	1	1	1	0	1	1	-0.07813																																																																																																																																																																																																																																																														
0	1	1	1	1	1	0	0	-0.0625																																																																																																																																																																																																																																																														
0	1	1	1	1	1	0	1	-0.04688																																																																																																																																																																																																																																																														
0	1	1	1	1	1	1	0	-0.03125																																																																																																																																																																																																																																																														
0	1	1	1	1	1	1	1	-0.01563																																																																																																																																																																																																																																																														
1	0	0	0	0	0	0	0	0																																																																																																																																																																																																																																																														
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Register Availability	<table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode ON, Idle Mode OFF, Sleep OUT</td><td>Yes</td></tr><tr><td>Normal Mode ON, Idle Mode ON, Sleep OUT</td><td>Yes</td></tr><tr><td>Partial Mode ON, Idle Mode OFF, Sleep OUT</td><td>Yes</td></tr><tr><td>Partial Mode ON, Idle Mode ON, Sleep OUT</td><td>Yes</td></tr><tr><td>Sleep IN</td><td>Yes</td></tr></table>	Status	Availability	Normal Mode ON, Idle Mode OFF, Sleep OUT	Yes	Normal Mode ON, Idle Mode ON, Sleep OUT	Yes	Partial Mode ON, Idle Mode OFF, Sleep OUT	Yes	Partial Mode ON, Idle Mode ON, Sleep OUT	Yes	Sleep IN	Yes																																																																																																																																																																																																																																																									
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Partial Mode ON, Idle Mode ON, Sleep OUT	Yes																																																																																																																																																																																																																																																																					
Sleep IN	Yes																																																																																																																																																																																																																																																																					
Default	<table><tr><th rowspan="2">Status</th><th colspan="4">Default Value</th></tr><tr><th>VCM_OUT[7:0]</th><th>VCM_REG_EN</th><th>VCM_REG[7:0]</th><th>nVM</th></tr><tr><td>Power ON Sequence</td><td>8'bXXXXXXXX</td><td>1'b0</td><td>8'b01100000</td><td>X</td></tr><tr><td>H/W Reset</td><td>8'bXXXXXXXX</td><td>1'b0</td><td>8'b01100000</td><td>X</td></tr></table>	Status	Default Value				VCM_OUT[7:0]	VCM_REG_EN	VCM_REG[7:0]	nVM	Power ON Sequence	8'bXXXXXXXX	1'b0	8'b01100000	X	H/W Reset	8'bXXXXXXXX	1'b0	8'b01100000	X																																																																																																																																																																																																																																																		
Status	Default Value																																																																																																																																																																																																																																																																					
	VCM_OUT[7:0]	VCM_REG_EN	VCM_REG[7:0]	nVM																																																																																																																																																																																																																																																																		
Power ON Sequence	8'bXXXXXXXX	1'b0	8'b01100000	X																																																																																																																																																																																																																																																																		
H/W Reset	8'bXXXXXXXX	1'b0	8'b01100000	X																																																																																																																																																																																																																																																																		

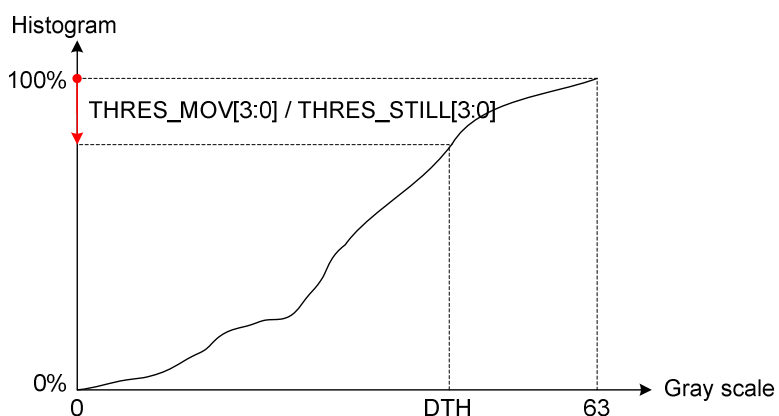
8.2.64. CABC Control 1 (C6h)

C6h	CABCCTRL9 (CABC Control 9)																								
	D/CX	RDX	WRX	D[15:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX												
Command	0	1	↑	XXXXXXXX	1	1	0	0	0	1	1	0	C6h												
1 st Parameter	1	1	↑	XXXXXXXX	SCD_VLINE[7:0]								XX												
2 nd Parameter	1	1	↑	XXXXXXXX	0	0	0	0	0	SCD_VLINE[10:8]			XX												
Description	SCD_VLINE [10:0]: This parameter is used set the display line per frame while partial mode ON.																								
	SCD_VLINE[8:0]											Display line													
	D10	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0														
	0	0	0	0	0	0	0	0	0	0	0	Setting prohibited													
	0	0	0	0	0	0	0	0	0	0	1	1 line													
	0	0	0	0	0	0	0	0	0	1	0	2 lines													
	0	0	0	0	0	0	0	0	0	1	1	3 lines													
	0	0	0	0	0	0	0	1	0	0	0	4 lines													
	:											:													
	:											:													
	0	0	1	1	1	0	1	1	1	0	1	477 lines													
	0	0	1	1	1	0	1	1	1	1	0	478 lines													
	0	0	1	1	1	0	1	1	1	1	1	479 lines													
	0	0	1	1	1	1	0	0	0	0	0	480 lines													
	Others											Setting prohibited													
	Restriction																								
	Register Availability	<table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode ON, Idle Mode OFF, Sleep OUT</td><td>Yes</td></tr><tr><td>Normal Mode ON, Idle Mode ON, Sleep OUT</td><td>Yes</td></tr><tr><td>Partial Mode ON, Idle Mode OFF, Sleep OUT</td><td>Yes</td></tr><tr><td>Partial Mode ON, Idle Mode ON, Sleep OUT</td><td>Yes</td></tr><tr><td>Sleep IN</td><td>Yes</td></tr></table>													Status	Availability	Normal Mode ON, Idle Mode OFF, Sleep OUT	Yes	Normal Mode ON, Idle Mode ON, Sleep OUT	Yes	Partial Mode ON, Idle Mode OFF, Sleep OUT	Yes	Partial Mode ON, Idle Mode ON, Sleep OUT	Yes	Sleep IN
Status		Availability																							
Normal Mode ON, Idle Mode OFF, Sleep OUT		Yes																							
Normal Mode ON, Idle Mode ON, Sleep OUT		Yes																							
Partial Mode ON, Idle Mode OFF, Sleep OUT		Yes																							
Partial Mode ON, Idle Mode ON, Sleep OUT		Yes																							
Sleep IN	Yes																								
Default	<table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power ON Sequence</td><td>11'b00111100000</td></tr><tr><td>S/W Reset</td><td>11'b00111100000</td></tr><tr><td>H/W Reset</td><td>11'b00111100000</td></tr></table>													Status	Default Value	Power ON Sequence	11'b00111100000	S/W Reset	11'b00111100000	H/W Reset	11'b00111100000				
	Status	Default Value																							
	Power ON Sequence	11'b00111100000																							
	S/W Reset	11'b00111100000																							
H/W Reset	11'b00111100000																								

8.2.65. CABC Control 2 (C8h)

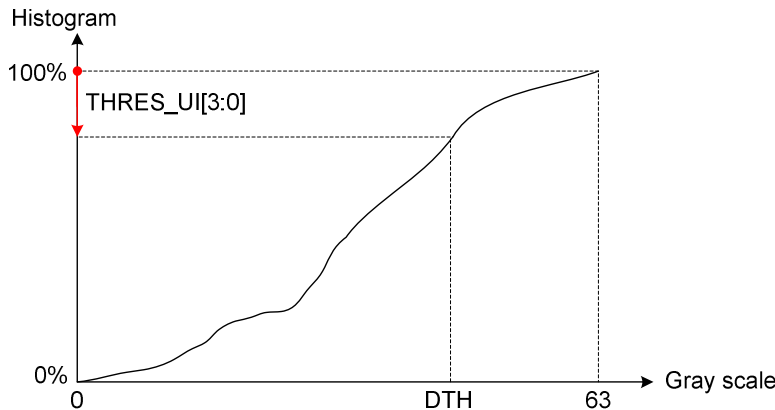
C8h	CABCCTRL1 (CABC Control 1)																															
	D/CX	RDX	WRX	D[15:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX																			
Command	0	1	↑	XXXXXXXX	1	1	0	0	1	0	0	0	C8h																			
1 st Parameter	1	1	↑	XXXXXXXX	0	0	0	0	0	LEDONR	LEDONPOL	PWMPOL	XX																			
Description	PWMPOL: The bit is used to define polarity of CABC_PWM signal.																															
	<table><tr><td>BL</td><td>LEDPWMPOL</td><td>CABC_PWM pin</td></tr><tr><td>0</td><td>0</td><td>Always low</td></tr><tr><td>0</td><td>1</td><td>Always high</td></tr><tr><td>1</td><td>0</td><td>Original polarity of PWM signal</td></tr><tr><td>1</td><td>1</td><td>Inversed polarity of PWM signal</td></tr></table>													BL	LEDPWMPOL	CABC_PWM pin	0	0	Always low	0	1	Always high	1	0	Original polarity of PWM signal	1	1	Inversed polarity of PWM signal				
	BL	LEDPWMPOL	CABC_PWM pin																													
	0	0	Always low																													
	0	1	Always high																													
1	0	Original polarity of PWM signal																														
1	1	Inversed polarity of PWM signal																														
LEDONPOL: This bit is used to control CABC_ON pin.																																
<table><tr><td>BL</td><td>LEDONPOL</td><td>CABC_ON pin</td></tr><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>LEDONR</td></tr><tr><td>1</td><td>1</td><td>Inversed LEDONR</td></tr></table>													BL	LEDONPOL	CABC_ON pin	0	0	0	0	1	1	1	0	LEDONR	1	1	Inversed LEDONR					
BL	LEDONPOL	CABC_ON pin																														
0	0	0																														
0	1	1																														
1	0	LEDONR																														
1	1	Inversed LEDONR																														
LEDONR: This bit is used to control CABC_ON pin.																																
<table><tr><td>LEDONR</td><td>Description</td></tr><tr><td>0</td><td>Low</td></tr><tr><td>1</td><td>High</td></tr></table>													LEDONR	Description	0	Low	1	High														
LEDONR	Description																															
0	Low																															
1	High																															
Restriction																																
Register Availability	<table><tr><td>Status</td><td>Availability</td></tr><tr><td>Normal Mode ON, Idle Mode OFF, Sleep OUT</td><td>Yes</td></tr><tr><td>Normal Mode ON, Idle Mode ON, Sleep OUT</td><td>Yes</td></tr><tr><td>Partial Mode ON, Idle Mode OFF, Sleep OUT</td><td>Yes</td></tr><tr><td>Partial Mode ON, Idle Mode ON, Sleep OUT</td><td>Yes</td></tr><tr><td>Sleep IN</td><td>Yes</td></tr></table>													Status	Availability	Normal Mode ON, Idle Mode OFF, Sleep OUT	Yes	Normal Mode ON, Idle Mode ON, Sleep OUT	Yes	Partial Mode ON, Idle Mode OFF, Sleep OUT	Yes	Partial Mode ON, Idle Mode ON, Sleep OUT	Yes	Sleep IN	Yes							
Status	Availability																															
Normal Mode ON, Idle Mode OFF, Sleep OUT	Yes																															
Normal Mode ON, Idle Mode ON, Sleep OUT	Yes																															
Partial Mode ON, Idle Mode OFF, Sleep OUT	Yes																															
Partial Mode ON, Idle Mode ON, Sleep OUT	Yes																															
Sleep IN	Yes																															
Default	<table><tr><td rowspan="2">Status</td><td colspan="3">Default Value</td></tr><tr><td>LEDONR</td><td>LEDONPOL</td><td>LEDPWMPOL</td></tr><tr><td>Power On Sequence</td><td>1'b0</td><td>1'b0</td><td>1'b0</td></tr><tr><td>SW Reset</td><td>No change</td><td>No change</td><td>No change</td></tr><tr><td>HW Reset</td><td>1'b0</td><td>1'b0</td><td>1'b0</td></tr></table>													Status	Default Value			LEDONR	LEDONPOL	LEDPWMPOL	Power On Sequence	1'b0	1'b0	1'b0	SW Reset	No change	No change	No change	HW Reset	1'b0	1'b0	1'b0
Status	Default Value																															
	LEDONR	LEDONPOL	LEDPWMPOL																													
Power On Sequence	1'b0	1'b0	1'b0																													
SW Reset	No change	No change	No change																													
HW Reset	1'b0	1'b0	1'b0																													

8.2.66. CABC Control 3 (C9h)

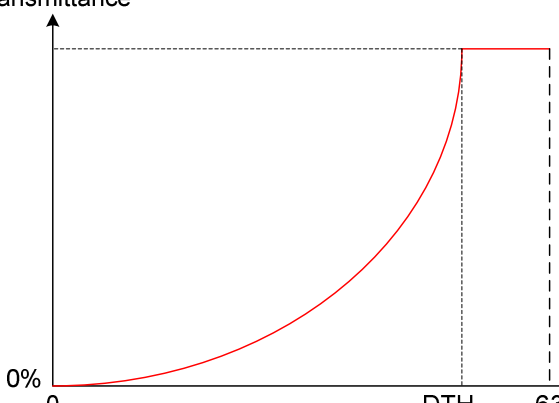
C9h	CABCCTRL2 (CABC Control 2)																																																																																																											
	D/CX	RDX	WRX	D[15:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX																																																																																															
Command	0	1	↑	XXXXXXXX	1	1	0	0	1	0	0	1	C9h																																																																																															
1 st Parameter	1	1	↑	XXXXXXXX	THRES_MOV[3:0]				THRES_STILL[3:0]				XX																																																																																															
Description	THRES_MOV [3:0]: This parameter is used to set the ratio (percentage) of the maximum number of pixels that makes display image white (data="63") to the total of pixels by image process in MOVING image mode. After this parameter sets the number of pixels that makes display image white, threshold grayscale value (DTH) that makes display image white is set so that the number of the pixels set by this parameter does not change.																																																																																																											
	<table><tr><th colspan="4">THRES_MOV[3:0]</th><th rowspan="2">Description</th></tr><tr><th>D3</th><th>D2</th><th>D1</th><th>D0</th></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td><td>99 %</td></tr><tr><td>0</td><td>0</td><td>0</td><td>1</td><td>98 %</td></tr><tr><td>0</td><td>0</td><td>1</td><td>0</td><td>96 %</td></tr><tr><td>0</td><td>0</td><td>1</td><td>1</td><td>94 %</td></tr><tr><td>0</td><td>1</td><td>0</td><td>0</td><td>92 %</td></tr><tr><td>0</td><td>1</td><td>0</td><td>1</td><td>90 %</td></tr><tr><td>0</td><td>1</td><td>1</td><td>0</td><td>88 %</td></tr><tr><td>0</td><td>1</td><td>1</td><td>1</td><td>86 %</td></tr></table>					THRES_MOV[3:0]				Description	D3	D2	D1	D0	0	0	0	0	99 %	0	0	0	1	98 %	0	0	1	0	96 %	0	0	1	1	94 %	0	1	0	0	92 %	0	1	0	1	90 %	0	1	1	0	88 %	0	1	1	1	86 %	<table><tr><th colspan="4">THRES_MOV[3:0]</th><th rowspan="2">Description</th></tr><tr><th>D3</th><th>D2</th><th>D1</th><th>D0</th></tr><tr><td>1</td><td>0</td><td>0</td><td>0</td><td>84 %</td></tr><tr><td>1</td><td>0</td><td>0</td><td>1</td><td>82 %</td></tr><tr><td>1</td><td>0</td><td>1</td><td>0</td><td>80 %</td></tr><tr><td>1</td><td>0</td><td>1</td><td>1</td><td>78 %</td></tr><tr><td>1</td><td>1</td><td>0</td><td>0</td><td>76 %</td></tr><tr><td>1</td><td>1</td><td>0</td><td>1</td><td>74 %</td></tr><tr><td>1</td><td>1</td><td>1</td><td>0</td><td>72 %</td></tr><tr><td>1</td><td>1</td><td>1</td><td>1</td><td>70 %</td></tr></table>					THRES_MOV[3:0]				Description	D3	D2	D1	D0	1	0	0	0	84 %	1	0	0	1	82 %	1	0	1	0	80 %	1	0	1	1	78 %	1	1	0	0	76 %	1	1	0	1	74 %	1	1	1	0	72 %	1	1	1	1	70 %
	THRES_MOV[3:0]				Description																																																																																																							
	D3	D2	D1	D0																																																																																																								
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1	1	1	1	70 %																																																																																																								
THRES_STILL [3:0]: This parameter is used to set the ratio (percentage) of the maximum number of pixels that makes display image white (data="63") to the total of pixels by image process in STILL mode. After this parameter sets the number of pixels that makes display image white, threshold grayscale value (DTH) that makes display image white is set so that the number of the pixels set by this parameter does not change.																																																																																																												
<table><tr><th colspan="4">THRES_STILL[3:0]</th><th rowspan="2">Description</th></tr><tr><th>D3</th><th>D2</th><th>D1</th><th>D0</th></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td><td>99 %</td></tr><tr><td>0</td><td>0</td><td>0</td><td>1</td><td>98 %</td></tr><tr><td>0</td><td>0</td><td>1</td><td>0</td><td>96 %</td></tr><tr><td>0</td><td>0</td><td>1</td><td>1</td><td>94 %</td></tr><tr><td>0</td><td>1</td><td>0</td><td>0</td><td>92 %</td></tr><tr><td>0</td><td>1</td><td>0</td><td>1</td><td>90 %</td></tr><tr><td>0</td><td>1</td><td>1</td><td>0</td><td>88 %</td></tr><tr><td>0</td><td>1</td><td>1</td><td>1</td><td>86 %</td></tr></table>					THRES_STILL[3:0]				Description	D3	D2	D1	D0	0	0	0	0	99 %	0	0	0	1	98 %	0	0	1	0	96 %	0	0	1	1	94 %	0	1	0	0	92 %	0	1	0	1	90 %	0	1	1	0	88 %	0	1	1	1	86 %	<table><tr><th colspan="4">THRES_STILL[3:0]</th><th rowspan="2">Description</th></tr><tr><th>D3</th><th>D2</th><th>D1</th><th>D0</th></tr><tr><td>1</td><td>0</td><td>0</td><td>0</td><td>84 %</td></tr><tr><td>1</td><td>0</td><td>0</td><td>1</td><td>82 %</td></tr><tr><td>1</td><td>0</td><td>1</td><td>0</td><td>80 %</td></tr><tr><td>1</td><td>0</td><td>1</td><td>1</td><td>78 %</td></tr><tr><td>1</td><td>1</td><td>0</td><td>0</td><td>76 %</td></tr><tr><td>1</td><td>1</td><td>0</td><td>1</td><td>74 %</td></tr><tr><td>1</td><td>1</td><td>1</td><td>0</td><td>72 %</td></tr><tr><td>1</td><td>1</td><td>1</td><td>1</td><td>70 %</td></tr></table>					THRES_STILL[3:0]				Description	D3	D2	D1	D0	1	0	0	0	84 %	1	0	0	1	82 %	1	0	1	0	80 %	1	0	1	1	78 %	1	1	0	0	76 %	1	1	0	1	74 %	1	1	1	0	72 %	1	1	1	1	70 %	
THRES_STILL[3:0]				Description																																																																																																								
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Restriction																																																																																																												

Register Availability	<table><tr><th>Status</th><th colspan="2">Availability</th></tr><tr><td>Normal Mode ON, Idle Mode OFF, Sleep OUT</td><td colspan="2">Yes</td></tr><tr><td>Normal Mode ON, Idle Mode ON, Sleep OUT</td><td colspan="2">Yes</td></tr><tr><td>Partial Mode ON, Idle Mode OFF, Sleep OUT</td><td colspan="2">Yes</td></tr><tr><td>Partial Mode ON, Idle Mode ON, Sleep OUT</td><td colspan="2">Yes</td></tr><tr><td>Sleep IN</td><td colspan="2">Yes</td></tr></table>			Status	Availability		Normal Mode ON, Idle Mode OFF, Sleep OUT	Yes		Normal Mode ON, Idle Mode ON, Sleep OUT	Yes		Partial Mode ON, Idle Mode OFF, Sleep OUT	Yes		Partial Mode ON, Idle Mode ON, Sleep OUT	Yes		Sleep IN	Yes	
	Status	Availability																			
	Normal Mode ON, Idle Mode OFF, Sleep OUT	Yes																			
	Normal Mode ON, Idle Mode ON, Sleep OUT	Yes																			
	Partial Mode ON, Idle Mode OFF, Sleep OUT	Yes																			
	Partial Mode ON, Idle Mode ON, Sleep OUT	Yes																			
Sleep IN	Yes																				
Default	<table><tr><th rowspan="2">Status</th><th colspan="2">Default Value</th></tr><tr><th>THRES_MOV[3:0]</th><th>THRES_STILL[3:0]</th></tr><tr><td>Power ON Sequence</td><td>4'b1011 b</td><td>4'b1011 b</td></tr><tr><td>S/W Reset</td><td>4'b1011 b</td><td>4'b1011 b</td></tr><tr><td>H/W Reset</td><td>4'b1011 b</td><td>4'b1011 b</td></tr></table>	Status	Default Value		THRES_MOV[3:0]	THRES_STILL[3:0]	Power ON Sequence	4'b1011 b	4'b1011 b	S/W Reset	4'b1011 b	4'b1011 b	H/W Reset	4'b1011 b	4'b1011 b						
			Status	Default Value																	
		THRES_MOV[3:0]		THRES_STILL[3:0]																	
		Power ON Sequence	4'b1011 b	4'b1011 b																	
		S/W Reset	4'b1011 b	4'b1011 b																	
H/W Reset	4'b1011 b	4'b1011 b																			

8.2.67. CABC Control 4 (CAh)

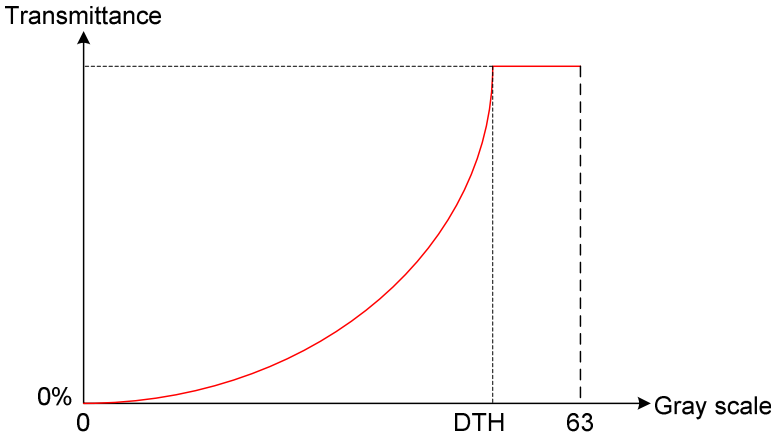
CAh	CABCCTRL3 (CABC Control 3)																																																																																																											
	D/CX	RDX	WRX	D[15:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX																																																																																															
Command	0	1	↑	XXXXXXXX	1	1	0	0	1	0	1	0	CAh																																																																																															
1 st Parameter	1	1	↑	XXXXXXXX	0	0	0	0	THRES_UI[3:0]			XX																																																																																																
Description	THRES_UI [3:0]: This parameter is used to set the ratio (percentage) of the maximum number of pixels that makes display image white (data="63") to the total of pixels by image process in USER INTERFACE mode. After this parameter sets the number of pixels that makes display image white, threshold grayscale value (DTH) that makes display image white is set so that the number of the pixels set by this parameter does not change.																																																																																																											
	<table><tr><th colspan="4">THRES_UI[3:0]</th><th rowspan="2">Description</th></tr><tr><th>D3</th><th>D2</th><th>D1</th><th>D0</th></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td><td>99 %</td></tr><tr><td>0</td><td>0</td><td>0</td><td>1</td><td>98 %</td></tr><tr><td>0</td><td>0</td><td>1</td><td>0</td><td>96 %</td></tr><tr><td>0</td><td>0</td><td>1</td><td>1</td><td>94 %</td></tr><tr><td>0</td><td>1</td><td>0</td><td>0</td><td>92 %</td></tr><tr><td>0</td><td>1</td><td>0</td><td>1</td><td>90 %</td></tr><tr><td>0</td><td>1</td><td>1</td><td>0</td><td>88 %</td></tr><tr><td>0</td><td>1</td><td>1</td><td>1</td><td>86 %</td></tr></table>					THRES_UI[3:0]				Description	D3	D2	D1	D0	0	0	0	0	99 %	0	0	0	1	98 %	0	0	1	0	96 %	0	0	1	1	94 %	0	1	0	0	92 %	0	1	0	1	90 %	0	1	1	0	88 %	0	1	1	1	86 %	<table><tr><th colspan="4">THRES_UI[3:0]</th><th rowspan="2">Description</th></tr><tr><th>D3</th><th>D2</th><th>D1</th><th>D0</th></tr><tr><td>1</td><td>0</td><td>0</td><td>0</td><td>84 %</td></tr><tr><td>1</td><td>0</td><td>0</td><td>1</td><td>82 %</td></tr><tr><td>1</td><td>0</td><td>1</td><td>0</td><td>80 %</td></tr><tr><td>1</td><td>0</td><td>1</td><td>1</td><td>78 %</td></tr><tr><td>1</td><td>1</td><td>0</td><td>0</td><td>76 %</td></tr><tr><td>1</td><td>1</td><td>0</td><td>1</td><td>74 %</td></tr><tr><td>1</td><td>1</td><td>1</td><td>0</td><td>72 %</td></tr><tr><td>1</td><td>1</td><td>1</td><td>1</td><td>70 %</td></tr></table>					THRES_UI[3:0]				Description	D3	D2	D1	D0	1	0	0	0	84 %	1	0	0	1	82 %	1	0	1	0	80 %	1	0	1	1	78 %	1	1	0	0	76 %	1	1	0	1	74 %	1	1	1	0	72 %	1	1	1	1	70 %
	THRES_UI[3:0]				Description																																																																																																							
	D3	D2	D1	D0																																																																																																								
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Restriction																																																																																																												
Register Availability	<table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode ON, Idle Mode OFF, Sleep OUT</td><td>Yes</td></tr><tr><td>Normal Mode ON, Idle Mode ON, Sleep OUT</td><td>Yes</td></tr><tr><td>Partial Mode ON, Idle Mode OFF, Sleep OUT</td><td>Yes</td></tr><tr><td>Partial Mode ON, Idle Mode ON, Sleep OUT</td><td>Yes</td></tr><tr><td>Sleep IN</td><td>Yes</td></tr></table>													Status	Availability	Normal Mode ON, Idle Mode OFF, Sleep OUT	Yes	Normal Mode ON, Idle Mode ON, Sleep OUT	Yes	Partial Mode ON, Idle Mode OFF, Sleep OUT	Yes	Partial Mode ON, Idle Mode ON, Sleep OUT	Yes	Sleep IN	Yes																																																																																			
Status	Availability																																																																																																											
Normal Mode ON, Idle Mode OFF, Sleep OUT	Yes																																																																																																											
Normal Mode ON, Idle Mode ON, Sleep OUT	Yes																																																																																																											
Partial Mode ON, Idle Mode OFF, Sleep OUT	Yes																																																																																																											
Partial Mode ON, Idle Mode ON, Sleep OUT	Yes																																																																																																											
Sleep IN	Yes																																																																																																											
Default	<table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power ON Sequence</td><td>4'b1011 b</td></tr><tr><td>S/W Reset</td><td>4'b1011 b</td></tr><tr><td>H/W Reset</td><td>4'b1011 b</td></tr></table>													Status	Default Value	Power ON Sequence	4'b1011 b	S/W Reset	4'b1011 b	H/W Reset	4'b1011 b																																																																																							
Status	Default Value																																																																																																											
Power ON Sequence	4'b1011 b																																																																																																											
S/W Reset	4'b1011 b																																																																																																											
H/W Reset	4'b1011 b																																																																																																											

8.2.68. CABC Control 5 (CBh)

CBh	CABCCTRL4 (CABC Control 4)																								
	D/CX	RDX	WRX	D[15:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX												
Command	0	1	↑	XXXXXXXX	1	1	0	0	1	0	1	1	CBh												
1 st Parameter	1	1	↑	XXXXXXXX	DTH_MOV[3:0]				DTH_STILL[3:0]				XX												
Description	DTH_MOV [3:0]: This parameter is used set the minimum limitation of grayscale threshold value in MOVING image mode.																								
	DTH_MOV[3:0]				Description	DTH_MOV[3:0]				Description															
	D3	D2	D1	D0		D3	D2	D1	D0																
	0	0	0	0	224	1	0	0	0	192															
	0	0	0	1	220	1	0	0	1	188															
	0	0	1	0	216	1	0	1	0	184															
	0	0	1	1	212	1	0	1	1	180															
	0	1	0	0	208	1	1	0	0	176															
	0	1	0	1	204	1	1	0	1	172															
	0	1	1	0	200	1	1	1	0	168															
	0	1	1	1	196	1	1	1	1	164															
DTH_OPT [2:0]: This parameter is used to set the minimum limitation of grayscale threshold value in STILL image mode.																									
DTH_STILL[3:0]				Description	DTH_STILL[3:0]				Description																
D3	D2	D1	D0		D3	D2	D1	D0																	
0	0	0	0	224	1	0	0	0	192																
0	0	0	1	220	1	0	0	1	188																
0	0	1	0	216	1	0	1	0	184																
0	0	1	1	212	1	0	1	1	180																
0	1	0	0	208	1	1	0	0	176																
0	1	0	1	204	1	1	0	1	172																
0	1	1	0	200	1	1	1	0	168																
0	1	1	1	196	1	1	1	1	164																
Transmittance  0%0DTH63→ Gray scale																									
Restriction																									
Register Availability	<table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode ON, Idle Mode OFF, Sleep OUT</td><td>Yes</td></tr><tr><td>Normal Mode ON, Idle Mode ON, Sleep OUT</td><td>Yes</td></tr><tr><td>Partial Mode ON, Idle Mode OFF, Sleep OUT</td><td>Yes</td></tr><tr><td>Partial Mode ON, Idle Mode ON, Sleep OUT</td><td>Yes</td></tr><tr><td>Sleep IN</td><td>Yes</td></tr></table>													Status	Availability	Normal Mode ON, Idle Mode OFF, Sleep OUT	Yes	Normal Mode ON, Idle Mode ON, Sleep OUT	Yes	Partial Mode ON, Idle Mode OFF, Sleep OUT	Yes	Partial Mode ON, Idle Mode ON, Sleep OUT	Yes	Sleep IN	Yes
Status	Availability																								
Normal Mode ON, Idle Mode OFF, Sleep OUT	Yes																								
Normal Mode ON, Idle Mode ON, Sleep OUT	Yes																								
Partial Mode ON, Idle Mode OFF, Sleep OUT	Yes																								
Partial Mode ON, Idle Mode ON, Sleep OUT	Yes																								
Sleep IN	Yes																								

Default		
	Status	Default Value
		DTH_MOV[3:0] DTH_STILL[3:0]
	Power ON Sequence	4'b1010 b 4'b1000 b
	S/W Reset	4'b1010 b 4'b1000 b
	H/W Reset	4'b1010 b 4'b1000 b

8.2.69. CABC Control 6 (CCh)

CCh	CABCCTRL5 (CABC Control 5)																																																																																																									
	D/CX	RDX	WRX	D[15:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX																																																																																													
Command	0	1	↑	XXXXXXXX	1	1	0	0	1	1	0	0	CCh																																																																																													
1 st Parameter	1	1	↑	XXXXXXXX	0	0	0	0	DTH_UI[3:0]			XX																																																																																														
Description	DTH_UI [3:0]: This parameter is used set the minimum limitation of grayscale threshold value in USER INTERFACE mode.																																																																																																									
	<table><tr><th colspan="4">DTH_UI[3:0]</th><th rowspan="2">Description</th></tr><tr><th>D3</th><th>D2</th><th>D1</th><th>D0</th></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td><td>252</td></tr><tr><td>0</td><td>0</td><td>0</td><td>1</td><td>248</td></tr><tr><td>0</td><td>0</td><td>1</td><td>0</td><td>244</td></tr><tr><td>0</td><td>0</td><td>1</td><td>1</td><td>240</td></tr><tr><td>0</td><td>1</td><td>0</td><td>0</td><td>236</td></tr><tr><td>0</td><td>1</td><td>0</td><td>1</td><td>232</td></tr><tr><td>0</td><td>1</td><td>1</td><td>0</td><td>228</td></tr><tr><td>0</td><td>1</td><td>1</td><td>1</td><td>224</td></tr></table>				DTH_UI[3:0]				Description	D3	D2	D1	D0	0	0	0	0	252	0	0	0	1	248	0	0	1	0	244	0	0	1	1	240	0	1	0	0	236	0	1	0	1	232	0	1	1	0	228	0	1	1	1	224	<table><tr><th colspan="4">DTH_UI[3:0]</th><th rowspan="2">Description</th></tr><tr><th>D3</th><th>D2</th><th>D1</th><th>D0</th></tr><tr><td>1</td><td>0</td><td>0</td><td>0</td><td>220</td></tr><tr><td>1</td><td>0</td><td>0</td><td>1</td><td>216</td></tr><tr><td>1</td><td>0</td><td>1</td><td>0</td><td>212</td></tr><tr><td>1</td><td>0</td><td>1</td><td>1</td><td>208</td></tr><tr><td>1</td><td>1</td><td>0</td><td>0</td><td>204</td></tr><tr><td>1</td><td>1</td><td>0</td><td>1</td><td>200</td></tr><tr><td>1</td><td>1</td><td>1</td><td>0</td><td>196</td></tr><tr><td>1</td><td>1</td><td>1</td><td>1</td><td>192</td></tr></table>				DTH_UI[3:0]				Description	D3	D2	D1	D0	1	0	0	0	220	1	0	0	1	216	1	0	1	0	212	1	0	1	1	208	1	1	0	0	204	1	1	0	1	200	1	1	1	0	196	1	1	1	1	192
	DTH_UI[3:0]				Description																																																																																																					
	D3	D2	D1	D0																																																																																																						
	0	0	0	0	252																																																																																																					
	0	0	0	1	248																																																																																																					
	0	0	1	0	244																																																																																																					
	0	0	1	1	240																																																																																																					
	0	1	0	0	236																																																																																																					
	0	1	0	1	232																																																																																																					
0	1	1	0	228																																																																																																						
0	1	1	1	224																																																																																																						
DTH_UI[3:0]				Description																																																																																																						
D3	D2	D1	D0																																																																																																							
1	0	0	0	220																																																																																																						
1	0	0	1	216																																																																																																						
1	0	1	0	212																																																																																																						
1	0	1	1	208																																																																																																						
1	1	0	0	204																																																																																																						
1	1	0	1	200																																																																																																						
1	1	1	0	196																																																																																																						
1	1	1	1	192																																																																																																						
																																																																																																										
Restriction																																																																																																										
Register Availability	<table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode ON, Idle Mode OFF, Sleep OUT</td><td>Yes</td></tr><tr><td>Normal Mode ON, Idle Mode ON, Sleep OUT</td><td>Yes</td></tr><tr><td>Partial Mode ON, Idle Mode OFF, Sleep OUT</td><td>Yes</td></tr><tr><td>Partial Mode ON, Idle Mode ON, Sleep OUT</td><td>Yes</td></tr><tr><td>Sleep IN</td><td>Yes</td></tr></table>													Status	Availability	Normal Mode ON, Idle Mode OFF, Sleep OUT	Yes	Normal Mode ON, Idle Mode ON, Sleep OUT	Yes	Partial Mode ON, Idle Mode OFF, Sleep OUT	Yes	Partial Mode ON, Idle Mode ON, Sleep OUT	Yes	Sleep IN	Yes																																																																																	
	Status	Availability																																																																																																								
	Normal Mode ON, Idle Mode OFF, Sleep OUT	Yes																																																																																																								
	Normal Mode ON, Idle Mode ON, Sleep OUT	Yes																																																																																																								
	Partial Mode ON, Idle Mode OFF, Sleep OUT	Yes																																																																																																								
Partial Mode ON, Idle Mode ON, Sleep OUT	Yes																																																																																																									
Sleep IN	Yes																																																																																																									
Default	<table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power ON Sequence</td><td>4'b0100 b</td></tr><tr><td>S/W Reset</td><td>4'b0100 b</td></tr><tr><td>H/W Reset</td><td>4'b0100 b</td></tr></table>													Status	Default Value	Power ON Sequence	4'b0100 b	S/W Reset	4'b0100 b	H/W Reset	4'b0100 b																																																																																					
	Status	Default Value																																																																																																								
	Power ON Sequence	4'b0100 b																																																																																																								
	S/W Reset	4'b0100 b																																																																																																								
H/W Reset	4'b0100 b																																																																																																									

8.2.70. CABC Control 7 (CDh)

CDh	CABCCTRL6 (CABC Control 6)																																																			
	D/CX	RDX	WRX	D[15:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX																																							
Command	0	1	↑	XXXXXXXX	1	1	0	0	1	1	0	1	CDh																																							
1 st Parameter	1	1	↑	XXXXXXXX	0	DIM_MOV[2:0]			0	DIM_STILL[2:0]			XX																																							
Description	DIM_STILL [2:0]: This parameter is used set the transition time of brightness level change to avoid the sharp brightness change on vision in still mode. DIM_MOV [2:0]: This parameter is used set the transition time of brightness level change to avoid the sharp brightness change on vision in still mode. <table><tr><th colspan="3">DIM_MOV[2:0]/DIM_STILL[2 :0]</th><th rowspan="2">Description</th></tr><tr><th>D2</th><th>D1</th><th>D0</th></tr><tr><td>0</td><td>0</td><td>0</td><td>1 frame</td></tr><tr><td>0</td><td>0</td><td>1</td><td>1 frame</td></tr><tr><td>0</td><td>1</td><td>0</td><td>2 frames</td></tr><tr><td>0</td><td>1</td><td>1</td><td>4 frames</td></tr><tr><td>1</td><td>0</td><td>0</td><td>8 frames</td></tr><tr><td>1</td><td>0</td><td>1</td><td>16 frames</td></tr><tr><td>1</td><td>1</td><td>0</td><td>32 frames</td></tr><tr><td>1</td><td>1</td><td>1</td><td>64 frames</td></tr></table> Note: As above picture DIM1[2:0] mean DIM_MOV[2:0] or DIM_STILL[2:0] or DIM_UI[2:0] in different mode.													DIM_MOV[2:0]/DIM_STILL[2 :0]			Description	D2	D1	D0	0	0	0	1 frame	0	0	1	1 frame	0	1	0	2 frames	0	1	1	4 frames	1	0	0	8 frames	1	0	1	16 frames	1	1	0	32 frames	1	1	1	64 frames
	DIM_MOV[2:0]/DIM_STILL[2 :0]			Description																																																
	D2	D1	D0																																																	
	0	0	0	1 frame																																																
	0	0	1	1 frame																																																
	0	1	0	2 frames																																																
	0	1	1	4 frames																																																
	1	0	0	8 frames																																																
	1	0	1	16 frames																																																
	1	1	0	32 frames																																																
1	1	1	64 frames																																																	
Restriction																																																				
Register Availability	<table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode ON, Idle Mode OFF, Sleep OUT</td><td>Yes</td></tr><tr><td>Normal Mode ON, Idle Mode ON, Sleep OUT</td><td>Yes</td></tr><tr><td>Partial Mode ON, Idle Mode OFF, Sleep OUT</td><td>Yes</td></tr><tr><td>Partial Mode ON, Idle Mode ON, Sleep OUT</td><td>Yes</td></tr><tr><td>Sleep IN</td><td>Yes</td></tr></table>													Status	Availability	Normal Mode ON, Idle Mode OFF, Sleep OUT	Yes	Normal Mode ON, Idle Mode ON, Sleep OUT	Yes	Partial Mode ON, Idle Mode OFF, Sleep OUT	Yes	Partial Mode ON, Idle Mode ON, Sleep OUT	Yes	Sleep IN	Yes																											
Status	Availability																																																			
Normal Mode ON, Idle Mode OFF, Sleep OUT	Yes																																																			
Normal Mode ON, Idle Mode ON, Sleep OUT	Yes																																																			
Partial Mode ON, Idle Mode OFF, Sleep OUT	Yes																																																			
Partial Mode ON, Idle Mode ON, Sleep OUT	Yes																																																			
Sleep IN	Yes																																																			
Default	<table><tr><th rowspan="2">Status</th><th colspan="2">Default Value</th></tr><tr><th>DIM_MOV[2:0]</th><th>DIM_STILL[2:0]</th></tr><tr><td>Power ON Sequence</td><td>4'b100 b</td><td>3'b011 b</td></tr><tr><td>S/W Reset</td><td>4'b100 b</td><td>3'b011 b</td></tr><tr><td>H/W Reset</td><td>4'b100 b</td><td>3'b011 b</td></tr></table>													Status	Default Value		DIM_MOV[2:0]	DIM_STILL[2:0]	Power ON Sequence	4'b100 b	3'b011 b	S/W Reset	4'b100 b	3'b011 b	H/W Reset	4'b100 b	3'b011 b																									
Status	Default Value																																																			
	DIM_MOV[2:0]	DIM_STILL[2:0]																																																		
Power ON Sequence	4'b100 b	3'b011 b																																																		
S/W Reset	4'b100 b	3'b011 b																																																		
H/W Reset	4'b100 b	3'b011 b																																																		

8.2.71. CABC Control 8 (CEh)

CEh	CABCCTRL7 (CABC Control 7)																																																			
	D/CX	RDX	WRX	D[15:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX																																							
Command	0	1	↑	XXXXXXXX	1	1	0	0	1	1	1	0	CEh																																							
1 st Parameter	1	1	↑	XXXXXXXX	DIM_MIN[3:0]				0	DIM_UI[2:0]			XX																																							
Description	DIM_UI [2:0]: This parameter is used set the transition time of brightness level change to avoid the sharp brightness change on vision in UI mode. <table><tr><th colspan="3">DIM_MOV[2:0]/DIM_STILL[2 :0]</th><th rowspan="2">Description</th></tr><tr><th>D2</th><th>D1</th><th>D0</th></tr><tr><td>0</td><td>0</td><td>0</td><td>1 frame</td></tr><tr><td>0</td><td>0</td><td>1</td><td>1 frame</td></tr><tr><td>0</td><td>1</td><td>0</td><td>2 frames</td></tr><tr><td>0</td><td>1</td><td>1</td><td>4 frames</td></tr><tr><td>1</td><td>0</td><td>0</td><td>8 frames</td></tr><tr><td>1</td><td>0</td><td>1</td><td>16 frames</td></tr><tr><td>1</td><td>1</td><td>0</td><td>32 frames</td></tr><tr><td>1</td><td>1</td><td>1</td><td>64 frames</td></tr></table> Note1: As above picture DIM1[2:0] mean DIM_MOV[2:0] or DIM_STILL[2:0] or DIM_UI[2:0] in different mode. Note2: As above picture DIM2[3:0] mean DIM_MIN[3:0]. DIM_MIN [3:0]: The parameter is used to set the imitation of minimum brightness change. If the parameter is large than the difference between target brightness and current brightness, then the brightness will not change.													DIM_MOV[2:0]/DIM_STILL[2 :0]			Description	D2	D1	D0	0	0	0	1 frame	0	0	1	1 frame	0	1	0	2 frames	0	1	1	4 frames	1	0	0	8 frames	1	0	1	16 frames	1	1	0	32 frames	1	1	1	64 frames
	DIM_MOV[2:0]/DIM_STILL[2 :0]			Description																																																
	D2	D1	D0																																																	
	0	0	0	1 frame																																																
	0	0	1	1 frame																																																
	0	1	0	2 frames																																																
	0	1	1	4 frames																																																
	1	0	0	8 frames																																																
	1	0	1	16 frames																																																
	1	1	0	32 frames																																																
1	1	1	64 frames																																																	
Restriction																																																				
Register Availability	<table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode ON, Idle Mode OFF, Sleep OUT</td><td>Yes</td></tr><tr><td>Normal Mode ON, Idle Mode ON, Sleep OUT</td><td>Yes</td></tr><tr><td>Partial Mode ON, Idle Mode OFF, Sleep OUT</td><td>Yes</td></tr><tr><td>Partial Mode ON, Idle Mode ON, Sleep OUT</td><td>Yes</td></tr><tr><td>Sleep IN</td><td>Yes</td></tr></table>													Status	Availability	Normal Mode ON, Idle Mode OFF, Sleep OUT	Yes	Normal Mode ON, Idle Mode ON, Sleep OUT	Yes	Partial Mode ON, Idle Mode OFF, Sleep OUT	Yes	Partial Mode ON, Idle Mode ON, Sleep OUT	Yes	Sleep IN	Yes																											
Status	Availability																																																			
Normal Mode ON, Idle Mode OFF, Sleep OUT	Yes																																																			
Normal Mode ON, Idle Mode ON, Sleep OUT	Yes																																																			
Partial Mode ON, Idle Mode OFF, Sleep OUT	Yes																																																			
Partial Mode ON, Idle Mode ON, Sleep OUT	Yes																																																			
Sleep IN	Yes																																																			
Default	<table><tr><th rowspan="2">Status</th><th colspan="2">Default Value</th></tr><tr><th>DIM_MIN[3:0]</th><th>DIM_UI[2:0]</th></tr><tr><td>Power ON Sequence</td><td>4'b0000 b</td><td>3'b010 b</td></tr><tr><td>S/W Reset</td><td>4'b0000 b</td><td>3'b010 b</td></tr><tr><td>H/W Reset</td><td>4'b0000 b</td><td>3'b010 b</td></tr></table>													Status	Default Value		DIM_MIN[3:0]	DIM_UI[2:0]	Power ON Sequence	4'b0000 b	3'b010 b	S/W Reset	4'b0000 b	3'b010 b	H/W Reset	4'b0000 b	3'b010 b																									
Status	Default Value																																																			
	DIM_MIN[3:0]	DIM_UI[2:0]																																																		
Power ON Sequence	4'b0000 b	3'b010 b																																																		
S/W Reset	4'b0000 b	3'b010 b																																																		
H/W Reset	4'b0000 b	3'b010 b																																																		

8.2.72. CABC Control 9 (CFh)

CFh	CABCCTRL8 (CABC Control 8)																																																																																																																																												
	DCX	RDX	WRX	D[15:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX																																																																																																																																
Command	0	1	↑	XXXXXXXX	1	1	0	0	1	1	1	1	CFh																																																																																																																																
1 st Parameter	1	1	↑	XXXXXXXX	PWM_DIV[7:0]							XX																																																																																																																																	
Description	PWM_DIV [7:0]: PWM_OUT output period control. This command is used to adjust the PWM waveform period of PWM_OUT. The PWM period can be calculated using the equation in the following. $f_{\text{PWM_OUT}} = \frac{18\text{MHz}}{(\text{PWM_DIV}[7:0] + 1) \times 255}$ <table><tr><th colspan="9">PWM_DIV[7:0]</th><th rowspan="2">f_{PWM_OUT}</th></tr><tr><th>D7</th><th>D6</th><th>D5</th><th>D4</th><th>D3</th><th>D2</th><th>D1</th><th>D0</th></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>70.58 KHz</td></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>35.29 KHz</td></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td><td>23.53 KHz</td></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>1</td><td>17.64 KHz</td></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td><td>14.11KHz</td></tr><tr><td colspan="9">:</td><td>:</td></tr><tr><td colspan="9">:</td><td>:</td></tr><tr><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>0</td><td>1</td><td>1</td><td>280.0Hz</td></tr><tr><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td><td>279.0 Hz</td></tr><tr><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>0</td><td>1</td><td>277.9 Hz</td></tr><tr><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>0</td><td>276.8 Hz</td></tr><tr><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>275.8 Hz</td></tr></table>													PWM_DIV[7:0]									f _{PWM_OUT}	D7	D6	D5	D4	D3	D2	D1	D0	0	0	0	0	0	0	0	0	70.58 KHz	0	0	0	0	0	0	0	1	35.29 KHz	0	0	0	0	0	0	1	0	23.53 KHz	0	0	0	0	0	0	1	1	17.64 KHz	0	0	0	0	0	1	0	0	14.11KHz	:									:	:									:	1	1	1	1	1	0	1	1	280.0Hz	1	1	1	1	1	1	0	0	279.0 Hz	1	1	1	1	1	1	0	1	277.9 Hz	1	1	1	1	1	1	1	0	276.8 Hz	1	1	1	1	1	1	1	1	275.8 Hz
	PWM_DIV[7:0]									f _{PWM_OUT}																																																																																																																																			
	D7	D6	D5	D4	D3	D2	D1	D0																																																																																																																																					
	0	0	0	0	0	0	0	0	70.58 KHz																																																																																																																																				
	0	0	0	0	0	0	0	1	35.29 KHz																																																																																																																																				
	0	0	0	0	0	0	1	0	23.53 KHz																																																																																																																																				
	0	0	0	0	0	0	1	1	17.64 KHz																																																																																																																																				
	0	0	0	0	0	1	0	0	14.11KHz																																																																																																																																				
	:									:																																																																																																																																			
	:									:																																																																																																																																			
1	1	1	1	1	0	1	1	280.0Hz																																																																																																																																					
1	1	1	1	1	1	0	0	279.0 Hz																																																																																																																																					
1	1	1	1	1	1	0	1	277.9 Hz																																																																																																																																					
1	1	1	1	1	1	1	0	276.8 Hz																																																																																																																																					
1	1	1	1	1	1	1	1	275.8 Hz																																																																																																																																					
PWM_OUT f_{PWM_OUT} ton toff CABC ON																																																																																																																																													
Note : The output frequency tolerance of internal frequency divider in CABC is ±10%																																																																																																																																													
Restriction	EXTC should be high to enable this command																																																																																																																																												
Register Availability	<table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode ON, Idle Mode OFF, Sleep OUT</td><td>Yes</td></tr><tr><td>Normal Mode ON, Idle Mode ON, Sleep OUT</td><td>Yes</td></tr><tr><td>Partial Mode ON, Idle Mode OFF, Sleep OUT</td><td>Yes</td></tr><tr><td>Partial Mode ON, Idle Mode ON, Sleep OUT</td><td>Yes</td></tr><tr><td>Sleep IN</td><td>Yes</td></tr></table>													Status	Availability	Normal Mode ON, Idle Mode OFF, Sleep OUT	Yes	Normal Mode ON, Idle Mode ON, Sleep OUT	Yes	Partial Mode ON, Idle Mode OFF, Sleep OUT	Yes	Partial Mode ON, Idle Mode ON, Sleep OUT	Yes	Sleep IN	Yes																																																																																																																				
Status	Availability																																																																																																																																												
Normal Mode ON, Idle Mode OFF, Sleep OUT	Yes																																																																																																																																												
Normal Mode ON, Idle Mode ON, Sleep OUT	Yes																																																																																																																																												
Partial Mode ON, Idle Mode OFF, Sleep OUT	Yes																																																																																																																																												
Partial Mode ON, Idle Mode ON, Sleep OUT	Yes																																																																																																																																												
Sleep IN	Yes																																																																																																																																												
Default	<table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power ON Sequence</td><td>8'b00011000</td></tr><tr><td>H/W Reset</td><td>8'b00011000</td></tr></table>													Status	Default Value	Power ON Sequence	8'b00011000	H/W Reset	8'b00011000																																																																																																																										
Status	Default Value																																																																																																																																												
Power ON Sequence	8'b00011000																																																																																																																																												
H/W Reset	8'b00011000																																																																																																																																												

8.2.73. NV Memory Write (D0h)

D0h	NVMWR (NV Memory Write)																														
	D/CX	RDX	WRX	D[15:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX																		
Command	0	1	↑	XXXXXXXX	1	1	0	1	0	0	0	0	D0h																		
1 st Parameter	1	1	↑	XXXXXXXX	0	0	0	PGM_ADR[4:0]					XX																		
2 nd Parameter	1	1	↑	XXXXXXXX	PGM_DATA[7:0]								XX																		
Description	This command is used to program the NV memory data. After a successful OTP operation, the information of PGM_DATA [7:0] will programmed to NV memory.																														
	PGM_ADR [4:0]: The select bits of ID2, ID3 and VMF[6:0] programming.																														
	PGM_ADR[4:0]					Programmed NV Memory Selection																									
	0	0	0	0	0	ID2 programming																									
	0	0	0	0	1	ID3 programming																									
	0	0	0	1	0	VMF[6:0] programming																									
	0	0	1	0	0	MDDI V1.2 programing																									
	Others					Reserved																									
	PGM_DATA [7:0]: The programmed data.																														
Restriction																															
Register Availability	<table><tr><th colspan="2">Status</th><th>Availability</th></tr><tr><td colspan="2">Normal Mode ON, Idle Mode OFF, Sleep OUT</td><td>Yes</td></tr><tr><td colspan="2">Normal Mode ON, Idle Mode ON, Sleep OUT</td><td>Yes</td></tr><tr><td colspan="2">Partial Mode ON, Idle Mode OFF, Sleep OUT</td><td>Yes</td></tr><tr><td colspan="2">Partial Mode ON, Idle Mode ON, Sleep OUT</td><td>Yes</td></tr><tr><td colspan="2">Sleep IN</td><td>Yes</td></tr></table>													Status		Availability	Normal Mode ON, Idle Mode OFF, Sleep OUT		Yes	Normal Mode ON, Idle Mode ON, Sleep OUT		Yes	Partial Mode ON, Idle Mode OFF, Sleep OUT		Yes	Partial Mode ON, Idle Mode ON, Sleep OUT		Yes	Sleep IN		Yes
Status		Availability																													
Normal Mode ON, Idle Mode OFF, Sleep OUT		Yes																													
Normal Mode ON, Idle Mode ON, Sleep OUT		Yes																													
Partial Mode ON, Idle Mode OFF, Sleep OUT		Yes																													
Partial Mode ON, Idle Mode ON, Sleep OUT		Yes																													
Sleep IN		Yes																													
Default	<table><tr><th rowspan="2">Status</th><th colspan="2">Default Value</th></tr><tr><th>PGM_ADR[4:0]</th><th>PGM_DATA[7:0]</th></tr><tr><td>Power ON Sequence</td><td>3'b000000</td><td>8'bXXXXXXXX</td></tr><tr><td>H/W Reset</td><td>3'b000000</td><td>8'bXXXXXXXX</td></tr></table>													Status	Default Value		PGM_ADR[4:0]	PGM_DATA[7:0]	Power ON Sequence	3'b000000	8'bXXXXXXXX	H/W Reset	3'b000000	8'bXXXXXXXX							
Status	Default Value																														
	PGM_ADR[4:0]	PGM_DATA[7:0]																													
Power ON Sequence	3'b000000	8'bXXXXXXXX																													
H/W Reset	3'b000000	8'bXXXXXXXX																													

8.2.74. NV Memory Protection Key (D1h)

D1h	NVMPKEY (NV Memory Protection Key)																								
	D/CX	RDX	WRX	D[15:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX												
Command	0	1	↑	XXXXXXXX	1	1	0	1	0	0	0	1	D1h												
1 st Parameter	1	1	↑	XXXXXXXX	KEY[23:16]								55h												
2 nd Parameter	1	1	↑	XXXXXXXX	KEY[15:8]								AAh												
3 rd Parameter	1	1	↑	XXXXXXXX	KEY[7:0]								66h												
Description	KEY [23:0]: NV memory programming protection key. When writing OTP data to D0h, this register must be set to 0x55AA66h to enable OTP programming. If D1h register is not written with 0x55AA66h, then NV memory programming will be aborted.																								
Restriction																									
Register Availability	<table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode ON, Idle Mode OFF, Sleep OUT</td><td>Yes</td></tr><tr><td>Normal Mode ON, Idle Mode ON, Sleep OUT</td><td>Yes</td></tr><tr><td>Partial Mode ON, Idle Mode OFF, Sleep OUT</td><td>Yes</td></tr><tr><td>Partial Mode ON, Idle Mode ON, Sleep OUT</td><td>Yes</td></tr><tr><td>Sleep IN</td><td>Yes</td></tr></table>													Status	Availability	Normal Mode ON, Idle Mode OFF, Sleep OUT	Yes	Normal Mode ON, Idle Mode ON, Sleep OUT	Yes	Partial Mode ON, Idle Mode OFF, Sleep OUT	Yes	Partial Mode ON, Idle Mode ON, Sleep OUT	Yes	Sleep IN	Yes
Status	Availability																								
Normal Mode ON, Idle Mode OFF, Sleep OUT	Yes																								
Normal Mode ON, Idle Mode ON, Sleep OUT	Yes																								
Partial Mode ON, Idle Mode OFF, Sleep OUT	Yes																								
Partial Mode ON, Idle Mode ON, Sleep OUT	Yes																								
Sleep IN	Yes																								
Default	<table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power ON Sequence</td><td>24'h55AA66h</td></tr><tr><td>H/W Reset</td><td>24'h55AA66h</td></tr></table>													Status	Default Value	Power ON Sequence	24'h55AA66h	H/W Reset	24'h55AA66h						
Status	Default Value																								
Power ON Sequence	24'h55AA66h																								
H/W Reset	24'h55AA66h																								

8.2.75. NV Memory Status Read (D2h)

D2h	RDNVM (NV Memory Status Read)																																											
	D/CX	RDX	WRX	D[15:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX																															
Command	0	1	↑	XXXXXXXX	1	1	0	1	0	0	1	0	D2h																															
1 st Parameter	1	↑	1	XXXXXXXX	X	X	X	X	X	X	X	X	XX																															
2 nd Parameter	1	↑	1	XXXXXXXX	ID2_CNT[3:0]				ID1_CNT[3:0]				XX																															
3 rd Parameter	1	↑	1	XXXXXXXX	VMF_CNT[3:0]				ID3_CNT[3:0]				XX																															
4 th Parameter	1	↑	1	XXXXXXXX	BUSY	0	0	0	0	0	0	MDDI_V12	XX																															
5 th Parameter	1	↑	1	XXXXXXXX	OTP_DATA[7:0]								XX																															
Description	PGM_CNT [1:0]: NV memory program record. The bits will increase “+1” automatically after writing the NV_VMF [5:0] to NV memory.																																											
	<table><tr><th colspan="4">ID1_CNT[3:0]/ID2_CNT[3:0] / ID3_CNT[3:0] / VMF_CNT[3:0]</th><th>Description</th></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td><td>No Programmed</td></tr><tr><td>0</td><td>0</td><td>0</td><td>1</td><td>Programmed 1 time</td></tr><tr><td>0</td><td>0</td><td>1</td><td>1</td><td>Programmed 2 times</td></tr><tr><td>0</td><td>1</td><td>1</td><td>1</td><td>Programmed 3 times</td></tr><tr><td>1</td><td>1</td><td>1</td><td>1</td><td>Programmed 4 times</td></tr></table>										ID1_CNT[3:0]/ID2_CNT[3:0] / ID3_CNT[3:0] / VMF_CNT[3:0]				Description	0	0	0	0	No Programmed	0	0	0	1	Programmed 1 time	0	0	1	1	Programmed 2 times	0	1	1	1	Programmed 3 times	1	1	1	1	Programmed 4 times				
	ID1_CNT[3:0]/ID2_CNT[3:0] / ID3_CNT[3:0] / VMF_CNT[3:0]				Description																																							
	0	0	0	0	No Programmed																																							
	0	0	0	1	Programmed 1 time																																							
	0	0	1	1	Programmed 2 times																																							
	0	1	1	1	Programmed 3 times																																							
	1	1	1	1	Programmed 4 times																																							
	MDDI_V12: The bit =0, mean MDDI Version 1.0, the bit=1, mean MDDI Version 1.2.																																											
	BUSY: The status bit of NV memory programming.																																											
<table><tr><th>BUSY</th><th>The Status of NV Memory</th></tr><tr><td>0</td><td>Idle</td></tr><tr><td>1</td><td>Busy</td></tr></table>													BUSY	The Status of NV Memory	0	Idle	1	Busy																										
BUSY	The Status of NV Memory																																											
0	Idle																																											
1	Busy																																											
OTP_DATA [7:0]: The programmed value of VMF parameter in NV memory. This function is only for read operation.																																												
Restriction																																												
Register Availability	<table><tr><th colspan="2">Status</th><th>Availability</th></tr><tr><td colspan="2">Normal Mode ON, Idle Mode OFF, Sleep OUT</td><td>Yes</td></tr><tr><td colspan="2">Normal Mode ON, Idle Mode ON, Sleep OUT</td><td>Yes</td></tr><tr><td colspan="2">Partial Mode ON, Idle Mode OFF, Sleep OUT</td><td>Yes</td></tr><tr><td colspan="2">Partial Mode ON, Idle Mode ON, Sleep OUT</td><td>Yes</td></tr><tr><td colspan="2">Sleep IN</td><td>Yes</td></tr></table>													Status		Availability	Normal Mode ON, Idle Mode OFF, Sleep OUT		Yes	Normal Mode ON, Idle Mode ON, Sleep OUT		Yes	Partial Mode ON, Idle Mode OFF, Sleep OUT		Yes	Partial Mode ON, Idle Mode ON, Sleep OUT		Yes	Sleep IN		Yes													
	Status		Availability																																									
	Normal Mode ON, Idle Mode OFF, Sleep OUT		Yes																																									
	Normal Mode ON, Idle Mode ON, Sleep OUT		Yes																																									
	Partial Mode ON, Idle Mode OFF, Sleep OUT		Yes																																									
	Partial Mode ON, Idle Mode ON, Sleep OUT		Yes																																									
Sleep IN		Yes																																										
Default	<table><tr><th rowspan="2">Status</th><th colspan="7">Default Value</th></tr><tr><th>MDDI_V12</th><th>ID3_CNT</th><th>ID2_CNT</th><th>ID1_CNT</th><th>VMF_CNT</th><th>BUSY</th><th>OTP_DATA</th></tr><tr><td>Power ON Sequence</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td></tr><tr><td>H/W Reset</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td></tr></table>													Status	Default Value							MDDI_V12	ID3_CNT	ID2_CNT	ID1_CNT	VMF_CNT	BUSY	OTP_DATA	Power ON Sequence	X	X	X	X	X	X	X	H/W Reset	X	X	X	X	X	X	X
	Status	Default Value																																										
		MDDI_V12	ID3_CNT	ID2_CNT	ID1_CNT	VMF_CNT	BUSY	OTP_DATA																																				
	Power ON Sequence	X	X	X	X	X	X	X																																				
H/W Reset	X	X	X	X	X	X	X																																					

8.2.76. Read ID4 (D3h)

D3h	RDID4 (Read ID4)																								
	D/CX	RDX	WRX	D[15:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX												
Command	0	1	↑	XXXXXXXX	1	1	0	1	0	0	1	1	D3h												
1 st Parameter	1	↑	1	XXXXXXXX	X	X	X	X	X	X	X	X	XX												
2 nd Parameter	1	↑	1	XXXXXXXX	0	0	0	0	0	0	0	0	00h												
3 rd Parameter	1	↑	1	XXXXXXXX	1	0	0	1	0	1	0	0	94h												
4 th Parameter	1	↑	1	XXXXXXXX	1	0	0	0	0	0	1	0	86h												
Description	Read IC device code.																								
	The 1 st parameter is dummy read period.																								
	The 2 nd parameter means the IC version.																								
	The 3 rd and 4 th parameter mean the IC model name.																								
Restriction																									
Register Availability	<table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode ON, Idle Mode OFF, Sleep OUT</td><td>Yes</td></tr><tr><td>Normal Mode ON, Idle Mode ON, Sleep OUT</td><td>Yes</td></tr><tr><td>Partial Mode ON, Idle Mode OFF, Sleep OUT</td><td>Yes</td></tr><tr><td>Partial Mode ON, Idle Mode ON, Sleep OUT</td><td>Yes</td></tr><tr><td>Sleep IN</td><td>Yes</td></tr></table>													Status	Availability	Normal Mode ON, Idle Mode OFF, Sleep OUT	Yes	Normal Mode ON, Idle Mode ON, Sleep OUT	Yes	Partial Mode ON, Idle Mode OFF, Sleep OUT	Yes	Partial Mode ON, Idle Mode ON, Sleep OUT	Yes	Sleep IN	Yes
Status	Availability																								
Normal Mode ON, Idle Mode OFF, Sleep OUT	Yes																								
Normal Mode ON, Idle Mode ON, Sleep OUT	Yes																								
Partial Mode ON, Idle Mode OFF, Sleep OUT	Yes																								
Partial Mode ON, Idle Mode ON, Sleep OUT	Yes																								
Sleep IN	Yes																								
Default	<table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power ON Sequence</td><td>ID4=24'h009486h</td></tr><tr><td>H/W Reset</td><td>ID4=24'h009486h</td></tr></table>													Status	Default Value	Power ON Sequence	ID4=24'h009486h	H/W Reset	ID4=24'h009486h						
Status	Default Value																								
Power ON Sequence	ID4=24'h009486h																								
H/W Reset	ID4=24'h009486h																								

8.2.77. PGAMCTRL(Positive Gamma Control) (E0h)

	PGAMCTRL (Positive Gamma Control)																								
	D/CX	RDX	WRX	D[15:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX												
Command	0	1	↑	XXXXXXXX	1	1	1	0	0	0	0	0	E0h												
1 st Parameter	1	1	↑	XXXXXXXX	0	0	0	VP0[4:0]					XX												
2 nd Parameter	1	1	↑	XXXXXXXX	0	0	VP1[5:0]					XX													
3 rd Parameter	1	1	↑	XXXXXXXX	0	0	VP2[5:0]					XX													
4 th Parameter	1	1	↑	XXXXXXXX	0	0	0	0	VP4[3:0]				XX												
5 th Parameter	1	1	↑	XXXXXXXX	0	0	0	VP6[4:0]				XX													
6 th Parameter	1	1	↑	XXXXXXXX	0	0	0	0	VP13[3:0]				XX												
7 th Parameter	1	1	↑	XXXXXXXX	0	VP20[6:0]					XX														
8 th Parameter	1	1	↑	XXXXXXXX	VP36[3:0]				VP27[3:0]				XX												
9 th Parameter	1	1	↑	XXXXXXXX	0	VP43[6:0]						XX													
10 th Parameter	1	1	↑	XXXXXXXX	0	0	0	0	VP50[3:0]				XX												
11 th Parameter	1	1	↑	XXXXXXXX	0	0	0	VP57[4:0]					XX												
12 th Parameter	1	1	↑	XXXXXXXX	0	0	0	0	VP59[3:0]				XX												
13 th Parameter	1	1	↑	XXXXXXXX	0	0	VP61[5:0]					XX													
14 th Parameter	1	1	↑	XXXXXXXX	0	0	VP62[5:0]					XX													
15 th Parameter	1	1	↑	XXXXXXXX	0	0	0	VP63[4:0]					XX												
Description	Set the gray scale voltage to adjust the gamma characteristics of the TFT panel.																								
Restriction																									
Register Availability	<table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode ON, Idle Mode OFF, Sleep OUT</td><td>Yes</td></tr><tr><td>Normal Mode ON, Idle Mode ON, Sleep OUT</td><td>Yes</td></tr><tr><td>Partial Mode ON, Idle Mode OFF, Sleep OUT</td><td>Yes</td></tr><tr><td>Partial Mode ON, Idle Mode ON, Sleep OUT</td><td>Yes</td></tr><tr><td>Sleep IN</td><td>Yes</td></tr></table>													Status	Availability	Normal Mode ON, Idle Mode OFF, Sleep OUT	Yes	Normal Mode ON, Idle Mode ON, Sleep OUT	Yes	Partial Mode ON, Idle Mode OFF, Sleep OUT	Yes	Partial Mode ON, Idle Mode ON, Sleep OUT	Yes	Sleep IN	Yes
Status	Availability																								
Normal Mode ON, Idle Mode OFF, Sleep OUT	Yes																								
Normal Mode ON, Idle Mode ON, Sleep OUT	Yes																								
Partial Mode ON, Idle Mode OFF, Sleep OUT	Yes																								
Partial Mode ON, Idle Mode ON, Sleep OUT	Yes																								
Sleep IN	Yes																								

8.2.78. NGAMCTRL (Negative Gamma Correction) (E1h)

	NGAMCTRL (Negative Gamma Correction)																								
	D/CX	RDX	WRX	D[15:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX												
Command	0	1	↑	XXXXXXXX	1	1	1	0	0	0	0	1	E1h												
1 st Parameter	1	1	↑	XXXXXXXX	0	0	0	VN0[4:0]					XX												
2 nd Parameter	1	1	↑	XXXXXXXX	0	0	VN1[5:0]					XX													
3 rd Parameter	1	1	↑	XXXXXXXX	0	0	VN2[5:0]					XX													
4 th Parameter	1	1	↑	XXXXXXXX	0	0	0	0	VN4[3:0]				XX												
5 th Parameter	1	1	↑	XXXXXXXX	0	0	0	RVN6[4:0]					XX												
6 th Parameter	1	1	↑	XXXXXXXX	0	0	0	0	VN13[3:0]				XX												
7 th Parameter	1	1	↑	XXXXXXXX	0	VN20[6:0]					XX														
8 th Parameter	1	1	↑	XXXXXXXX	VN36[3:0]				VN27[3:0]				XX												
9 th Parameter	1	1	↑	XXXXXXXX	0	VN43[6:0]					XX														
10 th Parameter	1	1	↑	XXXXXXXX	0	0	0	0	VN50[3:0]				XX												
11 th Parameter	1	1	↑	XXXXXXXX	0	0	0	VN57[4:0]					XX												
12 th Parameter	1	1	↑	XXXXXXXX	0	0	0	0	VN59[3:0]				XX												
13 th Parameter	1	1	↑	XXXXXXXX	0	0	VN61[5:0]					XX													
14 th Parameter	1	1	↑	XXXXXXXX	0	0	VN62[5:0]					XX													
15 th Parameter	1	1	↑	XXXXXXXX	0	0	0	VN63[4:0]					XX												
Description	Set the gray scale voltage to adjust the gamma characteristics of the TFT panel.																								
Restriction																									
Register Availability	<table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode ON, Idle Mode OFF, Sleep OUT</td><td>Yes</td></tr><tr><td>Normal Mode ON, Idle Mode ON, Sleep OUT</td><td>Yes</td></tr><tr><td>Partial Mode ON, Idle Mode OFF, Sleep OUT</td><td>Yes</td></tr><tr><td>Partial Mode ON, Idle Mode ON, Sleep OUT</td><td>Yes</td></tr><tr><td>Sleep IN</td><td>Yes</td></tr></table>													Status	Availability	Normal Mode ON, Idle Mode OFF, Sleep OUT	Yes	Normal Mode ON, Idle Mode ON, Sleep OUT	Yes	Partial Mode ON, Idle Mode OFF, Sleep OUT	Yes	Partial Mode ON, Idle Mode ON, Sleep OUT	Yes	Sleep IN	Yes
	Status	Availability																							
	Normal Mode ON, Idle Mode OFF, Sleep OUT	Yes																							
	Normal Mode ON, Idle Mode ON, Sleep OUT	Yes																							
	Partial Mode ON, Idle Mode OFF, Sleep OUT	Yes																							
	Partial Mode ON, Idle Mode ON, Sleep OUT	Yes																							
Sleep IN	Yes																								

8.2.79. Digital Gamma Control 1 (E2h)

E2h	DGAMCTRL (Digital Gamma Control 1)																								
	D/CX	RDX	WRX	D[15:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX												
Command	0	1	↑	XXXXXXXX	1	1	1	0	0	0	1	0	E2h												
1 st Parameter	1	1	↑	XXXXXXXX	RCA0[3:0]				BCA0[3:0]				XX												
:	1	1	↑	XXXXXXXX	RCAx[3:0]				BCAx[3:0]				XX												
16 rd Parameter	1	1	↑	XXXXXXXX	RCA15[3:0]				BCA15[3:0]				XX												
Description	RCAx [3:0]: Gamma Macro-adjustment registers for red gamma curve. BCAx [3:0]: Gamma Macro-adjustment registers for blue gamma curve.																								
Restriction																									
Register Availability	<table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode ON, Idle Mode OFF, Sleep OUT</td><td>Yes</td></tr><tr><td>Normal Mode ON, Idle Mode ON, Sleep OUT</td><td>Yes</td></tr><tr><td>Partial Mode ON, Idle Mode OFF, Sleep OUT</td><td>Yes</td></tr><tr><td>Partial Mode ON, Idle Mode ON, Sleep OUT</td><td>Yes</td></tr><tr><td>Sleep IN</td><td>Yes</td></tr></table>													Status	Availability	Normal Mode ON, Idle Mode OFF, Sleep OUT	Yes	Normal Mode ON, Idle Mode ON, Sleep OUT	Yes	Partial Mode ON, Idle Mode OFF, Sleep OUT	Yes	Partial Mode ON, Idle Mode ON, Sleep OUT	Yes	Sleep IN	Yes
Status	Availability																								
Normal Mode ON, Idle Mode OFF, Sleep OUT	Yes																								
Normal Mode ON, Idle Mode ON, Sleep OUT	Yes																								
Partial Mode ON, Idle Mode OFF, Sleep OUT	Yes																								
Partial Mode ON, Idle Mode ON, Sleep OUT	Yes																								
Sleep IN	Yes																								
Default	<table><tr><th rowspan="2">Status</th><th colspan="2">Default Value</th></tr><tr><th>RCAx[3:0]</th><th>BCAx[3:0]</th></tr><tr><td>Power ON Sequence</td><td>TBD</td><td>TBD</td></tr><tr><td>H/W Reset</td><td>TBD</td><td>TBD</td></tr></table>													Status	Default Value		RCAx[3:0]	BCAx[3:0]	Power ON Sequence	TBD	TBD	H/W Reset	TBD	TBD	
Status	Default Value																								
	RCAx[3:0]	BCAx[3:0]																							
Power ON Sequence	TBD	TBD																							
H/W Reset	TBD	TBD																							

8.2.80. Digital Gamma Control 2 (E3h)

E3h	DGAMCTRL (Digital Gamma Control 2)																								
	D/CX	RDX	WRX	D[15:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX												
Command	0	1	↑	XXXXXXXX	1	1	1	0	0	0	1	1	E3h												
1 st Parameter	1	1	↑	XXXXXXXX	RFA0[3:0]				BFA0[3:0]				XX												
:	1	1	↑	XXXXXXXX	RFAx[3:0]				BFAx[3:0]				XX												
64 rd Parameter	1	1	↑	XXXXXXXX	RFA63[3:0]				BFA63[3:0]				XX												
Description	RFax [3:0]: Gamma Micro-adjustment register for red gamma curve. BFAx [3:0]: Gamma Micro-adjustment register for blue gamma curve.																								
Restriction																									
Register Availability	<table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode ON, Idle Mode OFF, Sleep OUT</td><td>Yes</td></tr><tr><td>Normal Mode ON, Idle Mode ON, Sleep OUT</td><td>Yes</td></tr><tr><td>Partial Mode ON, Idle Mode OFF, Sleep OUT</td><td>Yes</td></tr><tr><td>Partial Mode ON, Idle Mode ON, Sleep OUT</td><td>Yes</td></tr><tr><td>Sleep IN</td><td>Yes</td></tr></table>													Status	Availability	Normal Mode ON, Idle Mode OFF, Sleep OUT	Yes	Normal Mode ON, Idle Mode ON, Sleep OUT	Yes	Partial Mode ON, Idle Mode OFF, Sleep OUT	Yes	Partial Mode ON, Idle Mode ON, Sleep OUT	Yes	Sleep IN	Yes
Status	Availability																								
Normal Mode ON, Idle Mode OFF, Sleep OUT	Yes																								
Normal Mode ON, Idle Mode ON, Sleep OUT	Yes																								
Partial Mode ON, Idle Mode OFF, Sleep OUT	Yes																								
Partial Mode ON, Idle Mode ON, Sleep OUT	Yes																								
Sleep IN	Yes																								
Default	<table><tr><th rowspan="2">Status</th><th colspan="2">Default Value</th></tr><tr><th>RFax[3:0]</th><th>BFAx[3:0]</th></tr><tr><td>Power ON Sequence</td><td>TBD</td><td>TBD</td></tr><tr><td>H/W Reset</td><td>TBD</td><td>TBD</td></tr></table>													Status	Default Value		RFax[3:0]	BFAx[3:0]	Power ON Sequence	TBD	TBD	H/W Reset	TBD	TBD	
Status	Default Value																								
	RFax[3:0]	BFAx[3:0]																							
Power ON Sequence	TBD	TBD																							
H/W Reset	TBD	TBD																							

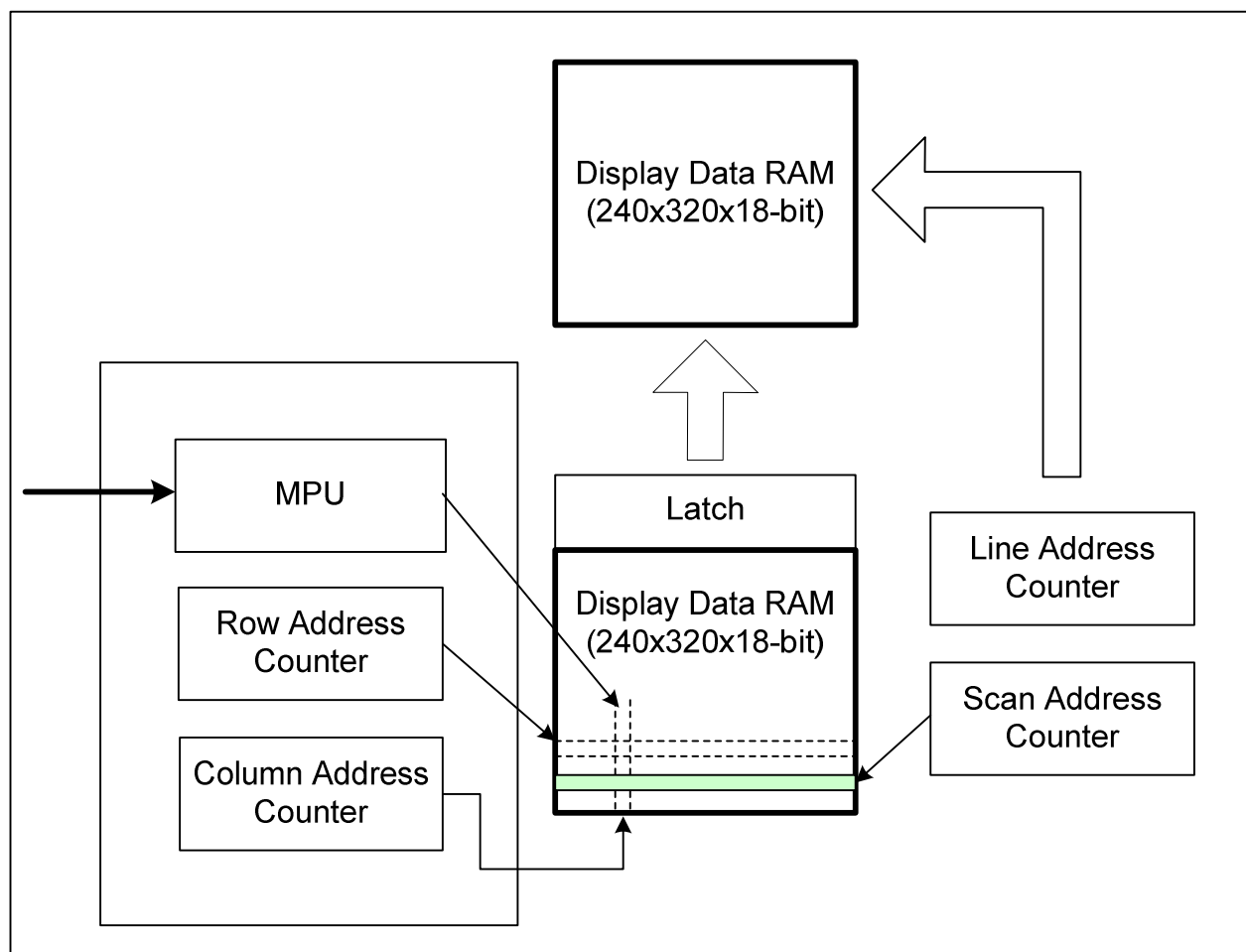
8.2.81. SPI Read Command Setting(FBh)

FBh	DGAMCTRL (Digital Gamma Control 2)																								
	D/CX	RDX	WRX	D[15:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX												
Command	0	1	↑	XXXXXXXX	1	1	1	1	1	0	1	1	FBh												
1 st Parameter	1	1	↑	XXXXXXXX	0	0	0	SPI_READ_EN	SPI_CNT[3:0]			XX													
Description	SPI_READ_EN: SPI read enable (see note). SPI_CNT [3:0]: SPI read parameter number (see note) <i>Note: Set “RFBh” once only usefull to read one parameter of register one time, the next read need to set “RFBh” again.</i>																								
Restriction																									
Register Availability	<table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode ON, Idle Mode OFF, Sleep OUT</td><td>Yes</td></tr><tr><td>Normal Mode ON, Idle Mode ON, Sleep OUT</td><td>Yes</td></tr><tr><td>Partial Mode ON, Idle Mode OFF, Sleep OUT</td><td>Yes</td></tr><tr><td>Partial Mode ON, Idle Mode ON, Sleep OUT</td><td>Yes</td></tr><tr><td>Sleep IN</td><td>Yes</td></tr></table>													Status	Availability	Normal Mode ON, Idle Mode OFF, Sleep OUT	Yes	Normal Mode ON, Idle Mode ON, Sleep OUT	Yes	Partial Mode ON, Idle Mode OFF, Sleep OUT	Yes	Partial Mode ON, Idle Mode ON, Sleep OUT	Yes	Sleep IN	Yes
Status	Availability																								
Normal Mode ON, Idle Mode OFF, Sleep OUT	Yes																								
Normal Mode ON, Idle Mode ON, Sleep OUT	Yes																								
Partial Mode ON, Idle Mode OFF, Sleep OUT	Yes																								
Partial Mode ON, Idle Mode ON, Sleep OUT	Yes																								
Sleep IN	Yes																								
Default	<table><tr><th rowspan="2">Status</th><th colspan="2">Default Value</th></tr><tr><th>SPI_READ_EN</th><th>SPI_CNT[3:0]</th></tr><tr><td>Power ON Sequence</td><td>1'b0</td><td>4'b0000</td></tr><tr><td>H/W Reset</td><td>1'b0</td><td>4'b0000</td></tr></table>													Status	Default Value		SPI_READ_EN	SPI_CNT[3:0]	Power ON Sequence	1'b0	4'b0000	H/W Reset	1'b0	4'b0000	
Status	Default Value																								
	SPI_READ_EN	SPI_CNT[3:0]																							
Power ON Sequence	1'b0	4'b0000																							
H/W Reset	1'b0	4'b0000																							

9. Display Data RAM

9.1. Configuration

The display data RAM stores display dots and consists of 345,600 bits (320x480x18 bits). There is no restriction on access to the RAM even when the display data on the same address is loaded to DAC. There will be no abnormal visible effect on the display when there is a simultaneous Panel Read and Interface Read or Write to the same location of the Frame Memory.



9.2. Memory to Display Address Mapping

In this mode, the content of the frame memory within an area where column pointer is 0000h to 013Fh and page pointer is 0000h to 01DFh is displayed.

To display a dot on leftmost top corner, store the dot data at (column pointer, page pointer) = (0,0)

